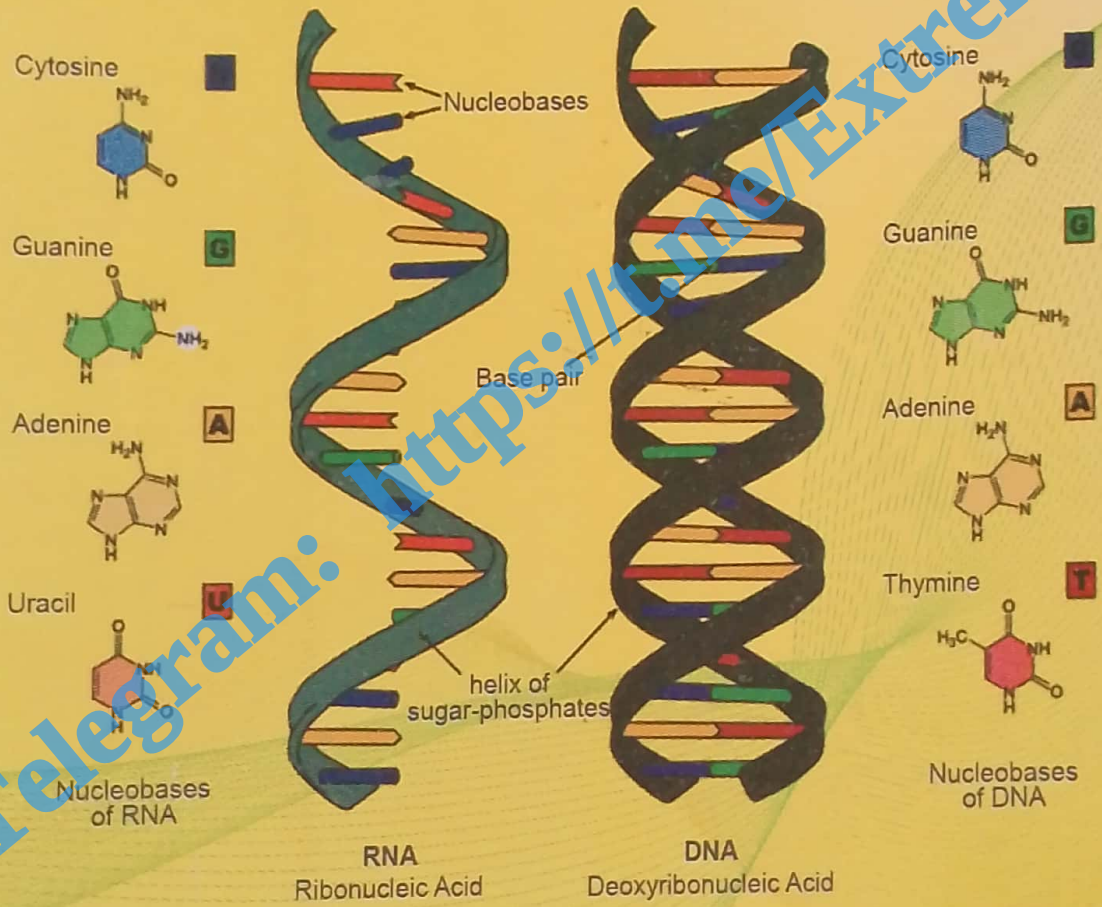
Extreme Series

BIOLOGY

GRADE 11 - 12



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Based on Common Currently Used
National and International Curriculum

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UNIT 1

1. Biology and Technology

1.1 Biology and Technology

Biology is the study of life : structures, functions, growth, origins, evolution and distribution of living organisms.

Technology is the application of scientific knowledge, skills, methods and process for the production of divices and tools for scientific investigations.

The blend of biology and technology forms biotechnology.

Biotechnology is the integration of natural and engineering sciences to achieve the application of organisms, cells, parts and moleculs to make product and services.

Examples

- Genetically modified Bacteria are used to produce industrial products such as antibiotics and insulin.
- Yeast cells are used to make bread and alcoholic drinks such as tella, beer and wine.

1.2 Learning from Nature

What is imitation (Biomimicry) ?

It is the emulation (copy) of biological structure, design and system for the purpose of solving complex human problems in engineering or technologies.

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Examples of technology imitated from nature are :

1. **House building design:** was imitated from Swallow (bird's) nest.
2. **Weaving technologies:** was imitated from spider's web.
3. **Building :** was imitated from mound architecture of termites.

   Examples

   The East gate center building in Zimbabwe (Harare) has chimneys, that keeps the room with constant temperature and humidity.
4. **Aircraft wing and their flight :** was imitated from wings of birds and bats
5. **Photograph camera:** was imitated from human eyes.
6. **The injection needle technology:** was imitated from mosquito's piercing proboscis.
7. **The sensors of electrical devices:** were imitated from biological neurons.
8. **Cutting saw :** was imitated from sharp teeth of animals.
9. **Synthetic bullet proof vests:** were imitated from the spin silks of spiders.
10. **Ceramics :** were imitated from shell fish exoskeletons.
11. **Robotic arms:** were imitated from an elephant's trunk.
12. **Submarine craft:** was imitated from the movement of shark and fish under the water.
13. **Hexagonal structure:** was imitated from bee's honey comb.
14. **Drone design:** was imitated from maple's seed whirling in air.
15. **Preserving live vaccine:** was imitated from Tardigrades (eight legged micro animals living in water), which can coat its DNA during unfavorable conditions to survive for hundreds of years.

Illustrative Examples

1. What is biotechnology ?

   A. is a blend of biology and technology

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B. is an integrated science

C. uses microorganisms for production

D. all

Explanation :- Biotechnology is :

- A blend (mixed) of biology and technology.
- An integrated science (natural science and engineering)
- An application of organisms (microorganisms) for production.

Answer : D

2. Identify the incorrect combination of technology and its imitation ?

A. a bullet train - beak of kingfisher

B. photograph camera - human eye

C. cutting saw - claws of animal

D. robotic arms - elephant trunk

Explanation

- A bullet train is imitated from diving beak of kingfisher.
- Photograph camera is imitated from human eye
- Cutting saw is imitated from sharp animal teeth not claws
- Robotic arms were imitated from elephant trunk.

Answer : C

3. The new inventing technologies need a blue print (design), which are imitated from biological ___ ?

A. Shape B. System C. Structure D. All

Explanation :- technological designs can be imitated from :

- Body parts (structure) of organisms or
- Design or shape of body parts or
- System (the way how different body parts work).

Answer : D

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1.2.1 The Benefits of Biology to Technology

Biology is a source of materials that enables us to imitate, design, adapt and develop modern technologies and solve complex human problems. This leads to the expansion of technologies.

Examples

- **Medicinal plants :** Biochemical molecules from plants are used to make medicines
- **Gene engineering and biosynthetic materials :** help to cure genetic disease and to make biosynthetic materials to repair damaged body tissues, respectively.
- **Building design :** it helps to Design buildings with efficient cooling and ventilation (imitated from mound architecture of termites).
- **Bullet train design :** imitated from the beaks of kingfisher when it dives into water.

1.2.2 Uses of Technology in Biology

It describes the application of technological tools to solve various human problems.

- What are the examples of biological studies that use technologies ?

1. **Biochemical studies:** it studies about biological molecules, found in living organisms such as carbohydrates, proteins, lipids and nucleic acids.
2. **Biomedical studies:** it deals about the chemical components of medicinal plants.
3. **Biophysical studies:** it is a science of using different instruments which help to collect biological data.
4. **Environmental studies:** it is a multidisciplinary field that deals about interaction between human with the environment.
5. **Bioinformatics:** it is a hybrid field that links biology, computer science and information technology to collect, store and analyze biological data by using different soft ware technologies (**e.g.** DNA sequencing)
6. **Biogeographical studies :** it is the study of the distribution of species

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along with geographic ecosystem.

List some examples of **technological devices** used to collect **biological data ?**

A pregnancy urine test kit : is used to check whether a woman is pregnant or not by detecting a hormone called **Human Chronic Gonadotrophin** (HCG) from blood or urine (pee).

Microscope (magnify small objects); Thermometer (to measure body temperature) ; **HIV test kit** (it is used check HIV in the blood) ; **Diabetic blood test kit** (measure blood sugar level) ; **Handheld Body Fat Calculator** (measure body fat)

Geographical Position System (GPS) :- is a device used to collect biological data (plant, animals) on land scape by mapping

Computer information Technology Scanning (CITS): investigate information on diseases from human body.

Computed Tomography Scan (CT scan) : used to detect and screen diseases from the body by using X - ray pictures ; **Position Emission Tomography (PET) :** detect disease through image, patients are let to swallow or injected radioactive dye prior to examination.

1.3 What are the impacts of Biology and Technology on the Society and the Natural World ?

The impactsof Biology :

Ensuring food security, improving health, supply of energy and clean water, industrial production, antibiotic production, to make technological devices, to make biological weapon, a predetermined sex limit on unborn children.

Note - making biological weapons and a predetermined sex limit on unborn children influence the society and the natural world negatively.

The impacts of Technology

Affect our natural world in two ways : **pollution** and **depletion of natural resources.**

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- Industrial emissions and effluents **pollute** clean air and drinking water.
- Pesticides and fertilizers cause toxicity and biohazards (*kill living* organisms, deplete natural resources).

Illustrative Examples

4. Which one of the following is a negative impact of biology on society and natural environment ?

A. improving health
B. food security
C. energy and clean water
D. biological weapons

Explanation :- the negative impacts of biology includes :

- Biological weapons: are the use of pathogenic microbes such as bacteria and viruses to harm people.
- A predetermined sex limit on unborn children (e.g. it may reduce the number of female in the society).

Answer : D

5. At what circumstances can a nurse confirms the presence of a pregnancy in a woman ?

A. when HCG decreases in the womb
B. when a one line appears on urine test kit
C. when a menstrual period flows on time
D. when HCG increases in the womb

Explanation

- A pregnancy urine test kit is used to check whether a woman is pregnant or not by detecting a hormone called **Human Chronic Gonadotrophin** (HCG) from blood or urine (pee).
- If a woman is posiitive to pregnancy, its HCG increases and two lines appear on urine test kit.

Answer : D

6. Identify the **incorrect** combination of technological devices and its importance ?

A. Handheld Body fat calculator - measures body fat
B. GPS - used to map places on earth

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C. Diabetic blood test kit - measures blood water level

D. Thermometer - measures body temperature

Explanation :- Diabetic blood test kit - measures human blood sugar level.

Answer : C

1.4 Ethical Issues in Biology

It refers to the rightness or wrongness of using biological discoveries by subjecting animal or plants to varies experiments / tests for the welfare of human.

1. 4. 1 Ethical Treatments of Plants and Animals During Biological Studies

Plants are primary producers and have decentralized functional system (no brain), subjecting plant species to severe conditions leads to a total disappearance.

What are unethical treatments of plants?

- Testing or consuming seeds in destructive way ; cutting tree terminals ; burning of forests and seedling ; restructuring the shapes of tree for aesthetics.

Animals have centralized functional system, have brain and nervous system, feel pain, have emotions and interest ; therefore it is unethical to harm them during experiments.

What are the principles (three Rs.) that researchers must use to reduce the impacts of unethical treatments of animals ?

1. **Reduction :** is minimizing the number of animals used in the experiments by improving techniques and sharing information with other.
2. **Refinement :** cleansing experiments to reduce suffering of animals through better medical care, improve living conditions, shorter time of tests.
3. **Replacement :** apply alternative techniques to substitute experimental animal.

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Examples

The use of computer simulated models, biosynthetic products, cell culture.

Supplementary Exercises

I. Matching Item : Match the Technologies Listed Under Column 'A' with Biological Nature from which they are Imitated Under Column 'B'

A	B
7. Weaving technologies	A. Elephant trunk
8. Drone technology	B. Spider web
9. Robotic arm	C. honey comb
10. Hexagonal structure	D. wings of birds
11. The injection needle technology	E. mosquito proboscis
12. Aircraft wing and their flight	F. maple seed whirling in air

II. Multiple choices Item : Choose the Best Answer from the Given Alternatives

13. The ethical principles, used to treat experimental animals by substtituting biosynthetic products rather than the animal itself is known as ___ ?

A. reduction
C. refining
B. replacing
D. all of the above

14. Which one of the following is an ethical way of using plants during experiments ?

A. cutting tree terminal
B. restructuring the shapes of tree for aesthetics
C. using seeds in sustainable way
D. burning of forests

15. An imaging technology of human body, which uses traces of radioactive dyes injected in the body to identify a disease, is ___ ?

A. microscope
B. computer information technology scanning

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C. position emission tomography

D. CT scan

16. A house design is imitated from ___ ?

A. wings of birds
C. swallow nest
B. dolphin and sharks
D. beaks of diving kingfisher

17. Which one of the following is not true about Bioinformatics ___ ?

A. it is a computer based science
B. can collect and interprets huge data at a time
C. can be used in DNA sequencing
D. has no problems of ethical issues

18. Identify the incorrect combination of technological devices and its importance ?

A. Microscope magnify small objects
B. GPS measures the hight of plants
C. Diabetic blood test kit measures blood sugar level
D. Thermometer measures body temperature

III. Short answer : Fill the Blank Space with Appropriate Terms

19. The integration of using organisms and genetic engineering sciences for the purpose of making different products that can solve human problems are called ___ ?

20. Another term of **imitation** of nature is ___ ?

21. The study of making drugs made from plants that used to cure different diseases is known as ___ ?

22. The study of biological molecules such as protein, lipid, nucleic acid and carbohydrate is called ___ ?

23. The computer technology, used to collect, store and analysis biological data such as DNA sequencing is called ___ ?

24. A technological device, used to take x - ray picture of the body to detect disease is ___ ?

25. List the three principles that help to reduce the pain of experimental animals that used for research ?

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26. The East Gate Center building in Harare / Zimbabwe is designed by imitating __ ?

27. List some unethical treatments of plants for the purpose of research and experiments ?

IV. Essay questions : Describe the Following Questions Breifly

28. How do technologies affect our natural world negatively ?

29. What are ethical issues in biology ?

30. Explain how biological organisms help human being in the discovery of technologies ?

31. If you are conducting a scientific reserch on the efficacy of a certain drug on rats ; how can you handle the pain, emotion and values of the rats in the laboratory ?

Answer of Supplementory Exercises

I. Matching :

7. B 8. F 9. A 10. C 11. E 12. D

II. Multiple choices :

13. B 14. C 15. C 16. C 17. D 18. B

III. Short Answers :

19. Biotechnology

20. Biomimicry

21. Biomedical studies

22. Biochemical studies

23. Bioinformatics

24. CT scan

25. Reducing, Replacing and Refining

26. Mound architecture of termites

27. Testing or consuming seeds in destructive ways ; cutting tree terminals ; burning of forests and seedling ; restructuring the shaps of tree for aesthetics.

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IV. Essay Type

28. Technologies affect our natural world in two ways : pollution and depletion of natural resources.

- Industrial emissions and effluents pollute clean air and drinking water.
- Pesticides and fertilizers cause toxicity and biohazards.

29. It refers to the rightness or wrongness of using biological discoveries by subjecting animal or plants to varies experiments / tests for the welfare of human.

30. Different technological designs are imitated from biological nature; therefore human beings have gained greater advantage from nature in the aspects of technological advancement.

31. During experimentation in the laboratory you should consider the ethical issues that help to reduce the pain of the rats by following the the three principles (reducing, replacing and refining).

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UNIT 2

2 . Animals

Animals are multicellular and heterotrophic organisms that lack cell walls.

Some animals, are called **vertebrates**, which have a backbone and other animals are called **invertebrates**, do not have a backbone.

- Invertebrates account for more than 95 percent of all animal species alive today.
- Most members of the animal kingdom have common characteristics, including sexual reproduction, movement, multicellular organization, and heterotrophy.

2.1 What are Invertebrate Animals?

Invertebrates : are animal that do not have a backbone. It makes up the greatest number of animals species.

- Most invertebrates display **radial or bilateral** symmetry
- Some invertebrate don't have body symmetry and true tissues, example, sponges
- They are **cold blooded** animals that can't regulate their body temperature
- Have **exoskeleton,** which is a rigid outer covering that protects the soft tissues of many animals

Circulatory system of invertebrate animals is : **open circulatory** system and **closed circulatory** system

i. **Open circulatory system -** in which circulatory fluid is pumped by the heart through vessels and into the body cavity and is then returned to the vessels.

ii. **Closed circulatory system -** blood is pumped by a heart and circulates through the body in vessel that form a closed loop.

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Examples of invertebrate animals includes earthworms, insects, spiders, snails, sponges, jellyfish, lobsters, crabs, sea stars, and squid.

Note :

A body plan describes an animal's shape, symmetry, and internal organization. Symmetry is a body arrangement in which parts that lie on opposite sides of an axis are identical. **Types of symmetry:**

i. **Radial symmetry** is a body plan in which the parts are organized in a circle around an axis. Example, cnidarians, sea anemone. Animals that have a top and bottom side, but no front, back, right, or left end are called

ii. **Bilateral symmetry** is a body plan in which the parts are organized into two similar halves on either side of a central plane. Most animals have a dorsal (back) and ventral (abdomen) side, an anterior (toward the head) and posterior (toward the tail) end.

iii. **Asymmetrical** is animal's body plan that does not display symmetry. Sponges have the simplest body plan of all animals, which are asymmetrical.

2.2 What are Vertebrates Animals?

Vertebrates are **chordates,** animals that have a **backbone.** They have specialized organ systems such as the circulatory systems, respiratory systems, nervous systems, and excretory systems :

- They are **bilaterally** symmetrical.
- The respiratory system consists of either **gills** or lungs
- The excretory system of vertebrates consist of **paired kidneys**
- They have a **centralized nervous system** with a brain and sensory organs (eyes, ears, nostrils).
- Classes of vertebrates include fishes, amphibians, reptiles, birds, and mammals
- All vertebrate classes except fishes spend part or all of their life on land
- Vertebrates have an **endoskeleton,** an internal skeleton made of bone and cartilage
- Vertebrates have a **closed circulatory** system with a multi - chambered heart
- Some vertebrate are **endothermic** (birds, and mammals) and some are

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ectothermic (fishes, amphibians, reptiles).

2.3 Reproduction in Animals

Reproduction is the process by which organisms reproduce or perpetuate its own kind to ensure species' continuity and survival. There are two types of reproduction : **sexual** and **asexual**

2.3.1 What is an A sexual Reproduction?

A sexual reproduction is a mode of reproduction in which a new offspring is produced by a single parent

- The new individual produced are **genetically** and **physically** identical to each other
- It involves a single individual and does not require the fusion of gametes from two parents
- It is more common among invertebrates than in vertebrates.
- Rapid multiplication. **Example**, bacteria can divide once every 20 minutes.
- **Budding** and **fragmentation** are the most common forms of asexual reproduction
- **Parthenogenesis** is a type of asexual reproduction, unfertilized eggs develop into new offspring as in some insects and vertebrates

2.3.2 What is Sexual Reproduction

Sexual reproduction is the production of offspring by the fusion of two gametes to form a diploid zygote which develop into mature organism.

- It requires the fusion of gametes from two parents, male and female
- The fusion of gamete is called **fertilization** to form a diploid cell known as the zygote
- Each of the male and female gametes in haploid (n)
- During fertilization the gametes are fused to forms diploid (2n)

There are two types of fertilization : **internal** and **extend** fertilization

1. Internal fertilization

In internal fertilization the male deposits the sperm cells inside the female vagina

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- The fusion of male gamete and female gamete is takes place in the reproductive organ of female
- Examples of animals that use internal fertilization includes : mammals, birds, insects, and reptiles.

2. External fertilization

In external fertilization both male and female release the gamete into the water.

- Zygote is formed in the water, outside the body of the organism
- A number of gametes are released into the water to increase the chance of fertilization.

Note: The female gametes are not amniotic, rather it is covered by jelly - like substance, because :

- It is used to protect the egg from drying out
- It helps eggs to stick together to escape from wave of water current

Examples of animals that use external fertilization includes : fish and frog

What is next after fertilization ?

At the start of fertilization, the sperm nucleus merges with the egg nucleus to form the diploid nucleus of the zygote.

The series of cell divisions that occurs immediately following fertilization is called **cleavage**

- During cleavage, mitotic divisions rapidly increase the number of cells, but the cells do not grow in size. The number of cells increases, from 2 to 4, then to 8, and so on.

As the number of dividing cells further increases, it forms a mass a hollow ball of cells called a **blastula.**

- The blastula begins to collapse in ward and reorganized to form **gastrula,** this process is called **gastrulation.**
- Gastrula is a multilayered embryo or germ layer (endoderm, mesoderm and ectoderm).
- Each of the germ layers formed during gastrulation develops into organs in a process called **organogenesis.**

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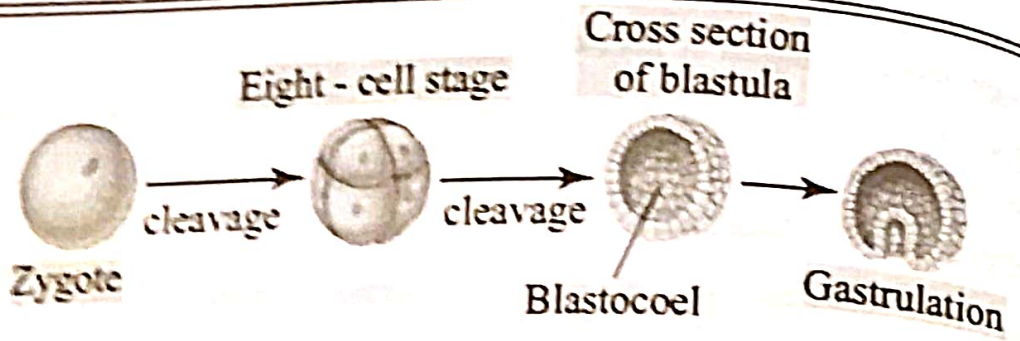

- The **endoderm** forms the lining of the urinary system, the reproductive system, and most of the digestive tract ; it also forms the pancreas, liver, lungs, and gills.
- The **ectoderm** forms the outer layer of skin, hair, nails, and the nervous system
- The **mesoderm** forms many body parts, including the skeleton, muscles, and the inner layer of skin, the circulatory system, and the lining of the body cavity.

Illustrative Examples

1. Sexual reproduction is characterized by all of the following, except ___

   A. Involves one parent

   B. Have diploid zygote

   C. have genetic variation

   D. fusion of male and female gametes

   Explanation :- sexual reproduction involves two parents male and female, have genetic variation, produce diploid zygote.

   Answer : A

2. In animal embryonic developmental stage that forms germ layer of cells is ?

   A. Cleavage

   B. blastula

   C. gastrula

   D. organogenesis

   Explanation :- Gastrula is a multilayered embryo or germ layer (endoderm, mesoderm and ectoderm)

   Answer : C

3. Which one of the following is true about vertebrate animal ?

   A. Body symmetry is radial

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B. mode of nutrition is autotrophs

C. most have external exoskeleton

D. have internal endoskeleton

Explanation :- vertebrates are bilaterally symmetrical, have endoskeleton, and heterotrophs.

Answer : D

4. Sexual reproduction differ from asexual reproduction in that:

A. Involves one parent

B. brings genetic variation

C. takes short period

D. the offspring looks like its parents

Explanation :- sexual reproduction involves two parents, there is genetic variation between the parent and offspring, and takes longer time.

Answer : B

Feature	**Sexual reproduction**	**Asexual reproduction**
Number of parent involves	One	Two
The type of animal mostly involves	Vertebrates	Invertebrates
Variation	Have genetic variation	Have no genetic variation
Time taken	Longer	Shorter

5. What is the difference between vertebrate and invertebrate animals ?

Explanation :-

Feature	**Vertebrate**	**Invertebrate**
Skeleton	endoskeleton	exoskeleton
Body symmetry	bilateral	Bilateral or radial
Circulatory system	closed	Open or closed
Mode of nutrition	heterotrophs	Parasitic, hetrotrophs

5. Arrange the following term in the correct order from simple to complex (gastrula, cleavae, and blastula) ?

Explanation : - cleavage, blastula and gastrula

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2.3.3 Reproduction in insects

The characteristics of insects include :

- Insects are the most diverse groups of animals
- The largest class of the phylum is arthropod
- Arthropods have segmented bodies, jointed legs, and exoskeletons
- They undergo sexual reproduction and internal fertilization

Examples of insects include flies, grasshoppers, lice, butterflies, bees a beetles.

Metamorphosis in insects:

After hatching from the egg, a young insect must undergo several mol before it reaches its adult size and becomes sexually mature.

A series of changes in the development of insects are called **metamorphosis** which is regulated by a hormone called **ecdyson.**

There are two types of metamorphosis : incomplete and complet

1. Complete metamorphosis :

It has four stages : egg, larvae, pupa and adult

- The immature form are called **larvae**
- The larvae stage is active and called the feeding stage
- The pupa stage is the **dormant** or inactive stage
- A pupa is a stage of development in which an insect changes from a larva to an adult.
- The adult stage is the reproductive stage
- Larvae don't resemble adults
- Most insects go through complete metamorphosis

Examples in insets with complete metamorphosis are : butterflies, moths, beetles, flies, ants, bees and wasps

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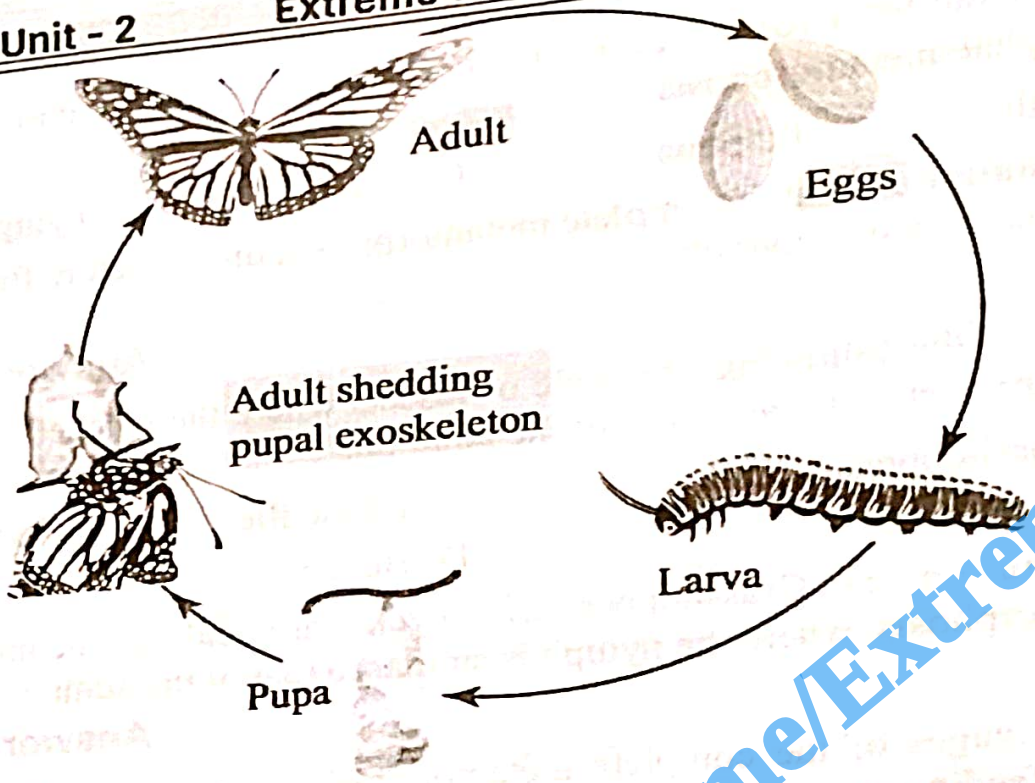
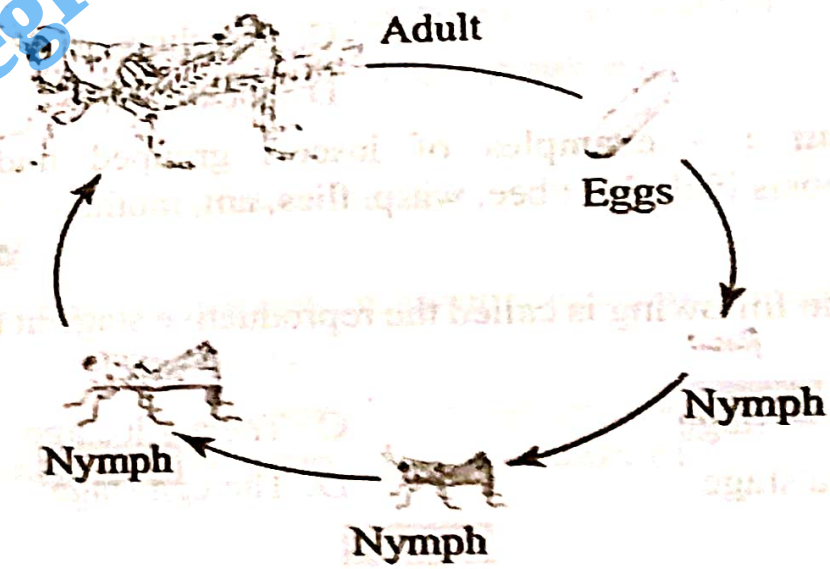


2. Incomplete metamorphosis :

Has three stages : egg, nymph, and adult

- The immature form is called nymph
- The nymph resembles the adult but it is smaller in size
- The nymph undergo a series of molting and finally form fully expanded wing in adult

Examples of insects with incomplete metamorphosis are: grasshopper, cockroach, termites



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Illustrative Example

7. What is the term for the immature form of an insect that undergoes incomplete metamorphosis ?

A. adult B. pupa C. infant D. nymph

Explanation :- in an incomplete metamorphosis, the immature form of insect is known as nymph.

Answer : D

8. Which of the following life cycle of insects where the hatched larvae are completely similar to that of the adult one ?

A. Grasshopper
B. Honey bees'
C. House flies
D. Butterfly

Explanation :- Grasshopper life cycle are called incomplete metamorphosis, where the nymph is similar to that of the adult.

Answer : A

9. Which stages of the complete metamorphosis in insect is called the active feeding stage and the inactive dormant stage, respectively?

A. larvae, pupa
B. pupa, larvae
C. adult, larvae
D. pupa, larvae

Explanation :- The larva stage is called the active, feeding stage, and the pupa stage is the inactive dormant stage.

Answer : A

10. Which of the following insects are grouped under complete metamorphosis ?

A. Bee
B. termite
C. grasshopper
D. cockroach

Explanation : - examples of insects grouped under complete metamorphosis includes : bee, wasp, flies, ant, moth.

Answer : A

11. Which of the following is called the reproductive stage in the insect life cycle ?

A. The larvae stage
B. The pupa stage
C. The adult stage
D. The egg stage

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Explanation :- the reproductive stage of insect is called the adult stage.

Answer : C

12. What is the difference between complete and incomplete metamorphosis

Explanation

Feature	Complete metamorphosis	Incomplete metamorphosis
Number of stage	4	3
Immature form	larvae	nymph
examples	Bee, wasp, flies	Grasshopper, cockroach, termites

What are the beneficial aspects of insects ?

Insects are useful in different aspects, such as in agriculture, food, industry, health and medicine

1. Insects in the field of agriculture :

It includes : pollinators, pest regulation, weed control, and soil fertility.

i. Insect as pollinator :

Insects have the potential to transfer male gametes (pollen) to the female gametes, resulting in pollination while foraging on flowering plants to obtain nectar or pollen. For instance, an estimated 35 % of crop production yielded in the world is a result of insect pollination.

ii. Insects as a pest regulation :

Some insects are useful for controlling agricultural pests. They attach and feed on other harmful insects that infect both animal and plants.

Such types of pest control is called **biological control**

For example :

- Mantis, lady beetles, ground beetles, rove beetles, flower bugs, lacewings and hover files.
- Stagmomantis insects, species of mantis feed on grasshoppers and caterpillars that damage crops.
- Chilomenes and a ladybird beetle, feed on aphids that damage cotton plants and destroys scale worms that are pests of orange and lemon trees respectively.

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- Epicauta, a blister beetle, eat up masses of the eggs of locusts

iii. Insects used for weed control and soil fertility :

- ☑ Insects also play a great role in feeding on unwanted weeds
- ☑ Improve soil aeration by creating channels for smaller organisms' to get water, and air.
  - It enhances the nutrient cycle
  - Improve soil structure and tilt
  - Decomposers recycle nutrients

2. Insects used as food

Many species of insects are being used as a food for people in many countries.

- ☑ Insects have a potential to become a valuable animal protein source
- ☑ There are over 1, 462 recorded species of edible insects in the world
- ☑ Most insects are consumed in Asia and Central America
  - For example, crickets, grasshoppers, beetle and moth larvae and termites
- ☑ **Grasshoppers** have been eaten in many parts of the world due to **high protein** sources
- ☑ A source of food for many birds, amphibians, reptiles, fish and mammals.
- ☑ Insect farming is a great potential to address the demand of food security for growing human population. Because insects are :
  - They are nutritious, with high protein, fat and mineral contents
  - Can be reared easily
  - Found everywhere
  - Reproduce quickly
  - High growth and feed conversion rates
  - Low environmental damage over their entire life cycle

3. Insects used as raw material for Industry :

Insects are being used to produce different materials at home and in industries for commercial purpose. For examples :

i. Production of Honey and Bee wax

- **Honeybees (Apis meliffera L)** produces honey and wax

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i. Production of silk

- Silk fiber is produced by silk worms (**Bombyx mori** and other silk worms), which is used for textile industry.

iii. Production of shellac :

- Shellac is commercial resin secreted by lac insects (**Laccifer lacca**). which is the commercially cultured.
- Shellac is still in use as dyes, inks, polishes, sealing waxes, and as stiffening agents in the fabrication of felt hats.

iv. Production of Cochineal :

- Cochineal pigment is extracted from insects such as **Dacylopicus coccus**
- The cochineal pigment is important for the intensity and Permanency of colors in painting.
- The cochineal pigment is also used in food colors, beverages, cosmetics (lipsticks) and art product

v. Production of Tannic acid :

- Tannic acid is a chemical compound used in dyeing leather in leather industries
- For tannin and manufacturing of some inks.
- Example, Tiny wasps in the family Cynipidae secrete some chemical and in response to this, the tree produces gall tissues that contain tannic acid.

4. Health and medicine :

Chemical products from some insects have medicinal value in treating diseases

Insect products have been used for medicine in human history. For example:

- Honey is applied to treat burns, chronic and post - surgical wounds.
- Bee and ant venomare used to treat joints pain
- Bee products can promote healthy immune systems, improve circulation and decrease inflammation

Blister beetles secrete **cantharidan**, used to treat viral infections because it prevents the reproduction of some viral particles.

- Cantharidan have a potential for treatment of cancerous tumors, which

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has been resistant to radiation and chemotherapy.

Several African cultures use **poultices** made from ground grasshoppers as pain relievers, especially for migraines.

What are the harmful aspects of insects ?

Most insects are useful ; however, there are also harmful to humans and animals.

Some insects are pests of plants, fruits, and grains in a store, spoiling products.

They feed on parts of plants and crops, such as leaves, stems, buds, flowers, fruits, and seeds ; reducing production.

- These insects include locusts, caterpillars, bugs, hoppers, aphids etc

Locusts are among the most destructive of all insect pests

- It is common in hot and dry regions
- Food shortage has further forced them to migrate from place to place
- They migrate in huge swarms, for several kilometers and destroying plants on their way

i. Some insects are pests for stored cereal grains. For example:

Rice Weevil (*Sitophilus orzae*); Lesser Grain Borer (*Rhyzopertha dom inica*); Rust Red flour Beetle : (Tribolium spp); Sawtooth Grain Beetle : (*Oryzaephilus surinamensis*); Flat Grain Beetle : (Cryptolestes spp.)

ii. Some insects are **vector of disease or intermediate hosts** for several pathogens and transfer disease. for example :

- Anopheles mosquitoes toes transfer malarial parasites, "Plasmodium, " from one person to another
- Culex mosquitos spread filariasis and transmit filarial worms from infected to healthy people
- The tsetse fly, *Trypanosoma gambiense*, also spreads the African sleeping sickness to the human population.
- The housefly (*Musca domestical*) spreads food and water - borne diseases to human populations.

Illustrative Examples

13. Which of the following is the most destructive of all insect pests ?

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A. Mosquito B. Thick C. Locusts D. Caterpillar

Explanation : - the most destructive of all insect on plants is locust, they move in the form of swarm.

Answer : C

14. Which of the following insect have more valuable protein sources for human ?

A. Grasshopper C. Crickets

B. Larvae of moths and termite D. Beetles

Explanation : - grasshopper is the best source of protein food among the other insects.

Answer : A

15. Insects are considered to be beneficial because they are _____ ?

A. Effective pollinator C. industrial raw material

B. Sources of food D. all is answer

Explanation : - the beneficial aspects of insects are : a source of food, pollinator, raw material for industries, etc.

Answer : D

16. List some benefits of insects with examples in ?

Explanation : - insects can be used in agriculture, food, medicine, and source of material for industries.

Examples	Benefits
a) Agriculture	Pollinator, biological pest control
b) Food	A source of protein, fat and mineral content
c) Industry	Insect products such as : silk, shellac, Cochineal, tannic acid
d) Medicine	Insect products such as : cantharidan, honey, poultices used to treat disease

17. Write examples of insect which transfer disease, and mention the type of disease with causative agents ?

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Explanation :-

Types of Insects	Disease causing agents	Type of disease
Anopheles mosquito	Plasmodium	malaria
Tsetse fly	trypanosoma	African sleeping sickness
Culex mosquitos	filarial worms	filariasis
House fly	bacterial	Food and water born disease

2.3.4 Reproduction in frog

The process of fertilization is begins with **courtship**, male frogs vocal call to attract females of the same species. Frogs such as *ptychadena harenna* and *Leptopelis ragazzi* are found in the Bale Mountains and Shoa forests, Ethiopia.

- The female responds only to the call from a male of the same species
- When a female approaches, the male frog climbs onto her back **(amplexus)** until she lays her eggs.
- When the female finally releases her eggs into the water, the male frog discharges his sperm over them, this is called **external fertilization.**

The reproductive system of the male frog includes:

- Two bean shaped testes located near the kidneys
- Sperm cells develop in the testes and pass through tubes to the kidneys and urinary ducts
- During mating, sperm leave the body through the cloacal opening
- Cloaca is serving as an exit cavity for excretion and reproductive organ

The reproductive system of the female frog includes:

- A pair of large ovaries containing thousands of tiny immature eggs lies near the kidneys
- The eggs enlarge, mature, and burst through the thin ovarian walls into the body cavity
- Cilia move the eggs forward into the funnel - like openings of the oviducts
- The eggs have no protective shell and membrane ; it is coated with a protective jelly like material to protect from drying out

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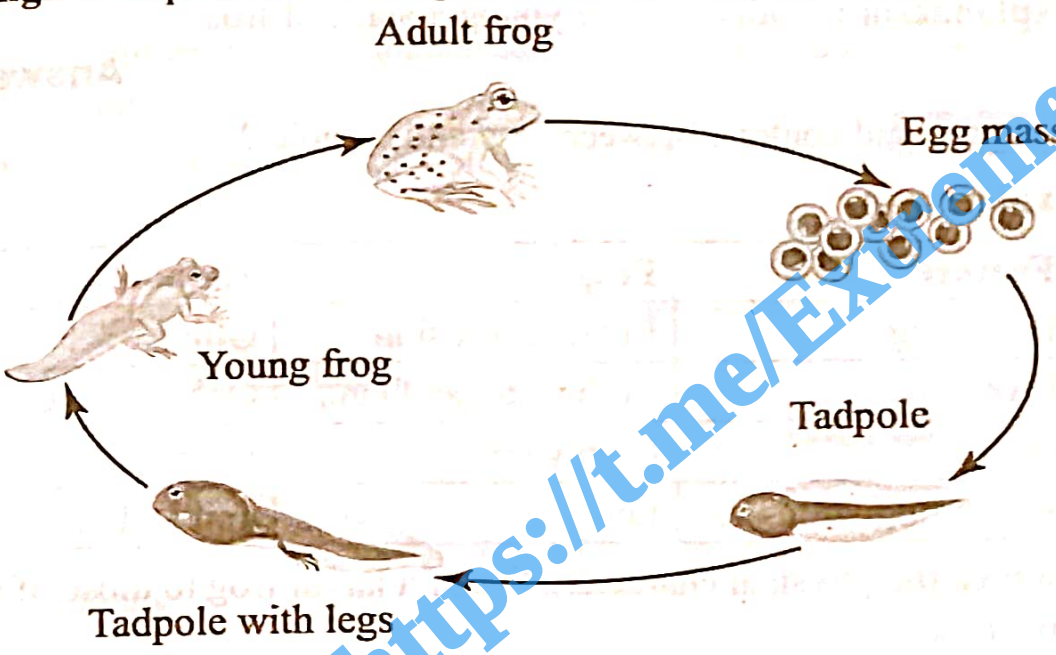
- The eggs exit by the cloaca to the external environment
- Within a few days of fertilization, the eggs hatch into tadpoles
- A newly hatched tadpole lives off yolk stored in its body
- Tadpole is totally **aquatic,** have **gills** and **tail**
- The tadpole grows and slowly changes from an **aquatic** larva into an **adult**
- The adult frog can live both in water and on land

During **metamorphosis,** the tadpole loses its **tail** and **gills** and grows legs and **lungs.** This process of change is called **metamorphosis.**



Illustrative Examples

18. Metamorphosis must take place before amphibians are able to do what ?

A. swim
B. live on land
C. respire with gills
D. feed themselves

Explanation :- in the metamorphosis of amphibians tadpole is changed into frog. That means the change is from water to land, before live on land, breath through lung, and develop leg.

Answer : B

19. Amphibians must lay eggs in water primarily for what reason ?

A. The eggs are not laid in nests
B. The eggs need oxygen from water
C. The eggs need protection from predators

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D. The eggs do not have membranes and a shell

Explanation :- The eggs of amphibians are not amniotic and have no shell membrane for protection. The eggs are covered by a jelly like substances to prevent drying out.

Answer : D

20. The younger or immature stage of frog, which lives in the water, is known as ___ ?

A. Larvae B. Tadpole C. Nymph D. Pups

Explanation : - tadpole is the younger stage of frog.

Answer : B

21. Compare and contrast between frog and tadpole ?

Explanation

Feature	Frog	Tadpole
Breathing	Lung, moist skin	Gills
Live in	Both water and land	Only in water
Feeding	Carnivore	Herbivore
Swimming structure	Leg	Tail and fin

22. Contrast the physical characteristics of a larval frog to those of an adult frog ?

Explanation : - Larval frog is called tadpole, have tail and fins, no leg, and have gills to breath and smaller in size. Adult frog have legs, lung to breath, and larger in size.

23. Explain how male frogs attract the attention of female frogs during mating season

Explanation : - during **courtship,** male frogs vocal call to attract females of the same species

2.3.5 Reproduction in Crocodile

Crocodiles are large semi - aquatic reptiles with lizard like appearance and carnivore, that live in different parts of the world. Crocodiles reproduce sexually involving both male and female parents

☑ Crocodiles are **oviparous,** meaning they lay eggs that are fertilized internally by the male

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- ☑ The **mating season** varies depending on the climate, but usually occurs in **spring** or **rainy months.**
- ☑ Males attract females and compete with each other for the right to mate, sometimes for hours
- ☑ Crocodiles are generally **polygynous,** which means that **males try to reproduce with as many females as possible.**
- ☑ The female digs a hole in the sand or soil and lies about 10 - 100 eggs, then covers them and guards them for 80 days. They lay hard, leathery eggs that enable them to protect their young.
- ☑ The temperature of incubation of egg determines the sex of the hatchlings : warmer nests produce males, cooler nests produce females.
  - ➥ Eggs incubated at 30°c will give birth to females, while at higher temperatures of 34°c will be males.
- ☑ The mother helps the babies out of the eggs and carries them to the water in her mouth, where they face many predators.

2.3.6 Reproduction in Birds

In the male bird, sperm is produced in two testes that lie anterior to the kidneys.

Sperm passes through small tubes called **vasa vas deferens** into the male 's cloaca.

- ➥ During mating, the male presses his cloaca to the female's cloaca and releases sperm

Most females have a **single ovary** located on the left side of the body.

The **oviduct** consists of the **infundibulum, magnum, isthmus, uterus, and vagina**

- ➥ The ovary releases eggs into a long funnel shaped oviduct, called **infundibulum** to engulf the yolk
- ➥ Fertilized eggs move down the magnum, which is the longest part of oviduct, where the **albumin** is made
- ➥ Then it move to isthmus, which is the shortest part of oviduct, where the **egg shell membrane** is made
- ➥ The egg move to the uterus, where they are enclosed in a protective covering and a **shell.**
- ➥ The egg passes out of the oviduct and into the cloaca

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- From the cloaca, it is expelled from the bird

During hatching the young chick inside the egg develops egg tooth to break out of the egg shell

- The young loses the egg tooth after hatching
- The chick also communicates with its parents a day or two before hatching, with some **vocal sounds.**

Nest Building and Parental Care in birds:

- Birds usually lay their eggs in a nest. Some birds do not use nests. Birds that make nests in an open area have **camouflaged eggs.**
- Nests hold the eggs conceal young birds from predators, provide shelter from the elements, and sometimes serve to attract a mate.
- One or both parents warm, or incubate, the eggs by sitting on them with featherless patch of skin on the abdomen called **a brood patch.**
- Birds rotate their eggs periodically to ensure an even distribution of warmth.
- Once the eggs hatch, the young usually receive extensive parental care.

2.3.7 Reproduction in Rat

Rats are the members of rodent families. The black rat **(Rattus rattus)** and the brown at **(Rattus norvegicus)** are among the most common types of rats species. They live every where that human populations have settled.

The black rat is predominantly live in warmer climates, and the brown rats are in temperate region.

Giant Mole rat (Tachyoryctes macrocephalus), is **endemic to Ethiopia** ; live in high altitude, Afro - alpine habitat such as the Bale Mountains.

The male reproductive structure of a rat consists of testes, scrotum, seminiferous tubules, epididymis, vasdeferens and penis with bacula.

The female reproductive structure of a rat consists of two ovaries, **oviducts, uterine horns** and **vagina with vulva.**

Pregnancy and Development in rats :

- Reproduction in rats is sexual and fertilization is **internal** in the body of a female.
- In brown - black rats, the age of sexual maturity is 3 - 4 months old, but in Giant mole rats it is 4 - 6 months

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- Gestation lasts an average of 22 - 24 days, and that of giant mole rats is 37 - 49 days
- Rats can have up to 7 - 12 pups in a single birth, but the number is fewer for giant mole rats
- Female rats can reproduce again 48 hours after giving birth.
- In three years, rats can give birth to up to **half a billion babies** under ideal conditions, but the pregnancy rate begins to decline when they turn one year old.

Parental care in rats :

- Before giving birth, pregnant females build their nests
- Rats prefer to build their nest in hidden and isolated places so their offspring are protected
- The new born or young of rat are called **pups or kittens**
- The new born rats are hairless, deaf with sealed eyelids offspring
- About 2 - 3 weeks after birth, their fur grows, their eyes open, and they are ready to leave the nest
- They feed on mothers milk

The female rats care pups regardless of which their true mothers are.

- If a mother dies, the other females will take over nursing her pups
- Male rats do not participate in the parental care

Illustrative Examples

24. Which of the following animals have hard, leathery eggs that enable them to protect their young ?

A. Frog
B. Crocodilians
C. Tad - pole
D. Rat

Explanation :- crocodilians lay hard, leathery eggs that enable them to protect their young.

Answer : B

25. Which of the following are not the characteristics of rats ?

A. They are mammal
B. Internal fertilization

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C. Have one ovary

D. Fertilization takes place in the oviduct

Explanation :- rat are mammals, have two ovaries, undergo internal fertilization, fertilization occur in the oviduct

Answer : C

26. During incubation, birds ensure an evenly distribution of warmth to the egg through ___ ?

A. Brood patch
C. sun light
B. rotating the egg
D. body temperature

Explanation : - during incubation, rotating the egg ensures an evenly distribution of heat.

Answer : B

27. The female reproductive organ of birds, where egg white (albumin) is made ?

A. isthmus
C. uterus
B. magnum
D. infundibulum

Explanation : - the magnum, which is the longest part of oviduct, where the albumin is made.

Answer : B

28. Sex determination in crocodile is determined by ____ ?

A. Temperature
C. environmental temperature
B. brood hatch
D. sex chromosome

Explanation : - crocodile have no sex chromosome ; the sex is determined by the temperature of egg incubation ; eggs incubated at 30°c will give birth to females, while at higher temperatures of 34°c will be males.

Answer : A

29. The main difference between rat and giant mole rat is that, rat is ___ ?

A. produce less pups
C. is larger in size
B. Larger gestation period
D. earlier sexual maturity

Explanation :- the rate is differ from giant mole rat in that, rat produces more pups, earlier sexual maturity smaller in size and lesser gestation period.

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Answer : D

2.4 Animal Behavior

Behavior is defined as an action that an individual carries out in response to a stimulus. Or it is the co - ordinated response of an organism to an internal or external stimulus. The study of animal behavior is called **ethology.**

Stimuli - is something in the environment that an organism might respond to, such as sound, smell, etc

Response - is the organism's reaction to the stimulus, such as blinking of eyes, withdrawal of hands from hot plate.

Types of Animal Behavior

Animal behavior can be categorized into two main types : **innate behavior and learned behavior**

2 . 4 . 1 What is Innate Behavior?

- Innate behavior is behavior that is present at birth or inborn
- It does not have to be learned
- The behavior that is generally programmed
- There are three types of innate behavior : **reflex action, instinctive, and orientation**

i. **Reflex actions** - is a sudden, involuntary response to stimuli.

- These are the simplest of the innate behaviors
- A single action is performed in response to a specific stimulus
- The messages about stimuli is processed in the spinal cord.

For example :

- The withdrawal reflex of hands from hot object
- knee jerk, sneezing
- blinking of eyes for dust

ii. **Instinctive behaviors** - these are often involving the most complex behaviors, inborn, stereotyped behavior pattern

- They are unique to each species

Once begun, the action pattern is carried out to completion , even if other stimuli intervene. Examples :

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- Web making in spiders,
- Nest - building in birds
- Swimming with dolphins
- Opening of mouth in chicks of many bird species when their mother returns to the nest
- Honeybees dance when they return to the hive after finding a source of food.

iii. **Orientational** - These are more complex behaviors, which move the animal from unfavorable conditions to favorable condition. It involves **kinesis** and **taxes.**

Kinesis - is a non - directional movement (response) in which the rate of movement is related to the intensity of the stimulus but not the direction of the stimulus

- Rate of movement increase with increasing the intensity of stimulus.

Example : the movement of woodlice in responding to changes in light intensity and humidity until they find in more humid or damp area.

Taxes - is a movement of the whole organism in response to an external directional stimulus

- The movement may be toward the stimulus (positive taxes) or away from the stimulus (negative taxis).

Taxic response can be seen to a wide variety of stimuli in the animal kingdom. Such as : photo taxis (response to light), geotaxis (to gravity), chemo taxes (to chemical), rheotaxis (resistance to movement).**Example of taxes :**

- Movement of euglena toward the source of light (positive photo taxes
- The earthworm moving away from light (negative photo taxes)
- Moths flying into the wind (positive rheotaxis).

2.4.2 What is a Learned Behavior?

Learned behavior is a behavior that is gained through experience or practice, repetition or trial and error

It is the strengthening of existing responses or the formation of new responses to existing stimuli that occurs because of practice or repetition

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Types of learned behavior

Learned behavior occurs in vast majority of animal groups, it can occur through : habituation, association (operant learning, classical), imprinting, latent learning, and insight learning.

i. Habituation :

Habituation is a type of learning in which an animal learns to ignore a repeated, irrelevant stimulus.

- When stimulus is repeated many times and nothing happen (no punishment or reward), then the stimulus is ignored.
- Habituation occurs when sensory system stops sending impulse to the brain

Example : animals learn to ignore the sounds of wind around them

- Falling leaves pass over the chick's head without consequence, the youngster learns not to react

ii. Sensitization :

Sensitization is learning to increase responses to a harmless stimulus when the stimulus occurs after a harmful stimulus.

- It is also referred to as **reverse tolerance,** is a non - associative learning process in which repeated administration of a stimulus results in the progressive amplification of a response.

Example, an alert animal escaped from a predator attack became more accustomed to frightened even to a fall of leaf

iii. Associative learning : classical and operant conditioning

In classical conditioning, an association is formed between some normal body function and a new stimulus. **Conditioning** is the process of learning by association

- These types of learning allow an animal to make an association between two stimuli.
- It was discovered by the Russian physiologist **Ivan pavlov,** who worked with dogs to develop the theory.

In classical conditioning the various stimuli and responses are :

- **The unconditioned stimulus (US)** - a stimulus that automatically triggers a response.

For example, when you smell a favorite food, you feel salivation.

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The smell of the food is the unconditioned stimulus.

- **The unconditioned response (UR)** - is unlearned response that occurs naturally to the unconditioned stimulus.

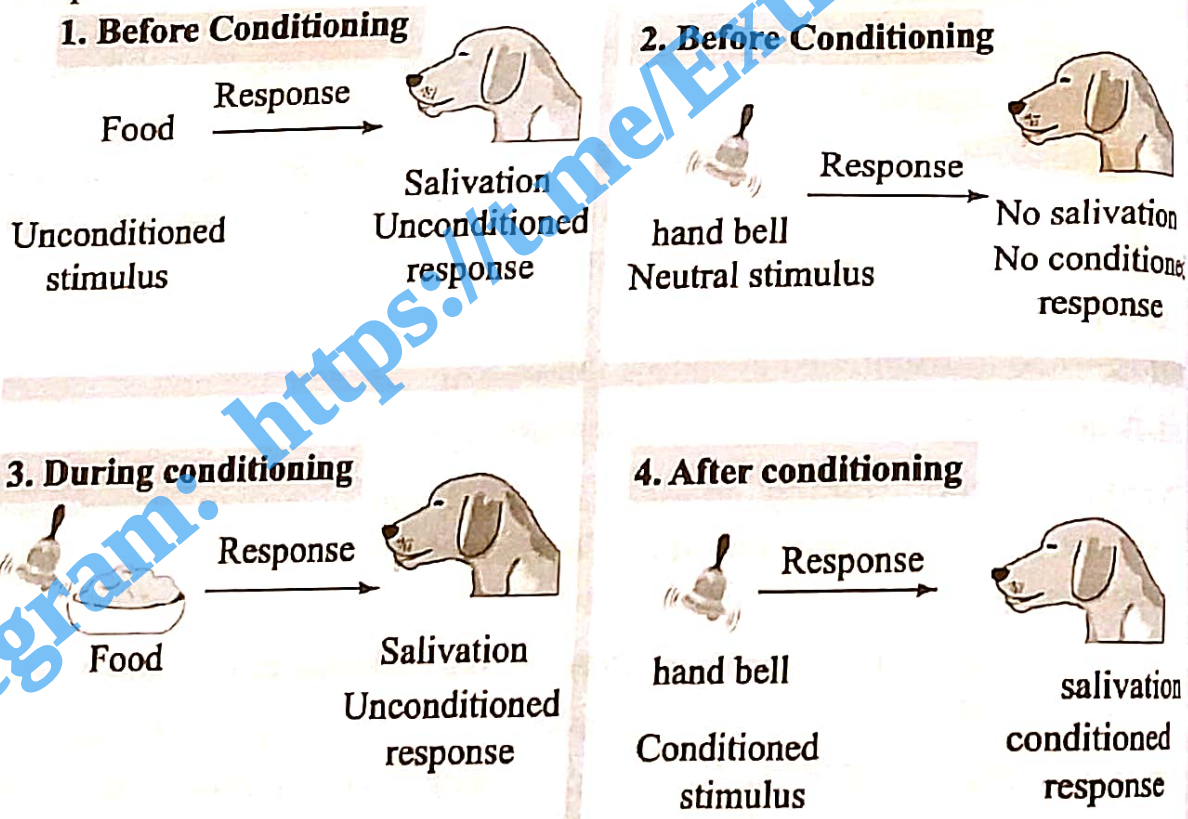
  For example, salivation for food is the unconditioned response.

- **The conditioned stimulus (CS)** - the neutral stimulus that does not initially produce the unconditioned response. But after association with the unconditioned stimulus, it triggers responses.

For example, the sound of bell causes salivation. The sound of bell is conditioned stimulus

- **The conditioned response (CR)** - is the learned response to the previously neutral stimulus after being associated with unconditioned stimulus.

For example, salivation to the sound of bell. Salivation is conditioned response.



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iv. Operant Conditioning Learning:

In operant conditioning, an animal learns a behavior to receive positive reinforcement or to avoid punishment. The likely hood of the repetition of the behavior is based on the consequences of that behavior.

- If the animal associates the outcome of behavior with reward, that

behavior is likely to be repeated

- If the animal associates the outcome of a behavior with punishment, it is less likely to be repeated

Note : The term 'operant conditioning' was first used by B , F Skinner. Skinner identified three types of responses :

- **Neutral operants :** responses from the environment that neither increase nor decrease the probability of a behaviour being repeated.
- **Reinforcers :** responses from the environment that increase the probability of a behavior being repeated, reinforcers can be either positive or negative..
- **Punishers :** responses from the environment that decrease the likelihood of a behavior being repeated

v. Insight learning

Insight learning is the ability to adapt past experiences that may involve different stimuli to solve a new problem. It is based on thought and reasoning.

- Finding the solutions to the problems which is not based on actual experience (as with trial and error) but on 'trials' occurring mentally.
- Much of the pioneering research on insight learning was carried out by Wolfgang Kohler, working with chimpanzees.

Example, a chimpanzee piles up boxes and climbs on it to reach a bunch of banana hanging from a ceiling

vi. Latent learning

Latent learning is learning that takes place when an animal explores new surroundings and learns them, without any immediate reward or punishment. The information may then be useful at another time.

- Example, a girl watching his mother baking injera on several occasions, finally made it by herself

Note: Imprinting is learning that establishes a parent - offspring bond during a critical period early in development, ensuring that the offspring recognizes the mother.

- It is one class of behavior that is determined by both **genes** and **learning**
- Imprinting occurs during a specific phase in an animal's development, called a sensitive period

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Illustrative examples

Match the examples listed under column 'A' with appropriate terms listed under column ' B'

A	B
1. Movement of cockroach away from source of light	A. Simple reflex
2. Non directional movement of woodlice in relation to light intensity	B. Positive photo taxis
3. Movement of Euglena toward the source of light	C. Kinesis
4. Web making in spider	D. Instinctive
5. Withdrawal of hands from hot object	E. Negative photo taxis

Answer

1. E 2. C 3. B 4. D 5. A

30. The unicellular organism, euglena, swims using its flagellum towards area of increased light intensity. What is kind of response to stimulus called ? (UEE, 2003 E.C)

A. Negative kinesis
B. Negative photo taxis
C. Positive kinesis
D. Positive photo taxis

Explanation :- Movements of euglena towards the source of light is positive photo taxis

Answer : D

31. Which one of the following is classified as an innate behavior ?

A. Latent learning
B. Insight learning
C. Reflex action
D. Sensitization

Explanation :- Reflex behaviour is inborn ; it is not acquired from experience.

Answer : C

32. To which kind of animal behavior does the spinning of a web by a spider be classified ? (UEE 2005 E. C)

A. Instinctive behaviour
B. Learned behaviour

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C. Experiential behaviour D. Accidertal behaviour

Explanation :- Web spinning in spider is innate behavior, which instinctive.

Answer : A

Mach the definitions listed under column 'A' with appropriate terms listed under column 'B'

A	B
1. A high level thought to solve problems using experience	A. Operant conditioning
2. After a harmful exposure to stimulus, you increased the responsiveness to harmless stimuli	B. Insight learning
3. Stop responding to a stimulus after a repeated exposure to it	C. Sensitization
4. The tendency of a repetition of a behavior based on its consequences	D. Habituation

Answer

1. B 2. C 3. D 4. A

33. A snail is moving along a bench, an experimenter taps the bench and the snail withdraws but after repeating the action the snail stop responding and just keeps on moving. The type of learning operated is

A. Sensitization C. insight

B. Latent D. habituation

Explanation :- The snail stop responding and ignoring to the stimulus after repeated exposure. The type of learning is habituation.

Answer : D

34. Young duckling first sees a researcher's red boots and then begins to follow him around. What kind of behaviour are the ducklings demonstrating ?

A. Trial and error learning C. Imprinting

B. Classical conditioning D. Operant conditioning

Explanation :- Imprinting is learning tat establishes a parent - offspring bond during a critical period early in development, ensuring that the offspring recognizes the mother.

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Answer : C

35. What do we call the learned behavior if a mouse that had just escaped from the attack of a cat, and jumped violently at a slight touch by a trivial object ?

A. Habituation
C. Imprinting
B. Latent learning
D. Sensitization

Explanation :- Sensitization is learning to increase responses to a harmless stimulus when that stimulus occurs after a harmful stimulus.

Answer : D

36. Suppose a boy, after watching his father mending shoes on several occasions, finally he made it by himself. What type of learned behavior is this ?

A. Operant conditioning
C. Latent learning
B. Classical conditioning
D. Habituation

Explanation :- Latent learning is learning that take place when an animal explores new surroundings and learns them, without any immediate reward or punishment. The information may then be useful at another time .

Answer : C

37. Which type of the following learned behaviour involves mental trial and error and finding solutions to problems without using an actual experience?

A. Insight learning
C. Operant conditioning
B. Latent learning
D. Sensitization

Explanation :- insight learning involves finding the solutions to the problems that is not based on actual experience (as with trial and error) but on 'trials' occurring mentally.

Answer : A

38. In pavlov's classical conditioning experiments on dogs, which one of the following is the unconditioned response ? (UEE, 2003 E. C)

A. The sound of the before presenting food
B. The sight of the food from far away
C. The salivation of the dog at the sight of food

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D. The salivation of the dog at the sound of the bell

Explanation : - Unconditioned response (UR) is a reaction to the natural (original) stimulus (US).

Answer : C

2.4.3 Patterns of Behavior

Animals need to obtain food, find places to live protect themselves, attract mattes, reproduce, and care for young. There are different patterns of behavior that help animals to accomplish these actions.

For example : behavioral cycles, reproductive behavior, social behavior, competition, territory and communication.

1. Behavioral cycles

Animals display a variety of cyclic behaviors that are synchronized with changes in their environment. These behaviors generally develop as a result of temperature changes, variations in availability of food, or likelihood of predation. It can be daily or seasonal cycles.

A daily biological cycle is called a **circadian rhythm**, synchronize within 24 hours

For **example**, sleep and wake cycle in human

- During the day, gorillas are busy foraging but during the night, they rest

Seasonal cycles : Some animals go into a period of inactivity and lowered by temperature during the winter when food is scarce.

For **example**, seasonal migration of birds, insects, and mammals from one habitat to another during different times of the year.

2. Territorial Behavior

Territorial behavior is behavior that animal to select and establish a territory to ensure that an animal is able to obtain sufficient resources for itself, its mate, and its offspring.

- **A territory** is an area that an animal or a group of animals occupies and defends from other members of the same species.
- Territorial animals threaten or attack intruders, but generally safer and uses less energy than fighting.

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An animal establishes a territory in many ways :

- By marking the boundaries with **pheromones** such as urine

  For example, Male dogs and lions use pheromones in their urine to mark their territory.

- By claiming an area with **vocal and display signals**

  For example, Male gorillas use display behavior to defend their territory by pounding on their chests and thumping the ground with their hands.

- Threatening by **visual / display cues**

3. Social behavior

Social behavior can be defined as any kind of interaction between two or more animals, usually of the same species.

- Social behavior has benefits including protection from predators and more success in foraging (getting food).

Examples of social animals are insects such as ants, termites, bees ; mammals such as lion, elephants, human, primates ; fish, penguin, etc.

4. Reproductive behavior

Reproductive behavior is a behavioral pattern of animals used for reproduction.

- It is the coordination of the timing and patterning of reproductive activity
- Reproductive behavior is vital for finding and selecting suitable mates, producing offspring, and rearing them successfully to independence.

For example, courtship (attracting a mate) that involves sounds visual displays or chemicals.

5. Communication behavior :

Communication behavior is a signal produced by one animal that result in some type of response in another.

There are many ways animals can communicate, including sight, sound, chemicals, tactile (touch), and possibly even language.

For example, ant uses pheromones, bird uses songs, whale uses song

6. Competition behavior:

It is a behavioral pattern of animals observed while competing for resources

Example, competition between animals for space, territory, water, mates

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and food

Competition occurs naturally between living organisms that coexist in the same environment.

Illustrative Examples

39. A daily cycle that occurs over 24 hour period of time is called ? (EUEE. 2011 E. C)

A. Circadian rhythm
B. Stimulus - response behavior
C. Kinesis behavior
D. Taxis behavior

Explanation : - circadian rhythm or daily cycle synchronizes within 24 hours.

Answer : A

40. Before a territorial disputes and actual fighting between animals, which of the following may occurs ?

A. Ritual fighting
B. Vocalizations
C. Threat display
D. All of the above

Explanation : - animal usually defend their territories by ritual fighting (fighting without physical contact), such as exaggerating body size, vocalization, threat display, release territorial pheromone.

Answer : D

41. Which one of the following is not a method of communication in animals to attract a mate

A. Uses of pheromone
B. Auditory or vocal
C. visual display and tactile
D. All of the above

Explanation : - animal can communicate through pheromone, vocal, visual display and tactile.

Answer : D

42. The activity that precedes and results in mating and reproduction is :

A. Social behavior
B. Courtship behavior
C. Learned behavior
D. Territorial behavior

Explanation : - courtship behavior is the behavior that precedes and results in reproduction.

Answer : D

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2.5 Homeostasis

Homeostasis is the process of maintaining balanced internal environment of the body all the time. For example temperature, water sugar, etc.

2.5.1 What is a thermoregulation?

The control of internal body temperature of animal is known as **thermoregulation.** Vertebrates regulate their body temperature in two different ways : poikilothermic (Ectothermic) and Homothermic (Endothermic)

1. **Ectothermic** - are animal that warms its body by absorbing heat from its surroundings.
   - ➥ They are also called poikilothermic
   - ➥ These are animals that have a body temperature similar to their environment.

   Examples of poikilothermic are reptiles, fishes, and amphibians
   - ➥ On hot day, burrow themselves into the ground, shade or cooler areas
   - ➥ On cold day, rest or bask in the sunlight, press their bodies close to a warm surface
2. **Endothermic** - are animals that keep their body temperature constant in spite of changing environmental temperatures
   - ➥ They are also called homothermic
   - ➥ Have a rapid metabolism, which generates heat needed to warm the body
   - ➥ Most endotherms have insulation, such as hair, feathers, or fat, to retain heat

   Examples of endotherms are mammals and birds

Note :

Temperature regulation is controlled by a part of the brain called **hypothalamus**. When the temperature of the environment changes (decreases or increases), signals are sent to the brain to alert the hypothalamus.

The hypothalamus then responds by activating the process of **vasodilation, vasoconstriction, shivering** and **sweating** to

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maintain the body temperature constant.

I. Physiological methods of temperature regulation in homothermic includes:

Sweating

When you are hot, sweat oozes out of the sweat glands and spreads over the surface of the skin. As the water evaporates it cools the skin, taking heat from the body.

Vasodilation

If the body temperature starts to go up, the blood vessels supplying the capillaries in your skin dilate, so that more blood flows through the capillaries. Your skin flushes and more heat is lost through radiation from the surface.

Vasoconstriction

If body temperature decreases, the blood vessels which supply your skin constricted to reduce the flow of blood through the capillaries. This reduces the heat lost through the surface of the skin, and makes you look paler, because more blood flows through the deeper blood vessels of your skin as a result.

Shivering and metabolic responses :

- When you shiver your muscles contract rapidly, more cellular respiration occurs. This releases some energy as heat which is used to raise the body temperature.

Panting and licking

Many mammals with thick, furry coats such as dog and cat have sweat glands in small areas of the skin such as the feet. So that panting allows water to evaporate from the moist surfaces of the mouth to cools them down.

Pilo erection (pulling the hairs upright)

- During cold the hair erector muscles is contracted to pull hair upright, trapping an insulating layer of air which is very effective at conserving heat.
- During hot, the hair erector muscles become relaxed that makes hair very flat against our skin, which is easier to lose heat by convection.

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Fat layer under the skin (subcutaneous fat)

- Under the surface of the skin there is an insulating layer of fat. This prevents unwanted heat loss.
- It is particularly noticeable in animals which live in very cold conditions, for example, seals and whales

II. morphological means of temperature regulation, Larger animals have smaller surface area to volume ratio, while smaller animals have larger surface area to volume ratio. Small sized animals' loss heat faster than larger animals.

For example, rat has larger surface area to volume ratio than elephant. So that the rat losses too much heat than the elephant. Therefore, the rat must eat too much (more than double of its body weight) to keep its body temperature constant.

- Big animals living in hot area have larger body extensions such as larger ear to lose heat and cool their body
- Animals with small ears and fur in cold areas help to minimize heat loss and keep their body warm.

III. Behavioral methods of temperature regulation such as:

Clothing - We wear warm clothes when the weather is cold, and wear cooler clothes when the external temperature is hot.

Seeking shade or shelter

Animals look for shade or shelter to keep them cool when it is hot and sunny. and look for shelter from cold, wet or windy conditions

Taking high - calorie food in cold conditions

Many mammals and birds eat high calorie food in cooled condition to keep them warm

Hiberuation :

- It is a period of deep sleep during cold season to prevent heat loss from the body.
- These animals eat a lot before hiding away in the nest or burrow.
- Their metabolic rate falls and so does their body temperature.
- They do not wake up until the warmer weather arrives and food for them to eat.

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Aestivation :

It is a period of escaping hot period for some animals by hides themselves underground or under a layer of mud and go into a deep sleep until conditions cool down again.

Wallowing or bathing :

Some animals wallowing in mud or bathing in water, the animals cover themselves in water and the water evaporates from the surface of their skin, cooling them down

Burning fire, central heating, air conditioning

- People can change the temperature of their environment by burning a fire or turning on the heating we can warm things up and reduce the heat we lose, keeping us warmer.
- Air conditioning is used in some buildings and vehicles to cool the air down if it gets too hot, so the people inside can lose more heat and keep themselves cool.

Illustrative Examples

43. Organisms that have constant internal body temperature are known as:

A. Cold blood
B. Homothermic
C. Poikilothermic
D. Environmental dependent

Explanation :- Homothermic animals have constant internal body temperature.

Answer : B

44. Which of the following mechanism is a physiological method of temperature regulation ?

A. Hibernation
B. Bathing
C. Sweating
D. Clothing

Explanation :- Sweating is physiological heat loss mechanism while ; bathing, clothing and hibernation are behavioral methods of temperature regulation.

Answer : C

45. Among the following temperature regulating mechanism which one are the physiological mechanisms of heat loss ?

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A. Vasoconstriction
C. eastivation
B. hibernation
D. vasodilation

Explanation :- vasodilation is a physiological means of heat loss from the body. Others are heat conserving mechanisms.

Answer : D

46. When an animal is placed in a hot environment, it loses heat through sweating and if placed in cold environment, it increases muscular activity to produce more heat. The animal in this thought is

A. Homothermic
C. Ectothermic
B. Poikilothermic
D. all are answer

Explanation :- Endothermic - are animals that keep their body temperature constant in spite of changing environmental temperatures.

Answer : A

47. Which of the following animal has variable body temperature ?

A. Cat B. Dog C. Zebra D. Lizard

Explanation :- examples of poikilothermic animal are - lizard, fish, amphibians.

Answer : D

48. What happens if the body temperature of animal is too low ?

A. The blood vessels supplying the capillaries constrict
B. The body increases sweating
C. The blood vessels supplying the capillaries dilate
D. The body stops shivering

Explanation :- during cold, blood vessels are constricted decreased sweating, shivering.

Answer : A

49. Which one of the following animal has the highest surface are to volume ratio than the others ?

A. elephant B. dog C. rat D. rabbit

Explanation :- small animals have the highest surface area to volume ratio.

Answer : C

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50. Which one of the following animal eats more food per Kg of its body weight per day ?

A. Rat B. lion C. giraffe D. elephant

Explanation :- small sized animal (large surface area to volume ratio), must eat more food to keep its body temperature constant.

Answer : A

2.5.2 What is Osmoregulation ?

It is the mechanism of maintaining solute and water in the body or maintaining constant osmotic condition in the body

- It involves regulation of the water content and solute concentration of the body fluid
- Body fluids - includes cellular fluid contents, blood plasma, tissue fluid and lymph

Based on asmoregulation animal can be: osmoconformers and Osmoregulators

Osmoconformers are organisms that change the concentration of their body fluid when the concentration changes in the environment. **Examples** are invertebrates like starfish, jellyfish and lobsters.

Osmoregulators: organisms that maintain an optimal body fluid concentration despite change in their surroundings. **Examples** are many vertebrates, including humans.

- In human and other animals, osmoregulation is brought about by osmoreceptors in the **hypothalamus,** which can detect changes in osmotic pressure.
- **Kidney** is the major organ of osmoregulation and vital in maintaining homeostasis in many animals.

☑ The kidney is the main organ responsible for osmoregulation in humans.

☑ The hypothalamus of the brain produces **Antidiuretic hormone (ADH)** and released by **pituitary gland** to control osmoregulation

When the water level in the body is high :

- Low ADH is produced
- Kidney reabsorbs very little water ; therefore the kidney releases large amount of **hypotonic urine,** urine containing high concentration of water.

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When the water level in the body is low :

- More ADH is produced,
- Kidney reabsorbs more water and retains water
- Kidney produces **hypertonic urine,** urine containing less water and yellow in color

2.5.3 How Does Our Body Regulate Blood Sugar ?

Glucose is a respiratory substrate of the body, which provides the body with energy to power cellular activities. Therefore, the body needs to regulate glucose at constant level in order to supply energy continuously.

- ☑ Pancreas is an organ of the body that plays a great role in the regulation of blood sugar
- ☑ Pancreas produces two hormones called **insulin and glucagon,** which are responsible for controlling the concentration of glucose in the blood.
- ☑ When blood glucose level is high, more **insulin** is released by the pancreas into the liver.
  - Insulin promotes the conversion of glucose into **glycogen,** which is stored in the liver
- ☑ When blood glucose level is low, more glucagon is released by pancreas into the liver.
  - Glucagon promotes the conversion of glycogen into glucose so that the lack of glucose is solved.

2.6 Feed back Mechanism

Feedback mechanism is physiological regulation systems that return the body to its normal internal state.

The components of feedback control mechanisms are : **receptor, control center** (brain) **and effector.**

- The **receptor** senses the change in the environment (stimulus)
- A signal is sent to the **control center** (the brain) to analyze the degree of deviation from normal state
- The brain sends impulse to an **effector** such as muscles or glands for contraction or relaxation or glands to secrete hormones, so that homeostasis is achieved.

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There are two types of feedback mechanisms. These are : Negative and positive feedback.

2.6.1 Negative feedback

It occurs when the activation of one component results in the deactivation of another.

- This means that if the level is too high from the normal, the body brings it down, and if the level is too low from the normal, the body lifts it up.
- The response counteracts the inappropriate change, restoring the steady state.
- The negative feedback brings you back to homeostasis

For example, if temperature rises, the receptor in the hypothalamus detect the change, then take corrective measures such as vasodilation, sweating to cool the body.

2.6.2 Positive feedback

It occurs when a change in a variable triggers a response that causes more change in the same direction

- It means, the activation of one component causes the activation of another
- Positive feedback mechanism accelerates a change rather than reversing it.
- The positive feedback take you further away from homeostasis

For example, during the birth of a baby. As the baby head pushes the uterus, the reflex action causes uterus to contract and relax again and again, so that the baby is born.

2.7 Renowned Zoologists in Ethiopia

Zoology is the study of animals in relation to their evolution, anatomy, physiology, behavior, habitats and health.

- Many zoologists from various Ethiopian universities have studied animals found though the country.
- Some researchers have invested their time and energy in studying animals in Ethiopia to improve zoological science and economy of the country. Such researchers are patriots for their country.

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Illustrative Examples

51. The mechanism of regulation of solutes and water in human is known as ___ ?

A. Thermoregulation
B. Homeostasis
C. Poikilotherms
D. Osmoregulation

Explanation :- osmoregulation is the regulation of he water content and solute concentration of the body fluid.

Answer : D

52. What does glucagon promote in the liver ?

A. Hydrolysis of glucagon
B. Condensation reaction glycogen
C) Hydrolysis of glycogen
D) Release of insulin

Explanation :- glucagon converts glycogen into glucose, which is known as hydrolysis of glycogen.

Answer : C

53. When the level of water in the blood is too much, which one of the following will be happen ?

A. More ADH is produced
B. kidney reabsorbs less water
C. urine become hypertonic
D. no change at all

Explanation :- When the water level in the body is high :

- ➥ Low ADH is produced
- ➥ Kidney reabsorbs very little water ; therefore :
- ➥ The kidney releases large amount of hypotonic urine, urine containing high concentration of water.

Answer : B

54. An increase in blood sugar level causes the release of insulin, and then sugar level decreases. Such feedback mechanism is __ ?

A. positive feedback
B. negative feedback
C. homeostatic imbalance
D. none of the above

Explanation :- Negative feedback control occurs when the activation of component results in the deactivation of another.

Answer : B

55. Arrange the following components of feedback control mechanism

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in the correct order (control center, receptor, effector) to balance the activity of the body, which is deviated from normal ?

Explanation : - receptor → control center → effector

Review Questions

1. Which group of animal is not an example poikilothermic (ectotherms) ?
   A. birds
   B. fish
   C. reptile
   D. amphibians
2. Which one of the following is not an example of a physiological method of heat control ?
   A. vasodilation
   B. sweating
   C. shivering
   D. hibernation
3. When the body temperature of animal drops from its normal, which one of the following will be happen to correct it ?
   A. Shivering
   B. sweating
   C. vasodilation
   D. bathing
4. Reproductive behavior consists of all, except ___ ?
   A. Selection of mate
   B. courtship
   C. rearing of young's
   D. defending for territories
5. Which one of the following is not an example of instinctive behavior ?
   A. Nest building
   B. knee jerk
   C. swimming of dolphin
   D. movement of cockroach to dark
6. Abel and Sisay are very close friends ; Abel is a smoker but Sisay is not. After a while Sisay becomes a smoker ; Abel always appreciates the smoking style of Sisay. In the above statement the appreciation of Abel is used as _____ ?
   A. positive reinforcer
   B. a negative reinforcer
   C. a neutral response
   D. a punisher
7. The mechanism of regulation of solute, the loss and gain of water in human is called ___ ?
   A. homeostasis
   B. thermoregulation
   C. osmoregulation
   D. homoeothermic

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8. Why do the hairs on the skin erected / stand/ during cold time ?
   A. fright
   B. to trap air to cool us
   C. to trap air to heat us
   D. to release air to cool us

9. An alert animal escaped from a predator attack became more accustomed to frighten even to a fall of leaf ; such type of learning is called ?
   A. Habituation
   B. insight
   C. sensitization
   D. classical

10. Fish learn to ignore leaves falling in to water is an example of :
   A. Habituation
   B. insight
   C. sensitization
   D. classical

11. Which of the following thermoregulatory responses would be considered as strictly behavioral response ?
   A. Shivering
   B. Sweating
   C. Pilloerection
   D. Basking in the sun

12. What will be happened as the size of an animal decreases ?
   A. The surface area to volume ratio decreases
   B. The rate of heat loss from the body increases
   C. The amount of food intake per day increases
   D. B and C

13. What is the reason that smaller mammals have higher metabolic rates ?
   A. Higher surface area to volume ratio
   B. Lower lifespans
   C. Lower surface area to volume ratio
   D. More frequent exposure to predator

14. One of the following is not true about operant conditioning ? (EUEE. 2014 E.)
   A. It is concerned with learned behaviour
   B. It is based on reward and punishment
   C. It is based on innate behaviour
   D. It is based on consequences of action

15. What is the behaviour that an animal learns to ignore a trivial stimulus

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that is repeated many times ? (EUEE, 2013 E. C)

A. Sensitization
B. Insight learning
C. Latent learning
D. Habituation

16. If a chimpanzee piles up boxes and climbs on it to reach a bunch of banana hanging from a ceiling, which behaviour is manifested ? (EUEE, 2011 E. C)

A. Operant conditioning
B. Trial and error learning
C. Latent learning
D. Insight learning

17. What is the significance of homeostasis ? It is important to

A. Increase the cellular activities of our body systems
B. denatures the enzymes and stop cell activities in our body
C. Burn food molecules to produce carbon dioxide
D. maintain the internal conditions of the body stable state

18. Which animal behaviour pattern is best illustrated by the famous Ivan Pavlov's dog ? (EUEE, 2011)

A. Habituation
B. Associative learning
C. Latent learning
D. Instinctive behaviour

19. When poikilothermic animals get too hot they ___ ?

A. Move into shade
B. Press their body in warm surface
C. Bask in the sun
D. Erect special sail of their skin

20. Which of the following organisms is a poikilothermic animal ?

A. Human B. Elephant C. Ape D. Fish

21. The process of organs formation in animals during embryonic development is

A. Gametogenesis
B. Organogenesis
C. Spermatogenesis
D. Parthenogenesis

22. Which genera of frog is indiginous to Ethiopia and not found embryonic development is

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A. Leptopelis **ragazzi**
B. Ranate **moraria**
C. Ptychadena **harenna**
D. All except B

23. Which of the following is not the economic importance of insects ?
A. Agents of pollinators
B. Biological pest control
C. Intermediate hosts for pathogens
D. None of the above

24. Which one of the following is different from the other ?
A. hibernation
B. vasodilation
C. shivering
D. sweating

25. Identify the method through which heat is not lost from the body ?
A. Vasoconstriction
B. Sweating
C. Vasodilation
D. Panting

26. Identify the term that describes what happens to blood vessels when the condition is hot :
A. Vasoconstriction
B. Sweating
C. Vasodilation
D. Shivering

27. Which of the following is the feeding stage in the complete metamorphosis of insect life cycle ?
A. The adult stage
B. The pupa stage
C. The egg stage
D. The larva stage

28. Which of the following is correct about the adaptation of snakes to high temperature ?
A. Basking in the sun
B. Sweating
C. Moving into shade
D. Vasoconstriction

29. How do the poikilothermic animals adapt to cold weather ?
A. Licking their body parts
B. Basking in the sun
C. Bathing in water
D. Wallowing by basking

30. Dilation of the blood vessels, supplying blood to the capillary in the skin is a response to

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A. Cold temperature
B. High acidity
C. Low oxygen concentration
D. Hot temperature

31. Which of the following physiological method of temperature regulation occurs in homoeothermic animals when temperature falls ?
A. Vasodilation
B. Licking
C. Panting
D. Vasoconstriction

32. Which of the following animal body is governed by the environmental temperature ?
A. Ostrich B. Lizard C. Lion D. Sheep

33. Which of the following is a physiological method of temperature regulation in homoeothermic ?
A. Hibernation
B. Bathing
C. Aestivation
D. Shivering

34. Latent learning is different from learning through operant conditioning in that
A. Is a kind of innate behaviour
B. Does not require reinforcement
C. Does not require any input from the environment
D. A kind of insight learning

35. Which of the following are the functions of pheromones ?
A. Define territories
B. Attract a mate
C. Trigger alarm
D. all are the answer

36. Which of the following behaviour is acquired through experience ?
A. Habituation
B. Kinesis
C. blinking of eye
D. Sneezing

37. Which of the following is not important body's heat conservation ?
A. Subcutaneous fat
B. Feather or fur covers
C. Vasoconstriction
D. Large surface area to volume ratio

38. When the size of cell increase which one of the following gets smaller ?

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A. The volume of the cell
B. The surface area of a cell
C. The surface area too volume ratio of a cell
D. The amount of oxygen needed to a cell

39. Among the following temperature regulating mechanism which one are the physiological mechanisms of heat loss ?

A. Vasoconstriction
B. hibernation
C. shivering
D. vasodilation

40. A daily cycle activity that occurs over 24 hour period of time is called (EUEE, 2011 E. C)

A. Circadian rhythm
B. Stimulus - response behavior
C. Kinesis behavior
D. Taxis behavior

41. A boy who had seen a snake crossing his way earlier, jumped up violently when grass touched his legs. Which behaviour is it (EUEE, 2011)

A. Conditioning
B. Habituation
C. Sensitization
D. Imprinting

42. An increase in blood sugar level causes the pancreas to release of the hormone insulin. Insulin lowers blood sugar level restoring the body to its original blood - sugar level by converting glucose to glycogen, this is an example of

A. Positive feed back method
B. Negative feed back method
C. Homeostasis imbalance
D. None of the about

43. Animal behavior such s threatening display through vocal, visual, and displays, is mostly associated with

A. Territorial behavior
B. Courtship behavior
C. Social behavior
D. Social behavior

44. Which of the following patterns of animal behavior is used the most for hunting and defending in groups ?

A. Communication
B. Social behavior
C. Territorial behavior
D. none of the above

45. One of the following patterns of innate behavior in animals, that considered as the simplest (EUEE, 2011 E)

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A. Instinctive behavior
B. Reflex action
C. Kinesis movement
D. Taxis movement

46. Which of the following statements about learned behavior is not true ?
A. it is fully functional the first time it is performed
B. varies individual to individual
C. it is modified by experience
D. it is developed by trial and error

47. In pavlov experiment on classical conditioning on dog, the ringing of the bell represents ___ ?
A. The condition response
B. The unconditioned response
C. The conditioned stimulus
D. The unconditioned stimulus

Short answer item : fills in the blank spaces with appropriate terms / words

48. The way in which animal interacts with other organism and physical environment is called ___ ?

49. The type of behavior that is genetically programmed and inborn is ___ , while ___ is acquired through experience and repetitions ?

50. A behavioral cycle, which is repeated every 24 Hrs. is known as ___, while ___ is yearly cycle ?

51. An area that contains food, mate and nest of animal which is defended against intruders is called _____such pattern of behavior is known as _____ ?

52. A chemical substance produced by animals, that used to influence the behavior of another animal of its own species is _____ ?

53. A part of a brain that control or regulate fluid and temperature of animal body is _____ ?

54. A hormone that converts excess glucose from the blood in to glycogen is ___, while glycogen in to glucose is ___ ?

55. Group the following heat control method as **physiological in cold, physiological in hot, behavioral, morphological**

a) Sweating b) Vasodilation c) vasoconstriction

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d) shivering g) bathing heat
e) Body size h) high calorie food j) insulation
f) pilo erection i) air condition and

56. A series of changes in the body structure of insects during development which is regulated by a hormone called _____ ?

57. Sex determination in crocodile is determined by _____ ; male at _____ and female at _____

58. The egg of a frog is covered by a jell like substances, this helps for _____

59. The type of asexual reproduction in some insects that directly formed from unfertilized egg is called ___ ?

60. In the body organization pattern of an animal, if he left and right sides of the body are mirror images of each other, it is called ___ ?

61. Who will take care of the young pups if the true mother of a young rat is died ?

Fill the following blanks with terms : unconditioned stimulus (US), unconditioned response (UR), conditioned stimulus (CS), conditioned response (CR), neutral stimulus (NS)

62. Before conditioned the pavlov dog doesn't respond to the sound of the bell (-----), but salivate for food (------) . After conditioning, the dog salivated (------) for the sound of a bell (------) ?

Essay type :

63. Discuss the effect of SA / V ratio of animal on temperature regulation ?

64. List the type of feedback control mechanism ?

65. Explain why animals living in tropics / hot/ area have body extension such as large ears than the highlanders ?

66. Explain why people living in desert are recommended to drink hot tea during hot time of a day ?

67. Arrange the following terms in a correct order from first to end whn animal responds to a change in an environment ? (stimulus, effectors, control center, receptors)

68. List the three types of innate behavior ?

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69. What is an adaptation of parental caring of some birds that lay egg on bare lands without nest ?

70. Explain the main reason why some animals lay hundreds of egg for external fertilization ?

Answer to the question : Multiple choices

1. A	9. C	17. D	25. A	33. D	41. C
2. D	10. A	18. B	26. C	34. B	42. B
3. A	11. D	19. A	27. D	35. D	43. A
4. D	12. D	20. D	28. C	36. A	44. B
5. B	13. A	21. B	29. B	37. D	45. B
6. B	14. C	22. D	30. D	38. C	46. A
7. C	15. D	23. C	31. D	39. D	47. C
8. C	16. D	24. A	32. B	40. A	

Short answer :

48. behavior

49. innate, learned

50. circadian rhythm

51. territory, territoriality

52. pheromones

53. hypothalamus

54. insulin, glucagon

55. **physiological :** sweating, vasodilation, insulation, vasoconstriction, shivering, pilloerection

Behavioral : bathing, hibernation, aestivation, high calorie food

Morphological : body size

56. Ecdyson or molting hormone

57. Temperature, male 34 % C, female 30^0C

58. Drying out, water current / wave

59. parthenogenesis

60. Bilateral symmetry

61. Another female rat from the family

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62. Neutral stimulus, unconditioned response ; conditioned response, conditioned stimuli

Essay type :

63. Larger animals have smaller surface area to volume ratio, while smaller animals have larger surface area to volume ratio. Small sized animals loss heat faster than larger animals

64. Negative and positive feedback

65. Animals living in low land or hot area are urged to loss heat from the body to balance their body temperature ; to do this body extensions such as large ear is very important

66. The hot tea stimulates the body to produce sweat, so that the body loses heat as the sweat evaporates.

67. Stimulus, receptor, control center and effector

68. Reflex, instinct, orientational

69. They lay camouflaged egg, which is similar to the object in the environment

70. To in crease the number of survival of eggs.

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UNIT 3

3. Enzyme

3.1 What are Enzymes ?

Enzymes are biological catalysts, molecules that speed up chemical reactions. Without enzymes, the reaction within the cell wouldn't happen fast enough to keep cells alive. All the enzymes in the cell involved in biochemical reactions called **metabolism.**

Metabolism is all the chemical activities of life including the breakdown (catabolism) and synthesis (anabolism) of molecule in the cell. Such process require enzyme to catalyze. It is the overall chemical reactions in the cell.

- Enzymes remain unchanged (not used up) at the end of the reactions. They neither affect the nature of the product formed.
- Enzymes are made of proteins.

3.2 Properties and Functions of Enzymes

The general properties of enzymes are the nature of both their physical and chemical properties

3.2.1 The physical properties of enzymes

The physical properties of enzymes include denaturation, solubility, colloids, bio - catalysts, precipitation, molecular weight, and enzyme activity.

Enzyme Denaturation

It is the process that changes the molecular structure / shape of enzyme and losing its function. It involves the breaking of many of the weak linkage, bonds within the protein molecule. This is brought about in various ways such as : high temperature (beyond optimum), change in PH, heavy metals,

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high salt concentrations, solvents and other reagents.

Solubility of Enzymes

Enzymes are soluble in a variety of solutions such as : water, salt, diluted glycerol and alcohol. It causes denaturation.

- Enzymes have hydrophilic R groups on the outside, making it soluble.
- Solvents have profound effect on the rate of reaction and selectivity.

The colloidal nature of enzyme

It is the tendency of having little or no dialysis cross the semipermeable membrane due to the large size or high molecular weight.

The biocatalyst property of Enzyme

It is the activity of enzymes in which very small quantities or a small amount of enzyme is enough to convert a large quantity of substrate and remain unchanged after the reaction.

Enzyme precipitation

It is the separation of enzymes for analysis using different aqueous or ethanol solvents

For example, acid and alkaline solution can cause enzyme precipitation, because proteins are amphoteric (possess amino and carboxylic functional groups).

Molecular weights

Enzymes are large protein biomolecules that hold polypeptide chains of various amino acid sequences.

In a single protein, there are about 200 - 300 peptide bonds that hold amino acids together. Thus, enzymes have high molecular weight.

Enzymatic activity

It is the general catalytic properties of an enzyme. It depends on factors such as temperature, pH, enzyme concentration an substrate concentration. Enzymes show the highest activity at optimum temperature and PH. A low concentration of enzymes and substrates slows down the enzymatic reaction.

3.2. 2 Chemical Properties of Enzymes

Enzyme chemical properties are sensitivity, regulations, specificity, catalysis and reversibility reactions.

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Heat and PH Sensitivity

- It is an enzymatic reaction to heat (temperatures) and pH (acidity and basicity) activated at optimum levels.
- Enzymes works best at optimum temperature and a narrow range of PH. A change in temperature or PH affects the activity of an enzyme

Regulation

It is the process of controlling the activity of enzymes by activator and inhibitor molecules.

- **Inhibitors** are any substance that can diminish the speed of an enzyme catalyzed reaction
- **Activators** are any substance binding on enzyme to increase the speed of a reaction

Catalysis

- It is the process of the acceleration of a chemical reaction by a catalyst
- Enzymes have high catalytic efficiency, they can transform about 10 - 10, 000 substrates per second
- The reactions catalyzed by the enzymes show a 10^3 - 10^8 time faster reaction rate in comparison to the non - catalyzed reactions.

Reversibility

- Enzymatic reactions catalyze biochemical reactions in both forward and reverse directions
- It is the ability of enzymes to catalyze various metabolic reactions

For example, build up new molecules (anabolism) or breaks down (catabolism)

Enzyme specificity

Enzymes are highly specific in nature, interacting with one or few substrates and catalyzing only one type of chemical reaction

Enzymes show different degrees of specificity

i. Bond specificity

- In this type of specificity, the enzyme acts on substrate that are similar in structure and contain the same type of bond.
- Example, amylase which acts on alpha 1 - 4 glcosidic bond in starch and glycogen, shows bond specificity.

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ii. Group specificity

In this type of specificity, the enzyme is specific not only the type of bond but also to the structure surrounding it

iii. Substrate specificity

➥ In this type of specificity, the enzyme acts only one substrate

Example, maltase, only acts on maltose, shows substrate specificity

iv. Optical / stereo - specificity

Enzymes are specific to substrate and also optical configuration

Illustrative Examples

1. What will be happen in living organisms if there is no enzyme ?

   Explanation :- With out enzymes, the reactions within the cell wouldn't happen fast enough to keep cells alive.

2. Which of the following is not true about enzymes ? Enzymes are :

   A. Biological catalyst

   B. used up in the reaction

   C. protein

   D. don't change the nature of the product

   Explanation :-

   ➥ enzymes are biological catalyst, speed up chemical reaction

   ➥ made of protein

   ➥ used again and again (not used up)

   ➥ don't change the nature of a product.

   Answer : B

3. The overall chemical reaction in living organism is known as ____?

   A. Anabolism

   B. catabolism

   C. metabolism

   D. none of the above

   Explanation :- metabolism is the sum of anabolism (building reaction) and catabolism (breaking down).

   Answer : C

4. The physical property of enzymes that is little or dialysis across the cell membrane due to its size is ____?

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A. biocatalyst
C. enzyme precipitation
B. colloids
D. denaturation

Explanation :- colloidal property of an enzymeis the tendency of having little or no dialysis cross the semipermeable membrane due to the large size or high molecular weight.

Answer : B

5. The specificity of enzyme can be explained in terms of _____ ?

A. bond specificity
C. substrate specificity
B. group specificity
D. all

Explanation :- Enzymes show different degrees of specificity. These are bond specificity, group specificity, substrate specificity, and optical / stereo specificity.

Answer : D

6. An enzyme sucrase, catalysis only sucrose molecule, such type of enzyme specificity is

A. Bond specificity
C. group specificity
B. substrate specificity
D. optical specificity

Explanation :- enzymes are specific, means each type of enzyme catalysis specific substrate.

Answer : B

7. Which of following is not the physical property of an enzyme ?

A. colloidal
C. heat and pH sensitivity
B. solubility
D. precipitation

Explanation :- the physical property of an enzyme includes colloidal, solubility, precipitation, solubility, denaturation, etc. Heat and pH sensitivity is an example of chemical property of an enzyme.

Answer : C

8. The chemical property of enzymes that is catalysis in both forward and back ward direction is _____ ?

A. reversibility
C. PH sensitivity
B. regulation
D. catalysis

Explanation :- reversibility - enzymatic reactions catalyze biochemical reaction in both forward and reverse directions.

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3.2.3 The Functions of proteins

Proteins are important to the structure and function of the cells facilitate chemical reaction. Here are some examples of the functions protien: catalysis, transport, support, defence, regulation, communicati motion and storage.

i. Enzymes

Protien act as catalysts, speeding up chemical reactions in the cel Enzymes are required for the digestion and growth of the cell.

ii. Structural support

Examples include : The collagen is structural protien of connective tiss in skin, tendons and ligaments.

iii. Transport material

Protiens transports material into and out of cells. For example hemoglobin found in red blood cells, transports oxygen in the blood of animals.

iv. Are body defensive

Antibodies are protective protiens which are part of immune respons toreduces the harmful effects of foreign protiens (antigens)

v. For muscle Moment

Contractile protiens, actin and myosin, cause muscles to move.

vi. Receptors or signals

vii. Hormons

Hormons are regulatory protiens. For example, the hormon insulin regulates the concentration of glucose in the blood ;

List some examples of enzymes and their functions in living organism ?

- **Sucrase** - break down sucrose in to glucose and fructose
- **Carbohydrase** - breakdown carbohydrate into sugar
- **Protease** - break own protein into amino acids

Answer :

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Enzyme	Function
Lipases	Split fat found in the blood, gastric juices, pancreatic secretions, intestinal juices, adipose (fatty) tissues and participate in digestions
Amylase	Amylase exists in saliva an helps in changing starches into sugars.
Maltase	Maltase exists in foods such as potatoes, pasta and beer and saliva breaks the sugar maltose into glucose.
Trypsin	Found in the small intestine, breaks proteins down into amino acids.
Lactase	Found in the small intestine, breaks lactose, the sugar in milk, into glucose and galactose.
Helicase	Unwinds DNA
DNA polymerase	An enzyme responsible for forming new copies of DNA in the form of nucleic acids molecules
Acetyl cholinesterase	Breaks down the neurotransmitter acetylcholine in nerves and muscles

Illustrative Examples

9. The type of an enzyme that catalysis the conversion of protein into amino acid in small intestine is

A. Amylase B. maltase C. trypsin D. lipase

Explanation :- Trypsin breaks protiens down into amino acid

Answer : C

10. Which of the following is not the function of protein ?

A. To make antibodies C. to make receptors

B. to make enzymes D. to make cholestrol

Explanation :-Proteins have many functions such as to make antibodies, enzymes, receptors, transport, structural, communication,etc.

Answer : D

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3.3 Mechanism of Enzyme Action

Enzymes have active site on a region where they bind with sub trate

☑ Active site is a small pocket or cliff on enzyme where the substrate is binded.

- ➥ It occupies less than 5 % of the total surface area of an enzyme
- ➥ It has specific shape due to tertiary structure of protein.
- ➥ The change in the shape of protein affects the shape of active site and function of protein

☑ The active site of an enzyme is shaped to allow :

- ➥ Binding with a particular substrate
- ➥ Binding in such a way that reaction can take place requiring less energy

☑ The reactant that binds at active site of enzyme is termed as **substrate.**

☑ When substrate binds to an active site of enzyme, it forms **E - S complex**

☑ Enzyme - substrate binding causes **reorientation,** due to a strain / tension of substrate on active site.

- ➥ E - S complex is formed at the transition state
- ➥ At transition state the bond between substrate is unstable and easily broken down ; then product is formed
- ➥ Enzyme - product complex is formed

Generally : $S + E \longrightarrow ES\text{-complex} \longrightarrow E\text{-}P\ \text{complex} \longrightarrow P + E$

For example, look at the following figure (hydrolysis of sucrose by sucrase)

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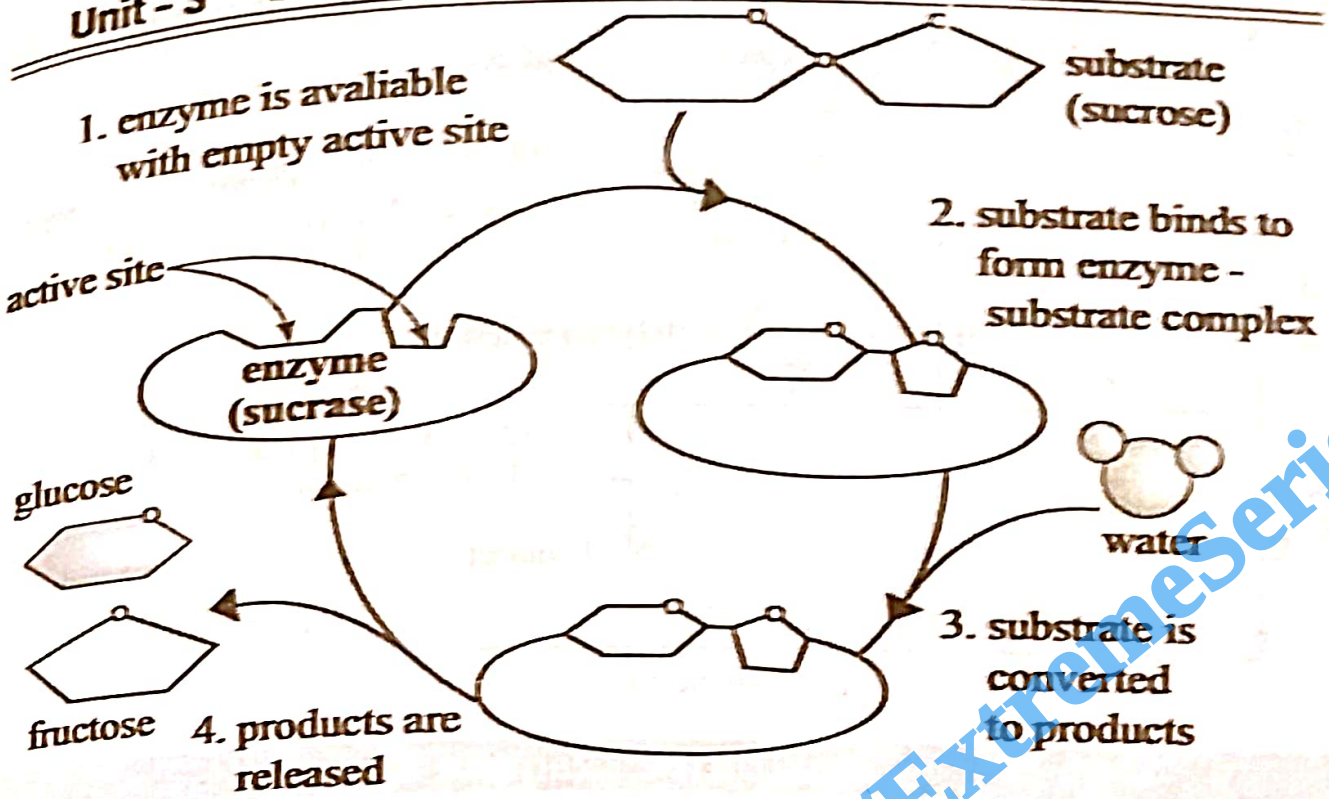

Note : The transition state is the intermediary state of the reaction, where molecules are neither a substrate nor a product.

- It represents the formation of ES - complex.
- Bonds are partially made / or broken, thus the molecules are less stable
- It has the highest free energy. Enzymes lower this energy

3.4 Activation Energy

In order for molecules to react, they must have sufficient energy. The least amount of energy needed for a chemical reaction to take place is known as activation energy (Ea).

- Enzyme acts on substrate and speed up chemical reaction by lowering the activation energy (Ea).
- In catalyzed reaction, Ea is lowered, so that the conversion time of reactant in to product is shortened
- in uncatalysed reaction, Ea is higher, which means reactant takes longer time to be converted to product

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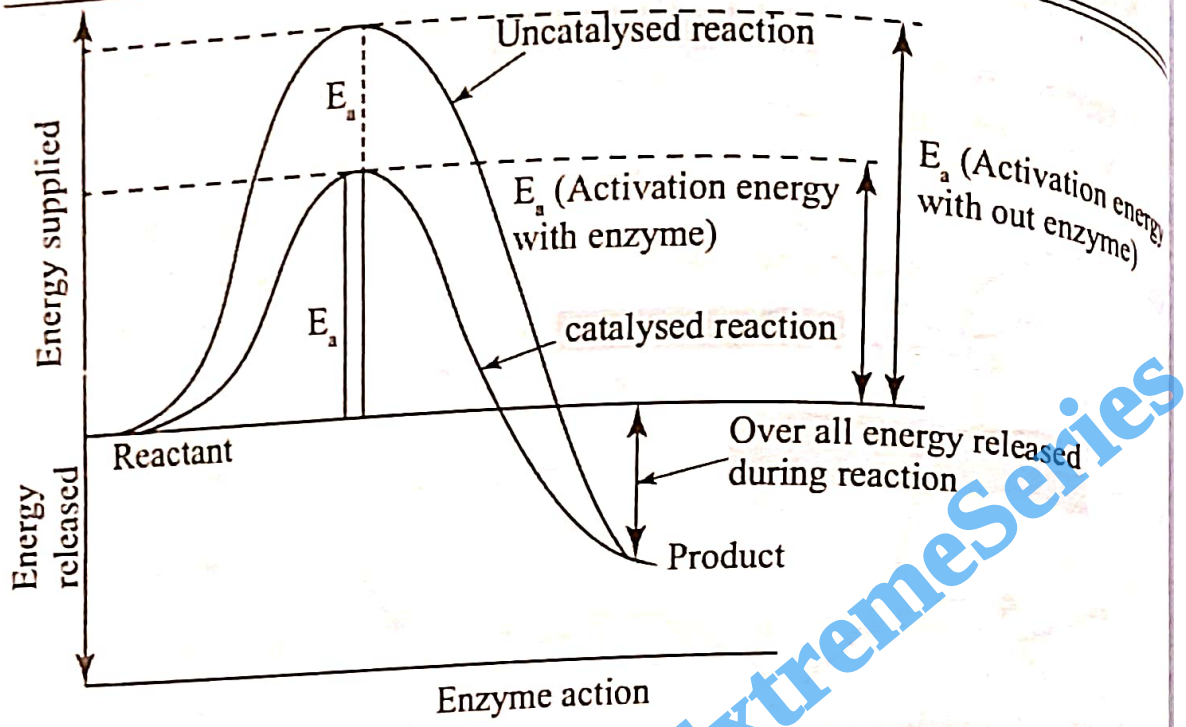

Illustrative Example

11. The region where an enzyme molecule binds with its substrate is :

A. Allosteric site
B. Active site
C. substrate
D. co - enzyme

Explanation :- the region where an enzyme molecule binds with its substrate is called active site

Answer: B

12. The energy that is required to start the chemical reaction is :

A. Chemical energy
B. Activation energy
C. Mechanical energy
D. Kinetic energy

Explanation :- activation energy is the energy which is required to start chemical reaction.

Answer : B

13. Explain about **transition state** in enzyme based chemical reaction ?

Explanation :- transition state :

- The transition state is he intermediary state of the reaction
- The molecules at transition state in neither a substrate nor product
- It represents the formation of ES - complex
- Bonds in substrate are partially made / or broken, thus the molecules

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are less stable

➥ It has the highest free energy. Enzymes lower this energy

3.5 Protein Structures

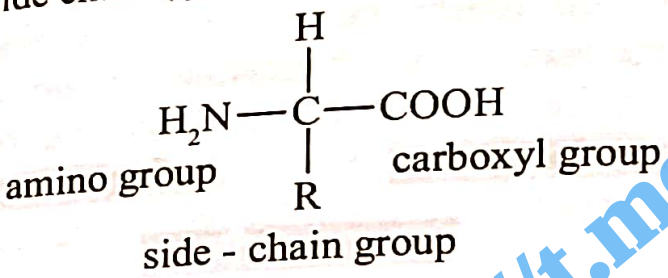
Protein are large molecules (polymers), made of smaller units known as **amino acids**. Each of the 20 amino acids found in cells has the same basic structure. A central carbon atom in the amino acid is bonded to four different chemical groups :

➥ A nitrogen - containing amino group (_ NH_2 or _ NH_3)

➥ A carboxyl group (_ COOH or _ COO _)

➥ A hydrogen atom (_ H)

➥ A side chain (_ R) that varies between the 20 amino acids



☑ A polymer of more than 50 amino acids sequences is called a **polypeptide.** A protein with fewer than 50 sequences of amino acids is **peptides.**

☑ Once a polypeptide is folded and becomes functional, the polypeptide is called **a protein.** Some proteins are made of more than one folded polypeptide chain.

☑ The specific relationship between proteins and other molecules is based on the **shape of the protein.**

☑ **Protein shape** is complex and is essential to **protein function.** For example : proteins control metabolism, transport, communication, structure, division, and many other aspects of cell function.

☑ If a protein unfolds, or denatures, it will no longer function.

Protein structure, or conformation, is fairly complex and is different levels : primary, secondary tertiary and quaternary.

I. Primary structure

The primary structure of a protein refers to the sequence, or order of **amino acid** in that protien. **20 amino acids** exist in cells, so the choice and ordering

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of amino acids makes the primary structure of one polypeptide different from that of another. The sequence of amino acids in each polypeptide is determined by the **DNA code** of the cell.

- Amino acids are attached to each other by **condensation reaction** that bonds the carboxyl group of one amino acid to the amino group of another amino acid.
- Two amino acid molecules react together in condensation reaction to produce a peptide bond & water is released.
- The bond between amino acids is called **a peptide bond.**

Because peptide bonds are covalent bonds, **strongest bonds** in cells. Increasing in temperature or changes in pH, can damage proteins, causing them to denature (unfold). But they don't lose their primary structure.

Note : *Humans need 20 amino acids to make the protein for healthy body structures and functions. Your body can manufacture 10 amino acids. The other 10 must be present in your food because your body can not manufacture them for itself ; these are referred to as* **essential amino acids.**

2. Secondary structure

As polypeptide chains fold up into their final structure, some areas of the chain form very regular folded patterns. These folded areas represent the secondary structure of the protein. Two patterns of folds are part of the secondary structure :

i. The alpha helix

- The helical structure is determined and maintained by the formation of **hydrogen bonds** between the back bones of the amino acids in successive turns of the spiral coil.
- The **hydrogen bond** forms between an **oxygen of carboxyl (CO)** of one amino acids and **hydrogen of a mine (NH)** group at the fourth amino acids residue.
- It is the basic structural unit of some **fibrous proteins** that make up wool, hair, skin, and nail
- It is elastic in nature due to **H** bonding and **helical** shape.

ii. Beta - pleated sheets

- The **H** bonding in beta pleated sheet takes place between different regions of a polypeptide chain that has turned back on itself.

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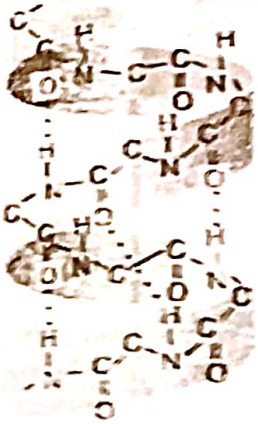
- Each chain is fully extended, but each has a zigzag structure the resulting "sheet" has an overall pleated conformation.
- Strong and flexible, but not elastic. This is because the distance between the pleat is fixed, determined by the strong covalent bond of the polypeptide back bone.

Note :

Protein may posses different types of secondary structures.
Some proteins have multiple alpha helices but no pleated sheets.
Other proteins have both structures, and or only pleated sheets.
For example :
Spider silk is a composite of proteins with alpha helix (provide elasticity) and others with beta pleated sheet conformation (providing strength).

Alpha helix



Beta - pleated sheets

3. Tertiary structure

The tertiary level of protein structure is the final **three - dimensional** shape of the polypeptide chain as a polypeptide chain **twists and folds** upon itself.

The final shape of any polypeptide chain is unique and will have specific areas that are necessary for the function of the protein.

The tertiary structure of a polypeptide chain is held together by **various types of bonds** between the atoms in the R - groups of the amino acids. These are : **Covalent, ionic, hydrogen** and **hydrophobic interaction.**

- **Covalent bounds :** When the R - groups from two cysteines come near each other in a folded polypeptide chain, they form a covalent bond called **a disulfide bridge** (_ S _ S _). **Disulfide bridges are strong and aren't lost when a protein denatures.**

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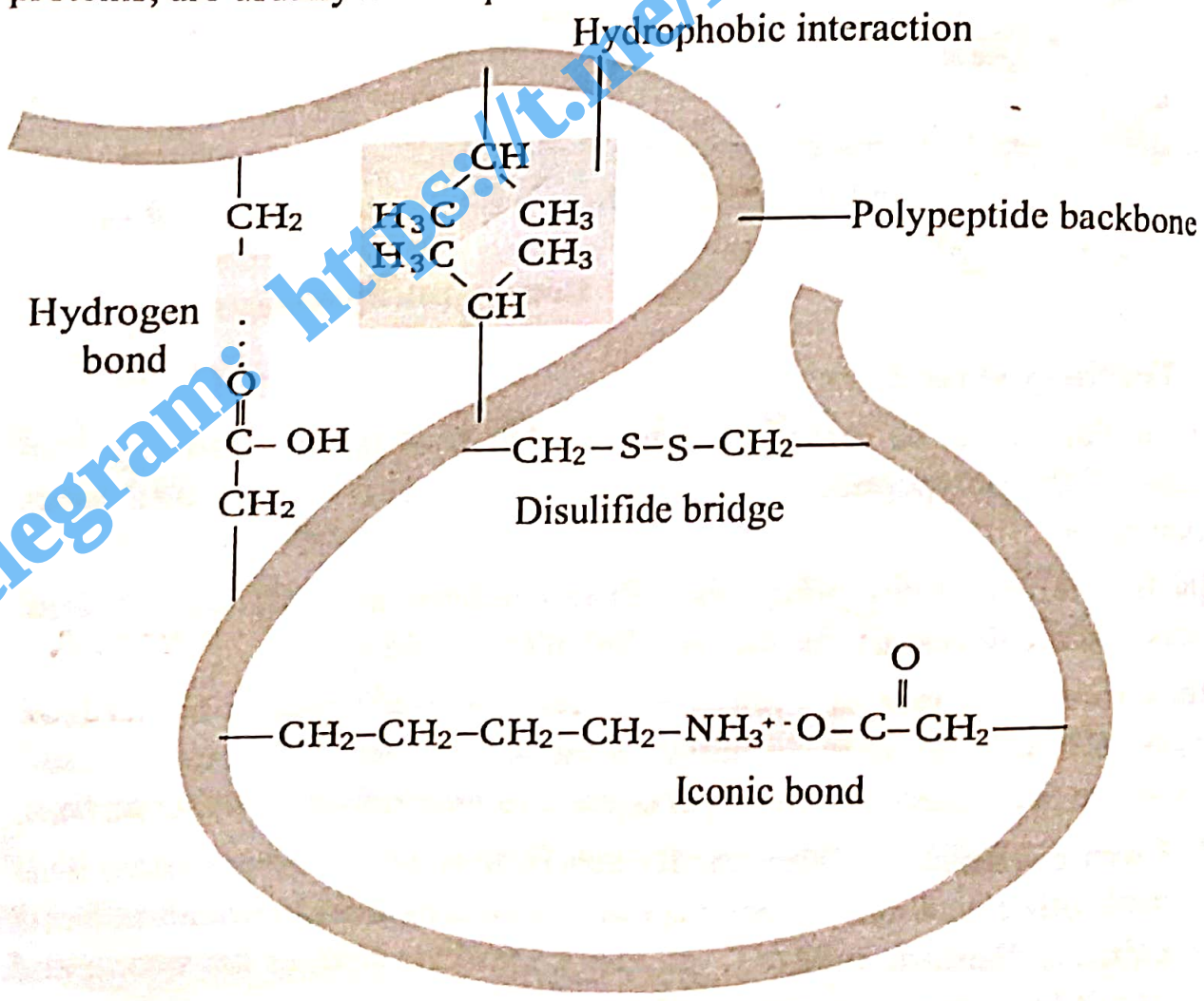
- **Ionic bonds :** can occur between an R groups with a unit of positive charge and one with a unit of negative charge. These ionic bonds are weak. They' ll be lost when a protein denatures.
- **Hydrogen bonds :** It is formed between R groups of certain amino acid. These weak bonds are lost when a protein denatures
- **Hydrophobic interactions :** result from the tendency of non - polar R groups to be excluded by the surrounding water and therefore to associate in the interior of the lobular structure. These weak bonds are lost when a protein denatures.

Based on overall three - dimensional shape, there are two catagories of enzyme

I. **Globular proteins** have an overall rounded or irregular shape. Many enzymes are globular proteins.

II. **Fibrous proteins** are long and cable - like

These proteins are important to the structure of the cell, like cytoskeletal proteins, are usually fibrous proteins.



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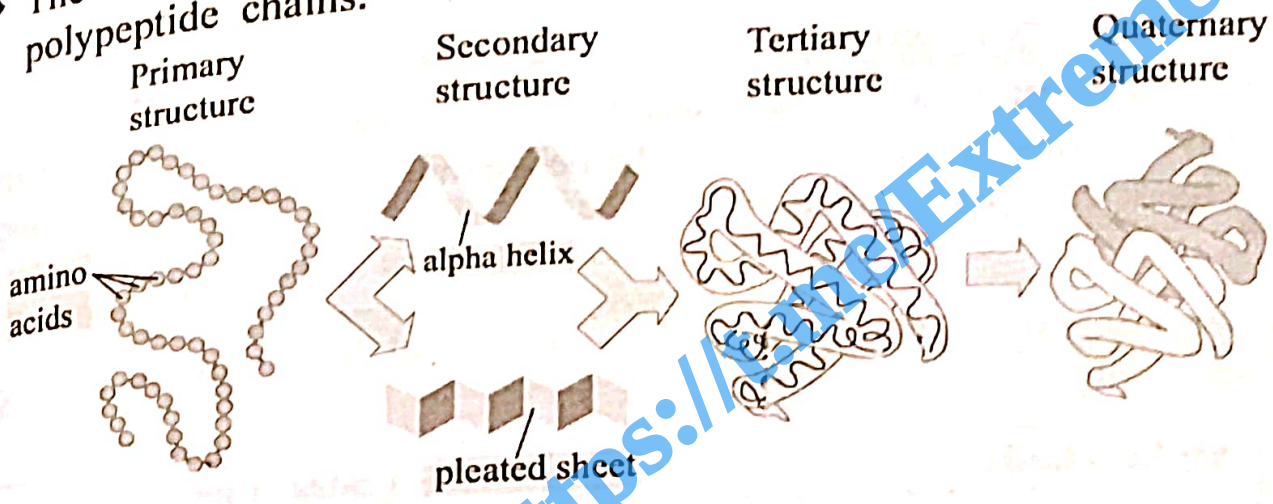
4. Quaternary structure

Some proteins are large and complex, consisting of **many polypeptide chains**. These proteins have quaternary structure. The bonds that hold multiple polypeptide chains together to make a protein are the same types of bonds that hold together the tertiary structure of proteins. For example :

- The **hemoglobin** that is carrying oxygen around in your blood right now is a protein with quaternary structure. In order to do its job for your body, each hemoglobin protein is made of four joined polypeptide chains, two of one type (alpha chains) and two of another (beta chains).
- The **collagen** that supports your skin is made of three identical polypeptide chains.



Illustrative Example

14. The sequences of amino acids in the polypeptide chain is :

A. primary structure
B. secondary structure
C. tertiary structure
D. quaternary structure

Explanation :- The primary structure of a protein is the **sequence of amino acids** in the polypeptide chain.

Answer : A

15. The name of chemical bond that maintain the tertiary structure of protein are :

A. Ionic bond
B. hydrogen bond
C. disulphide bridge
D. all

Explanation :- tertiary structure of a polypeptide chain is held together by **various types of bonds** . These are : covalent, ionic, hydrogen, and hydrophobic interaction.

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Answer : D

16. The best example of tertiary structure of a protein is :

A. enzyme
B. hemoglobin
C. collagen
D. silk

Explanation :- enzyme is a globular protein, which is a tertiary structure. Hemoglobin and collagen is a quaternary structure ; while a silk, spider web are secondary structure.

Answer : A

17. A protein structure, which is made of two or more polypeptide is :

A. Primary structure
B. secondary structure
C. tertiary structure
D. quaternary structure

Explanation :- quaternary structure of a protein is made of many polypeptides (more than two). The best example is hemoglobin, which is made of 4 polypeptide.

Answer : D

18. The comparison between alpha - helix and beta - plated sheet of secondary structure of protein is :

Alpha - helix	**Beta - plated sheet**
spiral / helical coil	Sheet like, zigzag
elastic in nature	Not elastic
Hydrogen bond is in between the first and the fourth amino acids residue at every turn	Hydrogen bond is at different regions of a polypeptide chain

19. Describe how only 20 amino acids can make so many different kinds of protiens.

Explanation :- By varying the number, and sequence of amino acid.

3.6 How Enzymes Work

There are two models of enzyme action ; the lock - and - key model, and the induced - fit model.

Both of these models suggest that the enzyme catalyses the reaction by lowering the activation energy. However, they differ in the way that they

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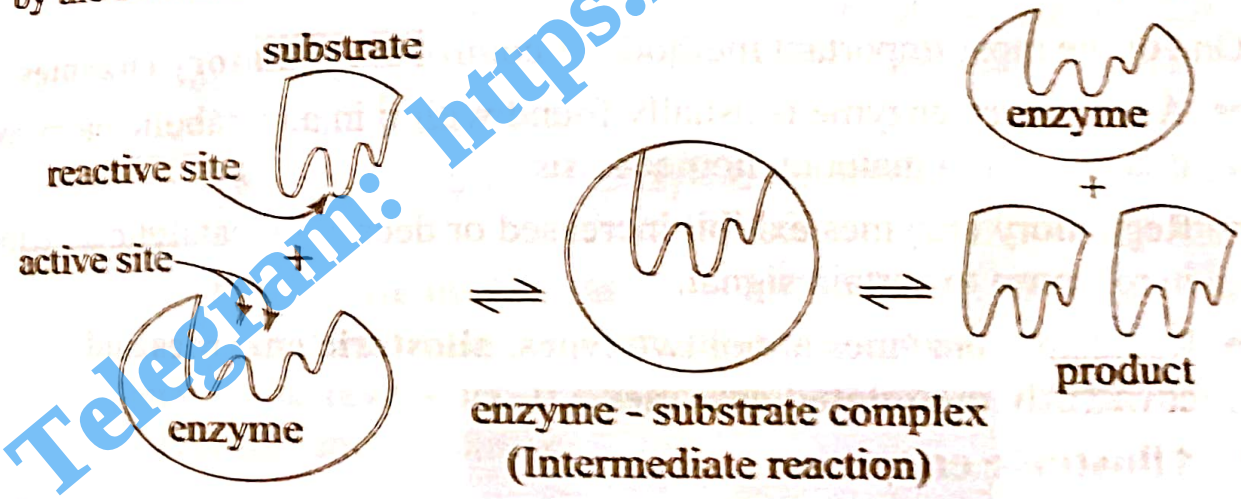
explain how this happens.

i. **Lock and key model :**

- The active site of an enzyme is rigid / static in its shape
- There is no change in the active site before and after a chemical reaction
- Active site and substrate are complementary
- Active site and substrate have non covalent interaction
- First proposed in 1894 by a German biochemist named fischer

ii. **Induced - modal :**

- According to this model substrate causes a change in enzyme, which causes the active site to change its shape to allow enzyme and substrate to bind. This is because
- The amino acid side chain that make up the active site mold into the precise positions
- Proposed by Koshland in 1958
- The active site and substrate become complementary after conformational change in the active site.
- Active site is not static, can change its conformation, which is tensioned by the substrate.



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Illustrative Examples

20. Which model that proposes the shape of the substrate molecules is complementary to that of the active site ?

A. Induced fit model
B. Lock and key model
C. Activation energy
D. None

Explanation : - The key and lock model - the active site and substrate

are complementary.

Answer : B

21. Compare and contrast the key - lock model and induced fit model :

Feature	key - lock model	Induced fit model
The active site	Is rigid / static	Is flexible
Substrate and active site	Are always complementary	Are not complementary before changes in the shape of active site
Acceptability	Currently not acceptable	Currently acceptable
First proposed in	1894 by Fischer	1958 by Koshland

22. What is the similarity between key - lock model and induced fit model :

Explanation : - both of these models suggest that the enzyme catalysis the reaction by lowering the activation energy

3.7 Enzyme Regulation

How do cells control the hundreds of reactions going on inside them ?

One of the most important methods of control is **regulatory enzymes**

- A regulatory enzyme is usually found within in a metabolic pathway; it is needed to maintain homeostasis
- Regulatory enzymes exhibit increased or decreased catalytic activity in response to certain signal.
- Regulatory enzymes are of two types, **allosteric enzymes** and **covalently modulated enzymes.**

I. Allosteric enzymes

These enzymes have another site, separate from the active site, to which another molecule can bind and act either as an **inhibitor** or as **activator.**

- It has two binding sites called active site (for binding substrate) and **allosteric site** (for binding effector).
- Enzymes activity can be activated or inhibited through non covalent interaction of the enzyme with effectors. This form of control is **allosteric regulation.**

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II. General and covalent modification

- Genetic modification of enzymes is to improve the properties enzymes and gain active and inactive forms.
- Covalent modifications is the covalent attachment of a molecule on enzymes and modify the activity of enzymes and other proteins. The donor molecule modify the properties of enzymes. **for example :**

Type of modification	Donor molecule	Example of modified enzyme	Function of enzyme
Phosphorylation	ATP	Glycogen phosphorylase	Glucose homeostasis, energy transduction

- Modification can target a single type of **amino acid** or multiple amino acids of an enzyme so that the chemical property is changed
- **Phosphorylation** is the transfer of phosphate from donor to recipients

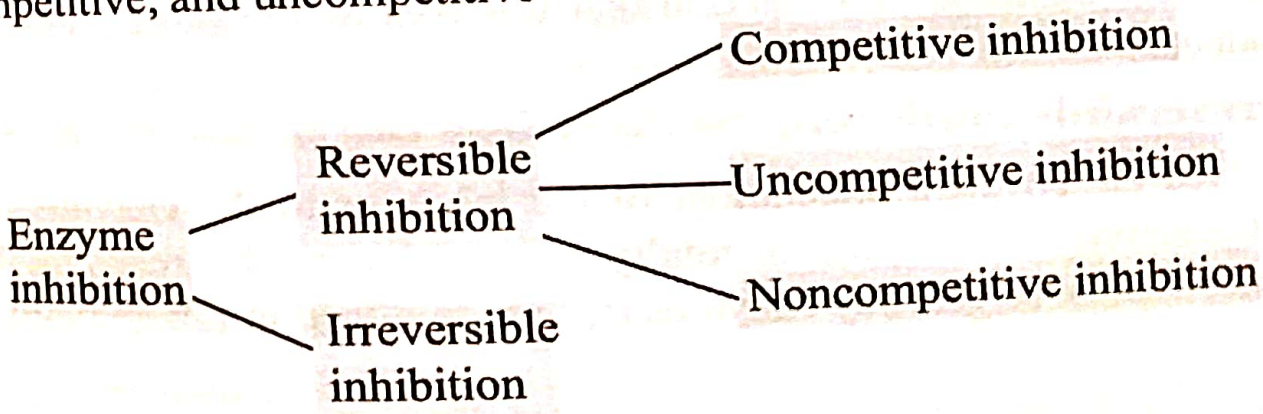
3.7.1 Enzyme Inhibition

Enzymes may also be deactivated, temporarily or permantly by enzyme inhibitors.

Enzyme inhibitors are any molecules that bind to enzyme and stop enzymes from working or reduce their catalytic power. The whole process is known as **Enzyme inhibition.**

There are two main types of enzyme inhibition : **irreversible** and **reversible.**

1. Reversible Enzyme inhibition : it is an inhibition of enzyme activity in which the inhibitor binds on enzyme non - covalently and dissociates from binding site reversibly. It can be grouped in to : competitive, non - competitive, and uncompetitive



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i. Competitive inhibition :

- Occurs when a molecule that is **similar in shape** to the **substrate** enters the **active site** and gets in the way of catalysis. In other words, the competitive inhibitor competes with the substrate for the active site.
- Formation of **E - S complex is reduced** while a new enzyme inhibitor complex **(E - I complex) is formed.**

ii. Non competitive or allosteric inhibition

- Occurs when a regulatory molecule binds to a site other than the active site of the enzyme. Binding sites other than the active site are called **allosteric sites**
- Molecules that bind to allosteric sites to regulate enzymes are called **non competitive inhibitors.**
- When non competitive inhibitors bind to alloteric sites, they cause slight shape changes in the enzyme. These shape changes affect the active site as well, which means the substrate can no longer bind to the enzyme.

Note : Feedback inhibition / end product inhibition :

- The most common way that cells regulate their metabolic pathways is though **feedback inhibition or end product** inhibition.
- During feedback inhibition, an end product of a pathway acts as the non competitive inhibitor of an enzyme in that pathway.
- If a cell has made enough of a product, then lots of those molecules will be floating around in the cell. The end product binds the allosteric site on the first enzyme and shut it down.
- When the cell uses up the product, there won't be any excess to bind to the enzyme, and production will start up again.

iii. Uncompetitive inhibitor

Such type of inhibitor binds only to the **E - S complex**, but not on free enzyme. It is common in a reaction that involves two or more substrates or products.

2. Irreversible inhibition

It involves the covalent attachment of the inhibitor to the enzyme. The catalytic activity of enzyme is totally lost, and can't be restored. For example, **aspirin** targets key enzymes that are involved in inflammation.

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23. Explain the meaning of end product inhibition ?

Explanation :- end product inhibition is a condition where the end product inhibits the enzyme controlling the first stage of a reaction sequence.

24. How does a non - competitive inhibitor reduce the rate of an enzyme catalyzed reaction ?

Explanation :- non - competitive inhibitor binds to some part of an enzyme other than the active site to change the shape of an enzyme so that the rate of catalysis decreases.

25. In a competitive inhibition, which one of the following factor determines the rate of reaction ?

A. temperature
B. number of activator
C. substrate concentration
D. pH value

Explanation :- in the competitive inhibition, the substrate and inhibitors have similar shape, they have equal chance of binding into the active site of an enzyme. So that, increasing the concentration of substrate will increase the rate because more substrate will bind into an active site than the inhibitor.

Answer : C

26. Competitive and non- competitive inhibitors are similar in :

A. both compete for active site
B. both inhibit the enzyme temporarily
C. both inhibit enzyme activity permanently
D. A and B

Explanation :- Competitive and non - competitive inhibitors are reversible type of inhibitor not permanent ; the former compete for active site but the later binds at allosteric site.

Answer : B

27. The type of inhibitor that binds only to the E - S complex, but not on free enzyme is known as ___ ?

A. Competitive
B. non - competitive inhibitors

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C. Uncompetitive inhibitors

D. coenzyme

Explanation :- uncompetitive type of inhibitor binds only to the E - S complex, but not on free enzyme.

28. What are allosteric enzymes ?

Answer : C

Explanation :- These enzymes have another site, separate from the active site, to which another molecule can bind and act either as an **inhibitor** or as **activator.**

29. Compar and contrast between reversible and irreversible inhibitor is

Reversible inhibitor	**Irreversible inhibitor**
Binds non - covalently on an enzyme	Binds covalentlyon an enzyme
It inhibits temporarily, and restored later	It inhibits permanently, and cannot be restored

30. List the three types of reversible inhibitor ?

Explanation :- competitive, non - competitive, and uncompetitive

31. The attachment of a molecule can modify the activity of enzymes and other proteins. The donor molecule modifies the properties of enzymes. Such type of enzyme is known as ____ ?

A. Covalently modulated enzymes

B. Genetically modified enzymes

C. Allosteric enzymes

D. B and C

Explanation :- Covalently modulated enzymes are enzymes which are formed by chemical modification, attachment of chemical groups on enzymes to change or improve enzyme activity .

Answer : A

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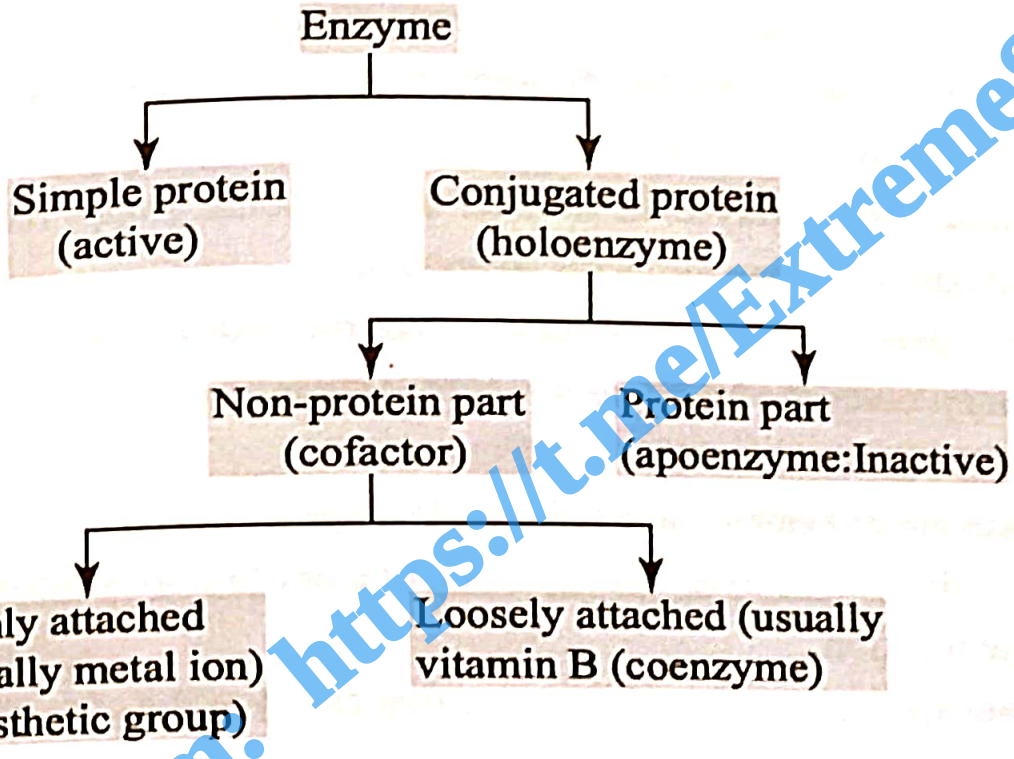
3.8 Enzyme Structural Classification

Enzyme activity is often influenced by the presence of other chemicals. Some of these can enhance an enzyme's activity called **cofactors**, they are non-protein component of an enzyme and may be organic molecules (called coenzymes), or inorganic molecule. Only some enzymes are exclusively made of protein, they are called **active enzymes** (simple protein) ; they don't need any addition of molecules to aid catalysis. However, many other enzymes need a non - protein addition known as co - factor to function properly.



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- ☑ The protein part of an enzyme is called **apoenzyme,** which is inactive (can't catalyze when it is alone)
- ☑ **Apoenzyme + cofactor = holoenzyme (conjugated protein)**, which is active for catalysis

Cofactor are of two types : coenzymes and prosthetic groups

i. Coenzymes

- ➥ Coenzymes are organic cofactor
- ➥ Loosely associated with an enzyme
- ➥ They transfer chemical groups, atoms, or electrons from one enzyme to another
- ➥ Many of them are derived from vitamins, example **DNA.**

ii. Prosthetic groups

- Prosthetic groups are organic molecules
- It is tightly bond to an enzyme
- Example : flavins, heme group containing iron
- Heme is a permanent part of catalase

Basic classification of enzymes

A systemic classification of enzymes has been developed by Internati Enzyme commission. This classification is based on the type of reacti catalyzed by enzymes. There are six major classes of enzymes. These a

1. **Oxidoreductase** - catalse the transfer of H and O atom or electron fr one substrate to another.
2. **Transferase** - catalyse the transfer of functional groups like methyl, phosphate from one molecule to another
3. **Hydrolases** - are enzymes that catalyze the hydrolysis of various bon which involves the breaking down of large molecules by addition of water
4. **Lyases** are enzymes that catalysis cleavage of bonds by means other than hydrolysis or oxidation in which two or more substrates are involv in one reaction.
5. **Isomerases** are a class of enzymes that catalyze the conversion of molecules from one isomer to another isomer.
6. **Ligases** are enzymes that catalyze the joining of two molecules to for new bond by hydrolysis of the di- phosphate bond in ATP.

Six major classes of Enzymes and Examples of Their Subclasses

	Classification	Distinguishing Feature
1.	**Oxidoreductases**	$A_{red} + B_{ox} \rightarrow A_{ox} + B_{red}$
	Oxidases	Use oxygen as an electron acceptor but do not incorporate it into the substrate
	Dehydrogenases	Use molecules other than oxygen (e.g NAD^+) as an electron acceptor
	Oxygenases	Directly incorporate oxygen into the substrate

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Peroxidases	Use H_2O_2 as an electron acceptor
2. Transferases	$A - B + C \rightarrow A + B - C$
Methyl transferases	Transfer one - carbon units between substrates
Aminotransferases	Transfer NH_2 from amino acids to keto acids
Kinases	Transfer PO_3 ~ from ATP to a substrate
Phosphorylases	Transfer PO_3 ~ from inorganic phosphate (p_x) to a substrate
3. Hydrolases	$A - B + H_2O \rightarrow A - H + B - OH$
Phosphatases	Remove PO_3 ~ from a substrate
Phosphodiesterases	Cleave phosphodiester bonds such as those in nucleic acids
Proteases	Cleave amide bonds such as those in proteins
4. Lyases	$A(XH) - B \rightarrow A - X + B - H$
Decarboxylases	Produce CO_2 via elimination reactions
Aldolases	Produce aldehydes via elimination reactions
Synthases	Link two molecules without involvement of ATP
5. Isomerases	$A \Leftrightarrow H_3O - A$
Racemases	Interconvert I, and D stereoisomers
Mutases	Transfer groups between atoms within a molecule
6. Ligases	$A + B + ATP \rightarrow A - B + ADP + Pi$

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Illustrative Example

32. All enzymes are exclusively made of protein ; they are called active enzymes so that they don't need any addition of molecules to aid catalysis.

A. true B. false

Explanation :- only some enzymes are made of exclusively protein (active enzymes). Most enzymes need an addition of chemical groups

to aid catalysis.

Answer : B

33. The class of enzyme involved in the reaction : $AH + NAD \rightarrow A + NADH$

A. Oxidoreductase
B. hydrolase
C. transferase
D. Ligase

Explanation : - oxidoreductase - catalyse the transfer of electron or hydrogen from one molecule to another.

Answer : A

34. An enzyme that catalybses the joining of two molecules by the formation of new bonds is :

A. Lyasase
B. ligases
C. isomerase
D. hydrolase

Explanation : - Ligase

Answer : B

35. An enzyme that catalyses the transfer of specific chemical group from one molecule to another

A. Lyasase
B. Ligases
C. isomerase
D. transferase

Explanation :- Transferase

Answer : D

36. Most of the co - enzymes are derived from :

A. vitamins
B. carbohydrate
C. lipids
D. protein

Explanation :- co - enzymes are a co - factor, derived from vitamins.

37. An apoenzyme is :

A. A protein part of congugated enzyme
B. none protein part of enzyme
C. It is inorganic molecule
D. it is a co - factor

Explanation :- apoenzynme is a protein part of conjugated protein.

Answer : A

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38. A co - factor that is bounded on enzymes tightly are known as :

A. Co - enzyme
C. NAD
B. Prosthetic groups
D. inorganic molecules

Explanation : - prosthetic groups are organic molecules such as heme (containing iron), which attach on enzymes firmly (tight) to enhance catalysis of enzymes.

Answer : B

39. According to international enzyme commission, enzymes are classified based on ____ ?

Explanation : - enzymes are grouped into 6 major classes based on the type of reaction they catalysis.

3. 9 Factor Affecting Enzyme Action

The functioning of enzymes can be affected by various factors. Some of these factors include : temperature. pH, substrate concentration, enzyme concentration, inhibitors, activators, end products, radiation, and water

I. Temperature

- When the temperature is raised, particles move around more quickly (have more kinetic energy).
- Then a substrate particle will collide with an enzyme molecule
- This puts strain on the bonds that hold the atoms in place. Bonds begin to break and, in the case of an enzyme, the shape of the molecule and the active site in particular, begin to change.
- The enzyme begins to lose its tertiary structure and **denature.**
- Increasing temperature up to optimum value will increase the rate of reaction (more E - S complex formed).

Note : Enzyme work maximum over a particular temperature known as optimum temperature

- Above the optimum temperature, the enzyme denatures very quickly to the point at which the shape of the active site has changed so much that an enzyme - substrate complex cannot form.
- When temperature is below optimum, enzymes are inactive / deactivated and slow down catalysis. But when the temperature is restored in to normal, the enzyme regains its catalytic power.

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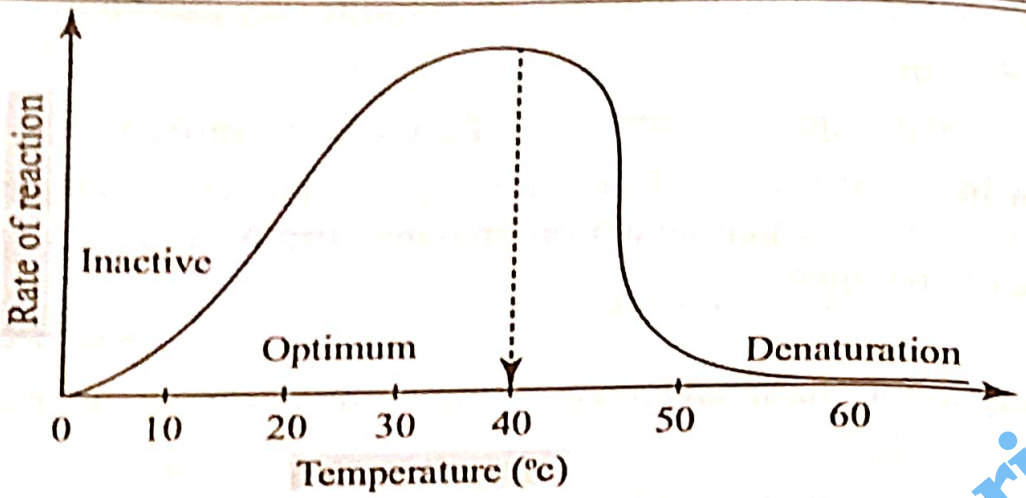
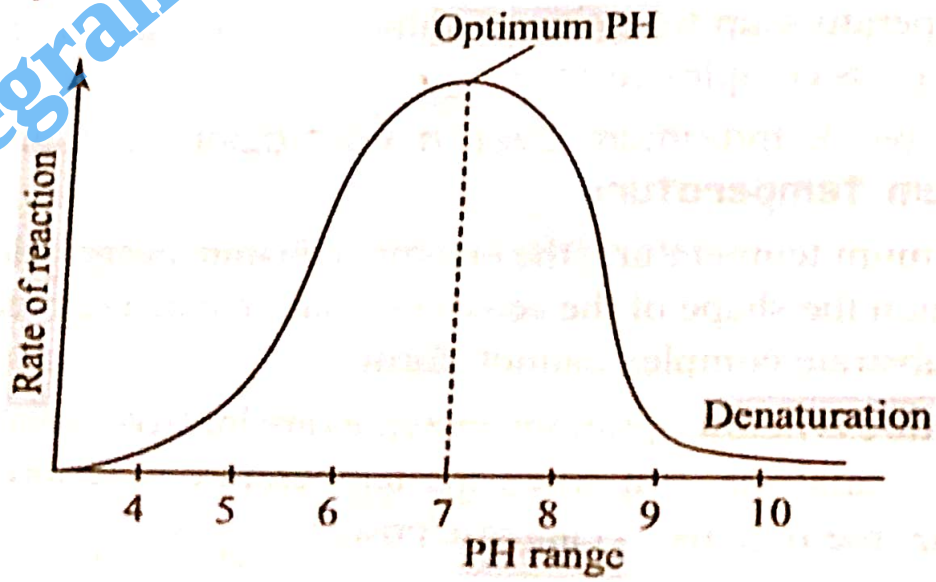


II. PH

- The pH scale is a measure of the hydrogen ion concentration of a solution or other liquid system
- The pH scale ranges from 0 to 14. Solutions with a pH of less than 7 are acidic, those with a pH of more than 7 are alkaline and a solution with a pH of exactly 7 is neutral.
- The majority of enzymes in most mammals function most efficiently within the pH range 5 - 9.
- Significant changes in pH can affect an enzyme molecule by breaking ionic bonds hat hold the tertiary structure in place ; this leads to denaturation of the enzyme molecule altering the charge on some of the amino acids that form the active site.
- High or low PH value than optimum value will cause ionization of enzyme which results in denaturation of enzyme



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II. Substrate concentrations

The activity of an enzyme depends on the number of substrate molecules per second that bind to form enzyme substrate complexes.

- A small number of substrate molecules mean few collisions and so only a few enzyme - substrate complexes form
- Increasing the concentration of the substrate means more collisions and more enzyme - substrate complexes. So, the overall rate of reaction is increased.
- At high substrate concentration, each enzyme molecule works at maximum turnover, each active site is binding with substrate molecules all the time.
- Increasing the substrate concentration beyond the saturation point will have no effect on the activity of the enzyme because all the active sites are occupied all the time.

Illustrative Examples

40. When an enzyme is denatured by heat or pH, which one of the following does it lose ?

A. The peptide bond
B. primary structure
C. secondary structure
D. tertiary structure

Explanation :- increasing in temperature beyond the optimum level or change in pH causes enzymes to lose its tertiary structure.

Answer : D

41. How do you describe the effect of substrate concentration on enzyme reaction ?

A. Rate of reaction increases as we increase substrate
B. At higher substrate concentration, the rate of reaction decreases
C. Substrate concentration does not have effect on chemical reaction
D. at higher substrate concentration, the rate of reaction remain constant

Explanation :- if all the active site of an enzyme is saturated (full), increasing substrate concentration does not increase the rate of reaction.

Answer : D

42. Change in pH affect an enzyme molecule by :

A. Breaking ionic bond that holds the tertiary structure

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B. Alter the charge of some amino acids in the active site

C. Increasing activation energy

D. A and B

Explanation : - Significant changes in pH can affect enzyme molecule by breaking ionic that hold the tertiary structure in place ; this leads to denaturation of the enzyme molecule altering the charge on some of the amino acids that form the active site.

Answer : D

43. Alteration of structural properties that causes alteration of the tertiary structure of protein is

A. Inhibition

B. inactivation

C. denaturation

D. deformation

Explanation :- as temperature increases beyond the optimum value, enzyme losses its conformation or shape (tertiary structure) so that it stops functioning. This known as denaturation.

Answer : C

44. At optimum temperature, enzyme controlled rate of reaction is

A. More ES - complex is formed

B. Rate of reaction becomes at its peak

C. Kinetic energy of molecules increases

D. All

Explanation :- Enzyme work maximum over a particular temperature known as optimum temperature. At this point more ES - complex formed, maximum rate of reaction, kinetic energy of molecule increases (more collision between enzyme and substrate).

Answer : D

3.10 What is Enzyme kinetics

Enzyme kinetics is the branch of biochemistry, that studies about the rate of enzyme catalyzed reaction. It plays a critical role in drug discoveries as well as elaborating the model of action of drugs. One of the most known models of enzyme kinetics is the **Michaelis - Menten model.**

According to this model the enzyme reversibly combined with substrate to form an ES - complex that subsequently yields product, regenerating the free enzyme.

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$E + S \rightarrow ES \rightarrow E + P$

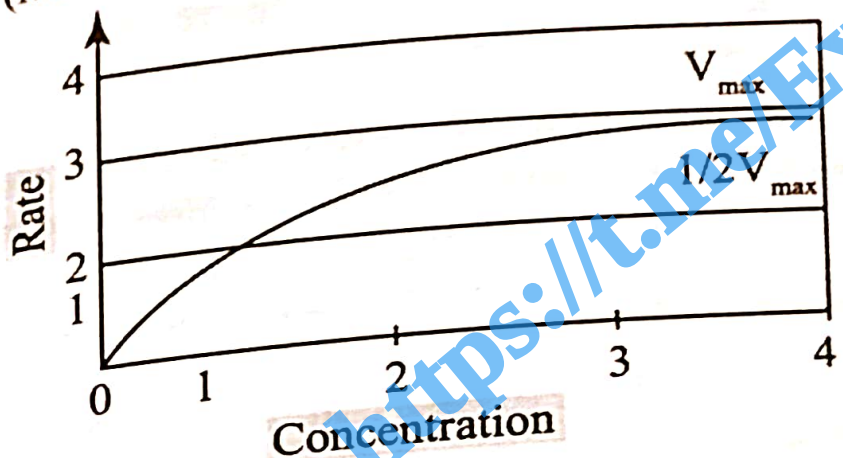
Michaelis - Menten equation : it is an equation which describes how reaction velocity varies with substrate concentration.

$$Vo = \frac{V_{max}[S]}{K_m + [S]}$$

- **Vo** = the initial velocity / rate reaction
- **Vmax** = the maximal velocity / maximum rate of reaction

[S] = the substrate concentration

KM = substrate concentration at half - maximal velocity (Michaelis constant).



4. Enzyme concentration

As the enzyme concentration increases, the rate of reaction will increase. Thus, the rate of reaction is directly proportional to the concentration of enzymes.

However, at a constant substrate concentration, further increasing the concentration of the enzyme will not increase the activity of the enzyme. Because, each enzyme works at maximum turnover, so the activity of the enzyme is likely to remain constant.

Turn over number of molecule is the number of substrates converted to product by one enzyme molecule per second at fully occupied (saturated) active site of on enzyme.

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3 . 11 Application of Enzymes in Industries and their Benefits

Enzymes are enabling various industries to guarantee the quality and stability of its products with increased production efficiency (100 - 1000 times efficient than inorganic catalyst).

- Enzymes are widely used in food, feed, textile, papermaking, leather and detergents, pharmaceutical and other industrial productions.
- It decreases production cost and energy consumption, environmentally friendly (reduce pollution).
- Here are some examples of enzymes and their functions that can be used in the industries :

Enzymes	Industrial application	Function
Cellulases	Biofuel industry	Breakdown cellulose into sugars and ferment to produce cellulosic ethanol
Ligninase	Biofuel industry	Pre-treatment of biomass for biofuel production
Protease, Amylase, Lipase	Biological detergent	Protease removes protein, amylase removes starch, and lipase removes fat or oil stains from laundry and dishware
Mannanase	Biological detergent	Remove food stains from the common food additive guar gum
Betaglucanase	Brewing industry	Improve beer filtration
Papayin	Culinary uses	Tenderize meat for cooking
Rennin	Dairy industry	It hydrolyses protein in the manufacture of cheese
Trypsin	Food processing	Manufacture hypo - allergenic baby foods
Cellulases, pectinase	Food processing	Clarify fruit juices

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Nuclease, DNA Ligase, polymerase	Molecular biology	Uses restriction digestion and polymerase chain reaction to create recombinant DNA

Illustrative Example

45. What is enzyme kinetics ?

Explanation :- Enzyme kinetics is the branch of biochemistry, that studies about the rate of enzyme catalyzed reaction.

46. Write some application of the study of enzyme kinetics ?

Explanation :- It plays a role in **drug discoveries** as well as the **model of action of drugs.**

47. Describe the Michaelis - Menten model ?

Explanation :- According to this model the enzyme reversibly combined with substrate to form an ES - complex that subsequently yields product, regenerating the free enzyme. $E + S \rightarrow ES \rightarrow E + P$

48. The Michaelis - Menten equation describes _____ ?

Explanation :- It is an equation which describes how reaction velocity varies with substrate concentration.

$$Vo = \frac{V_{max}[S]}{K_m + [S]}$$

49. Explain the reactions why enzymes are used in industries ?

Explanation :- Enzymes increase the production efficiency (100 1000) times. It decreases production cost and energy consumption, environmentally friendly (reduce pollution).

50. Select the wrong combination of enzyme with its function in the industries ?

A. Papain - tendering meat

B. rennin - tendering meat

C. ligninase - food production

D. mananase - removes stain from clothes

Explanation :- ligninase is an enzyme responsible for biofuel production.

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Answer : C

51. An enzyme that be used in molecular genetics is

A. Polmerase
B. DNA ligase
C. Nuclease
D. all

Explanation :- Molecular genetics studies about the genetic materials of organisms such as DNA and RNA. Enzymes responsible for making DNA, RNA are : polymerase, DNA ligase, Nuclease, RNA primase, etc

Answer : D

52. At a constant substrate concentration, increasing enzyme concentration will increases rate of reaction

A. True
B. False

Explanation :- Because, each enzyme works at maximum turnover, so the activity of the enzyme is likely to remain constant.

Answer : B

3. 11. 1 Malting in Ethiopian Tradition

Malting is the process of steeping, germinating and drying grain to convert it into malt. Germination and sprouting involve a number of enzymes such as (amylase, protease, etc.) to change seed into seedling, and the malt producer stops this stage of the process when the required enzymes are optimal.

- The malt is mainly used for brewing or whisky making, malt vinegar.
- It is the malt that provides colour, aroma, and flavour to every larger (beer).
- Various grains are used for malting such as: barley, sorghum, and wheat
- The most common enzymes used in the malting process are : beta-glucanase, alpha amylase, protease and beta amylase

Several equipment can be used to produce malt :

- Traditional floor malting germinates the grain in a thin layer on a solid floor, the grain is manually ranked, keeps the grain loose and aerated.
- In modern malt house the process is more automated, and the grain is germinated on a floor that is slotted to allow air to be forced through the grain bed.

Local alcoholic beverages in Ethiopia that uses malt includes : tella, korefe,

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borde, shameta, etc. They are used in occasions such as holidays, weeding, and celebrations.

3.11.2 Steps of Modern Malting

The process of malting involves **three main steps :** steeping, germinating and kilning / heating

1. **Steeping** - soaking the grain in order to awaken the dormant grain (break seed dormancy).
   - ➥ It is the first and critical step in creating quality malt
   - ➥ It begins with cleaning the kernel from seed
   - ➥ Soak the seed in water and let get oxygen
   - ➥ Existing enzymes in the seed stimulate the formation of new enzymes
   - ➥ It varies in different seeds type and size
   - ➥ It occurs over a period of 24 - 48 hours

2. **Germinating**
   - ☑ Growth and **modification** of grain
     - ➥ Modification occurs by breaking down of carbohydrate and protein, to produce simple molecules
     - ➥ The degree of modification can be checked by sense of smell, eyes, and hands
   - ☑ Supplied with moderate temperature, and regulation of moisture level
   - ☑ Rootlets and shoot emerge from kernel and form **'green malt'**
   - ☑ Takes 4 - 6 days
   - ☑ Periodic rotation of grains to avoid grain clumping

3. **Kilning**
   - ➥ It is the final step of malting process
   - ➥ Heat treatment dries the green malt prevent further germination.
   - ➥ If germination continue, all the starch reserve is used up so that the malt is useless
   - ➥ Moisture is initially removed from germinated grain, it is called **withering**
   - ➥ Enzymatic activity to produce malt colour, and flavour in ales and

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lager.

3. 12 Renowned Biochemist in Ethiopia

Biochemistry is the study of the chemical substances and processes that occur in living organism. **Biochemist** are scientist who are trained in biochemistry, they study chemical process and chemical transformations in living organisms.

- It comprises of : enzymology, molecular biology, immunology, structural biology, etc.
- Conduct research in the areas of : Agriculture, the effect of herbicides an pesticides on plants, cell biology, infectious disease, chromosome and RNA biology

Illustrative Example

53. What is malting ?

Explanation : - Malting is the process of steeping, germinating and drying grain to convert it into malt

54. Explain the importance of malt in the preparation of alcoholic drinks such as beer ?

Explanation : - it is the malt that provides colour, aroma, and flavor to every lager (beer).

55. Write the three steps of malting process ?

Explanation : - Steeping, germinating, and kilning

56. Which of the following enzyme is not produced during malting process ?

A. alpha amylase
B. beta glucanase
C. protease
D. cellulase

Answer: D

57. In malting process of barley, immersing the grain in to water and breaking the dormancy of the seed in to life is _____ ?

A. Kilning
B. steeping
C. germinating
D. none

Explanation : - the first step during malting process is called steeping. It involves soaking the grain into water in order to awaken the dormant

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grain (break seed dormancy).

Answer: B

58. Which of the following is not true about kilning process of malting ?

A. It is the last step of malting
B. Crucial step for colour, flavourof the malt
C. it involves drying of germinating grain
D. is the step of green malt formation

Explanation : - green mats is formed at germination.

Answer : D

59. In malting process different types of enzyme are produced at a stage of ___ ?

A. steeping
B. germinating
C. kilning
D. all

Explanation : - existing enzymes in the seed stimulate the formation of new enzymes as steeping .

Answer: A

Review Exercise on enzyme (Unit 3)

I. Matching item : match the class of enzymes in column "A" with the role of enzymes listed under column "B"

A	B
60. Oxidoreductase	A. catalyse the transfer of functional group from one molecule to another
61. Transferase	B. catalyse the transfer of electron from one molecule to another ($B + AH \rightarrow A + BH$)
62. Hydrolase	C. catalyse the building of large molecules using ATP and water
63. Lyasase	D. catalyse the breaking down of large molecules using water maltose + water → glucose + glucose)
64. Isomerase	E. catalyse the reactions such as $AB \rightarrow BA$
65. Ligase	F. catalyse the reactions such as $A = B + C \rightarrow A + BC$

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II. Multiple choice item : choose the correct answer from the given alternative

66. Which model proposes the shape of the substrate molecule complementary to that of the active site ?

A. Induced fit model
B. Lock and key model
C. Activation energy
D. None

67. Which class of enzyme is involved in breakdown of large molecule to smaller one ?

A. Hydrolases
B. Ligases
C. Transferase
D. Isomerase

68. To which class of enzyme do the digestive enzymes belong ?

A. Hydrolases
B. Isomerases
C. Esterases
D. Transferases

69. In competitive inhibition, the rate of inhibition depends on ___ ?

A. The reaction temperature
B. The enzyme concentration
C. The substrate concentration
D. The ratio of inhibitor to enzyme concentration

70. All are the enzymes produced during malting process, except ___ ?

A. alpha amylase
B. beta glucanase
C. protease
D. cellulase

71. The reason why small amount of each enzyme is needed in metabolic pathway is that enzymes _____ ?

A. Alter the nature of the end product
B. can be re -used over an over again
C. Are nonspecific in their function
D. Increase activation energy needed for the reaction

72. Many coenzymes are derived from :

A. proteins B. Lipids C. Hormones D. Vitamins

73. Which one of the following is wrong about a polypeptide in protein?

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A. Its' amino acid that held together by peptide
B. Formed by condensation reaction
C. can form primary, secondary and tertiary structure of protein
D. it can form quaternary structure

74. Which one of the following mechanisms is important in regulating metabolic pathways in cells ?
A. Irreversible inhibition
B. Allosteric inhibition
C. Enzyme denaturation
D. Competitive inhibition

75. Which class of enzymes is used for joining of two molecules by forming new bonds ?
A. Ligases
B. Transferase
C. Oxido - reductases
D. Isomerases

76. Which model that proposes the shape of the substrate molecules is complementary to that of the active site ?
A. Induced fit model
B. Lock and key model
C. Activation energy
D. None

77. The chemical bond that is found in the primary structure of protein is ___ ?
A. Hydrogen bond
B. peptide bond
C. Ionic bond
D. Disulfide bond

78. All amino acids have a central carbon atom attached with all of the following except _____ ?
A. Amino group
B. Carbonyl group
C. Carboxyl group
D. Hydrogen atom

79. Which one of the following proteins contains the most complex structure ?
A. Silk
B. Enzyme
C. Antibody
D. Hemoglobin

80. Non - protein component of enzymes made of metallic ions are ___ ?
A. Cofactor
B. Coenzyme
C. Prosthetic group
D. Conjugated protein

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81. Which of the following digestive enzyme works in strongly acidic medium ?

A. Amylase B. Pepsin C. lipase D. Erepsin

82. The energy that is required to start the chemical reaction is ____ ?

A. Chemical energy C. Mechanical energy
B. Activation energy D. Kinetic energy

83. One of the following molecules is the building unit of an enzyme ?

A. Amino acid C. Nucleotides
B. Glucose D. Fatty acids

84. In induced - fit model of an enzyme the change in shape of an active site of enzyme is stimulated by _____ ?

A. Active site C. Substrate
B. Allosteric site D. Enzyme

85. Which of the following is false about the active site of an enzyme ?

A. It occupy large portion of an enzyme
B. tensioned by substrate during catalysis
C. it is a catalytic center
D. complementary to its substrate

86. Which of the following is the common characteristic of all enzymes ?

A. They act inside cell only
B. They are proteins
C. They require co - factors
D. They operate at any PH values

87. In competitive inhibition, which one of the following factor determines the rate of inhibition ?

A. Temperature C. Enzyme concentration
B. Substrate concentration D. The inhibitor

88. Which region of human digestive tract contains enzyme that performs well at low PH ?

A. Mouth C. Small intestine
B. Stomach D. Large intestine

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89. In malting process of barley, immersing the grain in to water and breaking the dormancy of the seed into life is ___ ?

A. kilning
B. steeping
C. germinating
D. none

90. The physical properties of enzymes that is little or no dialysis across the cell membrane due to its size is _____ ?

A. biocatalyst
B. colloids
C. enzyme precipitation
D. denaturation

91. The chemical properties of enzymes that is catalysis in both forward and backward direction is _____ ?

A. reversibility
B. regulation
C. PH sensitivity
D. catalysis

92. The specificity of enzyme can be explained in terms of ___ ?

A. bond specificity
B. group specificity
C. substrate specificity
D. all

93. Select the wrong combination of enzyme with its function ?

A. Amylase ------ break protein
B. Helicase ------ unwinding DNA
C. Lipase ------- break fat
D. acetyl cholenstrases ------- break acetylcholine

94. ___ is cofactor, which is attached tightly (firmly) on enzymes and make it functional ?

A. Coenzyme (NAD, FAD)
B. apoenzymes
C. holoenzymes
D. Ca^{+}

95. Select the wrong combination of enzyme with its function in the industries ?

A. papain - tendering meat
B. rennin - curdle milk protein to make cheese
C. ligninase - biofuel production
D. mananase - removes stain from clothes
E. pectinase - clarify beer

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III. Short answer and essay : attempt all and write the answer to the questions properly

96. Draw the structure of amino acid and label the functional groups attached on it

97. In this biochemical reaction, A _ $E1$ → B ---- $E2$ → C → $E3$ → D D is excess and if it is used as an inhibition, it acts on enzyme slow down the reaction, such mechanism of enzyme regulation is called ------

98. What is the importance of enzyme kinetic ?

99. Write the six classes of enzymes ?

100. Write the different application of enzymes in industries

101. Compare and contrast lock - and - key and induced fit model

102. What is transition state in chemical reaction ?

103. Order the following terms in enzyme mechanisms of action ($E + S$, $ES_{complex}$, $E + P$, $ES_{product}$) ?

104. Why some enzymes need co - factors ?

105. Write the meaning of the following terms

A. Apo - enzyme **C. Co - factor**

B. Co - enzyme **D. Prosthetic group**

106. List the different factors that affect the activities of the enzymes

107. Explain how temperature affect enzyme controlled rate of reaction ?

108. Define the following terms : **a. turnover number b. activation energy c. metabolism**

109. What is malting, list the main enzymes used for malting ?

110. List some examples of enzymes and their function in industrial application ?

111. The color and flavor of malt is mainly achieved through malting process called _____ ?

112. List the four protein structure ?

113. Equal volume of Hydrogen peroxide is added to three test tubes A, B,

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C that contain liver, potato and carrot respectively ; the fastest rate of reaction is recorded in test tube A, justify your reason ?

114. A polypeptide chain, organized in to a shape by chemical bonds such as ionic bond, disulfide Bridge, H bond to make it functional such protein structure is called _____ or _____ ?

115. _____ and _____ is the type's secondary structure of protein ?

116. The most common example of covalent modification of enzyme is by addition of a chemical group known as _____ ?

117. _____ and _____ are the most common types of regulatory enzymes ?

118. What are end product inhibition and its importance ?

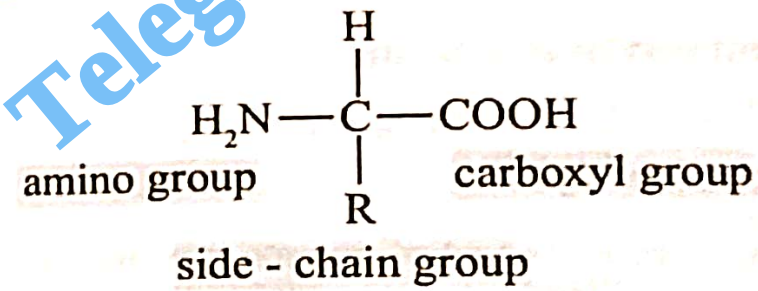
Answer

I. Matching Multiple choices

60. B	66. B	72. C	78. B	84. A
61. A	67. A	73. D	79. B	85. C
62. D	68. A	74. D	80. D	86. A
63. F	69. C	75. B	81. C	87. B
64. E	70. D	76. A	82. B	88. B
65. C	71. B	77. B	83. B	89. B
90. B	91. B	92. A	93. D	94. A
95. D				

III Short answer and essay :

96.



97. E1, end product inhibition

98. It plays a role in **drug discoveries** as well as the model **of action of drugs.**

99. Oxydoreductase, transferase, hydrolase, lyasase, isomerase, ligases

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100. Food processing, pulp, pharmaceutical, biofuel, diary, molecular genetics, biological detergent

101.

Feature	key - lock model	Induced fit model
The active site	Is rigid / static	Is flexible
Substrate and active site	Are always complementary	Are not complementary before changes in the shape of active site
Acceptability	Currently not acceptable	Currently acceptable
First proposed in	1994 by Fischer	1958 by Koshland

102. The transition state is the intermediary state of the reaction.

- The molecules at transition state is neither a substrate nor product
- It represents the formation of ES - complex
- Bonds in substrate are partially made / or broken, thus the molecules are less stable.
- It has the highest free energy. Enzymes lower this energy

103. $E + S \rightarrow ES$ complex $\rightarrow EP$ complex $\rightarrow E + P$

104. Only some enzymes are exclusively made of protein, they are called **active enzymes** (simple protein); they don't need any addition of molecules to aid catalysis. However, many other enzymes need a non - protein addition known as **co - factor** to function properly.

105. **Apoenzyme :** the protein part of an enzyme

Cofactor : the non - protein part of an enzyme

Co - enzyme : the non - protein part of an enzyme, derived from vitamins, attach on enzymes loosely

Prosthetic group : the non - protein part of an enzyme, attach firmly on enzymes

106. Temperature, pH, substrate concentration, enzyme concentration, effectors

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107. As temperature increase up to optimum, rate of reaction also increases. Too much temperature denatures enzyme activity. Too low temperature inactivate it

108. **Turn over number:** is the number of substrate molecule transformed per minute by a single enzyme molecule

Activation energy : a minimum amount of energy required to start chemical reaction

Metabolism : the sum of anabolism and catabolism ; the overall chemical reaction in living organism

109. Malting is the process of steeping, germinating and drying grain to convert it into malt. Germination and sprouting involve a number of enzymes such as (amylase, protease, etc). The most common enzymes used in the malting process are beta - glucanase, alpha amylase, protease and beta amylase

110.

Cellulases	**Breakdown cellulose into sugars to get ethanol**
Ligninase	Pre - treatment in biofuel production
Protease, Amylase, Lipase	Protease removes protein, amylase removes starch, and lipase removes fat or oil stains from laundry and dishware
Mannanase	Remove food stains from the common food additive guar gum
Betaglucanase	Improve beer filtration
Papayin	Tenderize meat for cooking
Rennin	It hydrolyses protein in the manufacture of cheese
Trypsin	Manufacture hypo - allergenic baby foods
Cellulases, pectinase	Clarify fruit juices
Nuclease, DNA Ligase, Polymerase	Uses restriction digestion and polymerase chain reaction to create recombinant DNA

111. Kilning

112. Primary, secondary, tertiary and quaternary

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113. Liver has more catalase enzyme to speed up the breakdown of hydrogen peroxide

114. Tertiary or quaternary

115. alpha helix, beta plated sheet

116. Phosphate

117. allostric enzyme, covalently modified

118. During feedback inhibition, an end product of a pathway acts as the non competitive inhibitor of an enzyme in that pathway. It maintain homeostasis of body cells to perform its activity.

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UNIT 4

4. Genetics

Genetics is a science of heredity that study about the genetic make - up of an organism, and how traits/ character of an organism transfers from parent to offspring

The Genetic Materials (DNA and RNA)

Genetic material is the hereditary substance in the cell. It could be DNA or RNA. It carries all information specific to an organism. Almost all organisms have DNA as the main hereditary material, but some viruses have RNA.

4.1 The Structure and Function of DNA

DNA is an acronym for deoxyribonucleic acid ; it is a nucleic acid, used as the storages of the genetic material of most organisms. Some of the features/ characteristics of DNA are :

- ☑ It is a polymer, made of smaller units called nucleotide
- ☑ Nucleotide is made of **phosphate, pentose sugar** and **nitrogenous base**
- ☑ The **phosphate** and **sugar** make up the **back bone** of vertical ladder
- ☑ DNA is made of two long polynucleotide strands

The two strands are connected to each other by **hydrogen bond** in between bases. Each strand shows **polarity :**

- ➥ If one strand is **5' end to 3'** then the other will be the opposite (**3' end to 5'**).
- ➥ One strand is complementary to the other

- ☑ The nitrogenous bases form the **ladder's rungs**

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- ☑ The nitrogenous bases could be : **purine** (Adenine and Guanine) and **pyrimidine** (Cytosine and Thymine)
- ☑ The nitrogenous bases pairing rules are : ' **A**' is always paired with 'T' , and 'C' is with '**G**'
  - ➥ In DNA molecule the number of **A + C = T + G**
- ☑ The model of double helix DNA was discovered by **J. Watson and F. Crick**

4.2 The Structure and Function of RNA

RNA is an acronym of ribonucleic acid. It is a complex compound that functions in cellular protein synthesis and the storage of genetic material in some viruses. There are three types of RNA that are involved in protein synthesis these are : mRNA, tRNA, and rRNA

- ➥ **Messenger RNA (mRNA) :** carries instruction from DNA to ribosome about which amino acids will be in the polypeptide
- ➥ **Transfer RNA (tRNA) :** it picks up specific amino acids from the cytoplasm and deliver it to ribosome
- ➥ **Ribosomal RNA (rRNA) :** make up the bulk of ribosome itself

The Comparisons Between DNA and RNA

Feature	DNA	RNA
Full name	Deoxyribonucleic acid	Ribonucleic acid
Nitrogenous bases	A, T, C and G	A, U, C and G
Stability	Very stable, less prone to mutation	Less stable, high prone to mutation
Size	Very large	Smaller
Found in	Nucleus and mitochondria	Nucleus, cytoplasm and ribosome
replication	Self	Depend on DNA
Sugar	Deoxyribose	Ribose
Number of strand	2 : double strand with base pairing	1 : single - stranded

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Base pairing	A with T ; C with G	A with U ; C with G

4.3 Genes and Chromosomes

Both genes and chromosomes play an equally important role in holding the genetic information. The chromosome has many genes and these genes hold the information to build specific proteins.

What are the difference between gene and chromosome ?

- **Gene** is a small section of chromosome that determine a specific trait of an organism ; it is a functional unit of heredity
- Genes are arranged on a chromosome, one next to another, much like beads on a necklace
- Each gene has its own position on the chromosomes called **locus / loci**
- **Chromosomes** are a large thread like structures, made of DNA and a protein called **histone**
- Chromosomes exists in pair, one from the mother and the other from father called **homologous pair**

4.3.1 DNA Replication

DNA replication is a process by which DNA makes a copy of itself during cell division. It is essential for transfer of genetic information from one cell to another. DNA replication is **a semiconservative process,** because :

- Each strand of DNA used as a **template** for the synthesis of a new complementary strand
- Each newly made DNA strand is a combination of **one old** and **one new.**

What are the stages of DNA replication ?

1. **Unwinding** / unzipping of DNA strands

- Hydrogen bond in between nitrogen bases are broken down by **helicase** enzyme
- The two strands of DNA are opened

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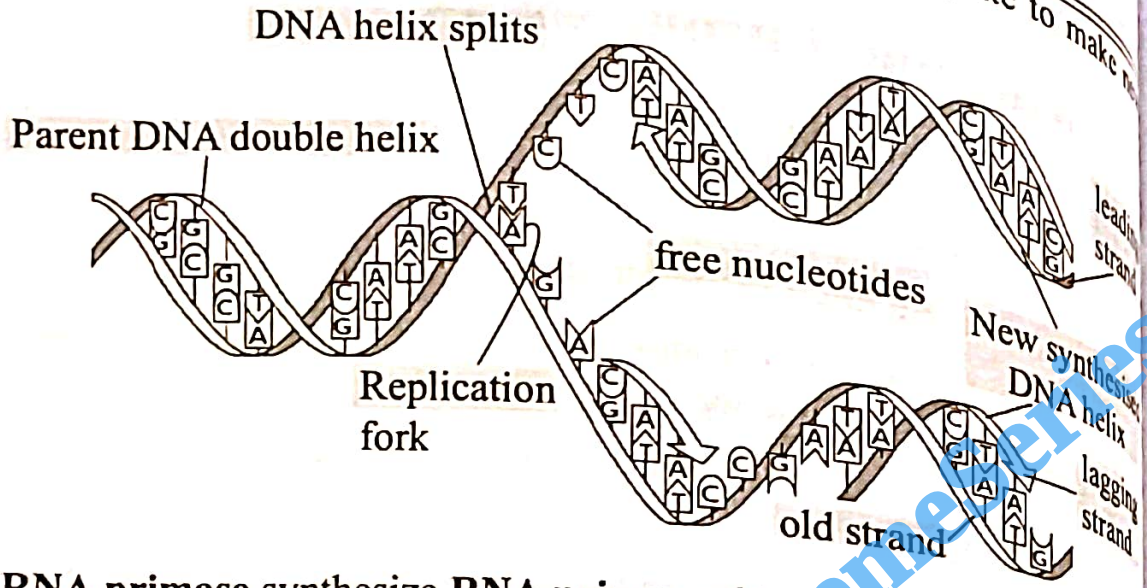
2. Each of the old strands of DNA is used as a template to make n strands



RNA primase synthesize **RNA primer,** which is complementary to D strand, to initiate DNA polymerase' s nucleotide assembly. The prime serve as starting points for copying the DNA

DNA polymerase add free nucleotides in the growing region of bo **lagging** and **leading** strand

- **Leading strand :** a new DNA strand growing in the direction of rep cation fork
- **Lagging strand :** a new DNA strand growing in the opposite directio of replication fork
- **Replication fork :** a Y shaped point formed due to unwinding of t two DNA strands.

DNA ligase joins fragments of DNA by **phosphodiester bond**

Topoisomerase prevents super coiling at the region a head of replicatio fork

3. Two new identical DNA are formed, which are similar to the origin DNA
   - The newly synthesized DNA becomes twisted to form a doub helix.

Illustrative Example

1. The main difference between DNA and RNA is that, RNA is __?

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A. Stable molecule
C. smaller in size
B. is double stranded
D. contain deoxyribose sugar

Explanation : - RNA is a single stranded, smaller in size ; contain ribose sugar, not stable molecule.

Answer : C

2. Which of the following enzyme is used to assemble free nucleotide in to the growing region of DNA ?

A. DNA polymerase
C. DNA helicase
B. topoisomerase
D. ligase

Explanation :- there are man enzymes that are associated with DNA replication. These are :

Enzymes used for DNA replication	**Functions**
DNA polymerase	Assembly of free nucleotides
Topoisomersase	Prevent super coiling of DNA strands
DNA helicase	Unwinding / unzipping of DNA strands
DNA ligase	Join fragments of DNA by phosphodiester bond
RNA primase	Synthesis of RNA primer

Answer : A

3. Which of the following enzyme and its function is mismatched in the process of DNA replication ?

A. DNA polymerase - assembly of free nucleotides

B. Primase - synthesis of RNA primer

C. Topoisomerase - prevents super coiling

D. DNA helicase - join fragments of DNA

Explanation :- see the above table

Answer : D

4. A type of RNA molecule that carries DNA instruction from nucleus to

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ribosome for protein synthesis is __ ?

A. tRNA B. mRNA C. rRNA D. All

Explanation :-

- **tRNA :** it picks up specific amino acids from the cytoplasm and deliver it to ribosome
- **mRNA :** are carries instruction from DNA to ribosome for protein synthesis.
- **rRNA :** It is a part of ribosome itself.

Answer : B

5. When a DNA molecule is replicating which one of the following is not true ?

A. The process is a semiconservative

B. The new DNA formed has one new and old strands

C. DNA polymerase assembled free nucleotides

D. The new daughters DNA are identical to each other but not identical to parent DNA

Explanation :- DNA replication refers to the process of making an identical copy of DNA molecule

Answer : D

6. Which of the following is not true about a DNA molecule ?

A. the strands are 5' - 3' and 3' - 5' ends

B. the strands are antiparallel

C. has complementary bases

D. similar bases

Explanation :-

- DNA has two antiparallel strands
- If one strand has 5' - 3' end the other strand will be 3' - 5' ends
- The nitrogen bases on the two strands are complementary.

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Answer : D

7. If one DNA strand has the base sequences of CGAAT, then the complementary strand will have ___ sequences ?

A. GCUUA
B. GCTTA
C. ATTCG
D. CGAAT

Explanation :-

- The nitrogen bases on the two DNA strands are complementary (C - G, and A - T)
- If one strand has CGAAT, the complementary strand will be GCTTA.

Answer: B

8. Given that (1. chromosome 2. DNA, 3. Gene, 4. Nucleotide) ; the correct order from biggest to smallest is ___ ?

A. 1 2 3 4
B. 2 1 3 4
C. 3 1 2 4
D. 1 2 4 3

Explanation :- the genetic materials from complex to simple is Chromosome → DNA → Gene → Nucleotide

Answer: A

9. If a DNA strands has a total of 100 nitrogen bases ; among them `A` is 20, then the number of `G` is ____ ?

A. 40 B. 30 C. 20 D. 60

Explanation :-

- In the DNA molecule the number of A and T are equal ; and C and G are equal
- The sum of A + T + C + G = 100

$2A + 2G = 100$

$40 + 2G = 100 \quad G = 30$

Answer : B

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10. Which of the following bases pairs with uracil in the RNA molecule ?

A. adenine
B. guanine
C. thymine
D. cytocine

Explanation :- Nitrogen bases in RNA molecule are : A, C, G, and U. by base pairing rules, U is paired with A.

Answer : A

11. Why you say that DNA replication is semiconservative Process?

A. Each new DNA molecule contains one new strand and one old one
B. Every other new DNA molecule contains two old strands
C. Each new DNA strand contains two new strand
D. Every other new DNA molecule contains one new strand and one old one.

Explanation :- DNA replicate and every new cells gets a copy. During DNA replication each old strand gives rise to a new strand.

Answer : D

4.4 The Process of Cell Division

Cell division : it is the process by which a cell divides to form a copy of it. The importance of cell division is :

- For growth, development, repair old tissues, reproduction (genetic exchange), to control doubling of chromosomes.

Cell cycle : it is a time bounded and ordered series of events to produce new cells by division. It has two major phases : interphase and mitosis

1. **Interphase :** the period of the cell cycle during which activities such as cell growth and protein synthesis occur without visible sign of cell division ; it is a stage of preparation of a cell prior to divide. Interphase has three stages : **G1, S, and G2**

G0 phase : is a resting phase (no cellular division nor preparation to divide) but perform normal cellular function.

G1 (first gap) :

- It is a stage to collect nucleotides for S - phase,

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- Cellular activities such as respiration takes place,
- Cell increase in size due to growth,
- It is the longest phase of interphase

S phase : It is a stage of replication (synthesis) of chromosome / DNA (sister chromatids are formed)

G2 phase (second gap) : it is a stage of duplication of organelles, restoring energy lost at S phase, it is the final stage of preparation of a cell to start division. The cell grows much larger.

2. **Mitosis :** it is a type of cell division in which one cell (mother cell) divide to produce two identical daughter cells.

- It occurs in the somatic cell / body cells (somatic cell are that are found in all parts of our body, **except** in reproductive / germ cells).
- In multicellular it used for growth, development and repair of old tissues
- It has two processes : **karyokinesis** (division of nucleus) and **cytokinesis** (division of cytoplasm).
- The karyokinesis has four stages : **prophase, metaphase, anaphase and telophase**

(i) Prophase :

The main activities at prophase stage are :

- Chromatids are condensed (thick or short) and visible by light microscope
- Spindle fibers are formed by centriole and move to opposite poles
- Nuclear membrane disappear
- It is the first and the longest duration in mitosis

(ii) Metaphase :

The main activities at metaphase stage are :

- Chromosomes are aligned at the center / metaphase plate of the cell
- Spindle fibers are connected to each chromosomes

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(iii) Anaphase :

The main activities at anaphase stage are :

- Centromere holding the chromatids together breaks up
- Sister chromatids are separated and move to opposite poles
- It is the shortest duration in time

(iv) Telophase :

The main activities at telophase stage are :

- Chromosomes are at opposite poles
- Chromosoms are uncoil / unraven
- Nuclear membrane reappears

Cytokinesis :

- It is the division of cytoplasm of a cell
- It is called **cell plate formation in plants**
- It is called **cleavage furrow in animal cell**

Meiosis :

It is a type of cell division that occur in the germ cell to produce gamete sex cells (sperm and egg cells). A cell divides to produces **four daughter** cells, each with half number of chromosomes as the parent cell (it is reduction division). Each of the daughter cells shows genetic variations has two successive nuclear divisions : **meiosis I and meiosis II**

I. Meiosis I :

- A cell divide and produces two haploid (n) daughter cells
- It has four stages : prophase I, metaphase I, anaphase I, and telophase

Prophase I :

The main activities at prophase I stage are :

- Chromosomes are condensed, visible by light microscope
- Spindle fibers are formed by centrioles
- Pairing of homologous pair (bivalent), which is known as synapsis

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- There is a point of contact in between bivalent during synapsis, which is called **chiasmata**
- An exchange of genetic materials in between bivalent is called **crossing over**
- Crossing over is the major source of **genetic variation** in the population
- Takes the longest duration of time (85 - 95 % of meiosis)

Metaphase I :

The main activities at metaphase I stage are :

- Homologous chromosomes // bivalent move to the cell center
- Bivalent are line up randomly on either sides of the center. This **random assortment of chromosomes** is another source of genetic variation in the population. Only chance determine which of the pair is passed into any given gamete.
- Spindle fiber from one pole attach to one chromosome and spindle fibers from opposite pole attach to the other pair

Anaphase I :

The main activities at anaphase I stage are :

- Spindle fibers are shortened
- Bivalent // homologous chromosomes are separated and move to opposite poles
- but sister chromatids are not separated

Telophase I :

The main activities at telophase I stage are :

- Spindle fibers disappeared
- Nuclear membrane reappear
- Cytokinesis, a complete division of cytoplasm produces two cells

2. Meiosis II

Meiosis II is the second step of meiotic division which involves the halving of chromosome number to produce haploid cells (n). In meiosis II, the 4

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phases are, again analogous to mitosis. The four phase of meiosis II a

Prophase II

- Chromosomes are condensed
- Nuclear membrane disappeared
- Centriole duplicated and spindle fiber formed

Metaphase II :

- Sister chromatids line up at the cell center / metaphase plate
- Spindle fibers are attached to each sister chromatids.

Anaphase II :

- Sister chromatids are separated and move to opposite poles
- Separated sister chromatids becomes individual chroomosome

Telophase II :

- Chromosomes are at opposite poles
- Spindle fibers are disappeared
- Cytokinesis divides a cell in to two
- Nuclear membrane reappears in each cell

Illustrative Examples

12. What are the main difference between mitosis and meiosis ?

Feature	Mitosis	Meiosis
Division occur in	Somatic / body cell	Germ cell / reproductive cells
Sequences to complete the process	One	Two
Synapsis and crossing over	No synapsis and crossing over between homologous pair	There is synapsis and crossing over

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Genetic variation	No genetic variation among daughter cells	Genetic variation among daughter cells
Used for	Growth, repair old tissues	Producing sex cells / gametes/
How many cells are produced per division	Two cells	Four cells

13. When homologous are separated during meiosis, only chance determine which of the pair is passed into any give gamete. This finding is known as the

A. Law of chance

B. Law universal inheritance

C. Law of segregation

D. Law of independent assortment

Explanation :- Bivalent are line up randomly on either sides of the center at metaphase I of meiosis. This random assortment of chromosomesis the source of genetic variation in the population.

Answer : D

14. In what stage of the cell cycle is the DNA copied ?

A. G1 B. S C. G2 D. G0

Explanation :- interphase has three stages

G1 phase : collect nucleotides, growth

S phase : synthesis of DNA

G2 phase : duplication of organelles, restoring energy lost at S phase.

Answer : B

15. How many chromosomes would a cell have during metaphase I of meiosis if it has 12 chromosomes during interphase ?

A. 6 B. 12 C. 24 D. 36

Explanation :- at metaphase I of meiosis, the cell doesn't complete division ; so that the 12 chromosomes remain in the cell.

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Answer: B

16. The longest and shortest duration of time in the cell cycle of mitosis respectively is ___ and ___ ?

A. prophase / anaphase
C. telophase / metaphase
B. anaphase / prophase
D. metaphase / prophase

Explanation :- In the cell cycle of mitosis, **prophase** is the longest **duration,** while **anaphase** is the shortest.

Answer: A

17. What might happen if cytokinesis were omitted from the cell cycle ?

A. The daughter cell would die
B. The daughter cell would not have nuclei
C. the cell would loss its mitochondria
D. the cell would not divide into two daughter cell

Explanation :- Cytokinesis is the process that divides the cytoplasm of a cell in to two during cell division at telophase stage. If it is omitted, the cell will not be divided.

Answer : D

18. Which of the following is not happening at the end of meiosis II ?

A. Four haploid daughter cells are produced
B. Four sex cells are produced
C. Each daughter cell has the same type and number of chromosome
D. Each daughter cell has genetic variation

Explanation : - Each of the 4 daughters cells have genetic variation / ,have genetic differences.

Answer : C

19. The process of exchanging a section of chromosomes in between homologous pairs/ bivalents during prophase I in meiosis ?

A. Crossing over
B. Synapsis

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D. Duplication

C. Chiasmata

Explanation :-

- At prophase I of meiosis the pairing between bivalents are called **synapbsis**
- The point of contact between the bivalents is called **chiasmata**
- An exchange of a section of chromosomes in between the bivalents is called **crossing over.**

Answer: A

20. During meiosis crossing over occurs in between _____ ?

A. Sister chromatids

B. non - homologous chromosomes

C. non - sister chromatids

D. All

Explanation :-

Crossing over is takes place in between **bivalents** or **homologous pair** or **non - sister chromatids** in prophase I of meiosis

Answer : C

21. Homologous chromosomes are separated and move to opposite poles during ___ phase of meiosis ?

A. Metaphase I

B. Anaphase I

C. anaphase II

D. Metaphase II

Explanation :-

- Separation of **homologous chromosomes** takes place at anaphase I ;
- Separation of **sister chromatids at anaphase II.**

Answer : B

22. The main source of genetic variation among sexually reproducing organism are ?

A. Crossing over

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B. Arrangement of chromosomes at metaphase

C. Asexual reproduction

D. A and B

Explanation :-

The source of genetic variation among sexually reproducin organisms are : crossing over at prophase I, independent assortment and arrangement of chromosomes at metaphase I in meiosis cell division.

Answer: D

4. 4 . 1. Protein Synthesis

Protein synthesis refers to the biological process were by amino acids are assembled by peptide bonding into specific polypeptide sequences in accord with genetic blue print encoded by DNA. Protein is a polymer / large molecule made of a monomer / smaller unit called **amino acid.**

- One protein molecule is made of at least 50 or more amino acids
- Each type of protein is made in accordance with instruction from DNA
- Life can't exist without proteins (such as enzymes, hormones, etc)
- It takes two steps, **transcription** and **translation.**

1. Transcription :

Transcription is the process of copying a segment of DNA into RNA or the synthesis of mRNA from DNA in the nucleus.

Steps of transcription process :

- Unwinding / unzipping of DNA strands
- Breaking down of hydrogen bond in between complementary bases by **DNA helicase**
- One of unzipped DNA strand acts as a template to make complementary mRNA. Nucleotides are freely available in the nucleus to make mRNA. **Example,** if the DNA base triplet is **GCA,** then the mRNA codon will be **CGU.**
- Once the mRNA has formed, it moves away from DNA template and leaves the nucleus to ribosome.

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2. Translation :

It is the conversion of the codes on mRNA in to protein. Translation involves decoding of a mRNA and using its information to build a chain of amino acids or protein.

- It begins with the initiation codon / **start codon (AUG)** on mRNA that codes for amino acids **methionine.**
- It ends with the **stop codons (UAA, UAG and UGA)** on mRNA, which signaling the termination of protein synthesis.
- The process is takes place in the cytoplasm, particularly at the ribosome.

Steps of translation process :

- mRNA attached to ribosome with exposed codons
- Ribosome uses the mRNA as a blue print to synthesize protein
- Each tRNA carries one particular amino acid from cytoplasm to ribosome
- The tRNA's anticodon matched up with the codon on mRNA so that the amino acids are placed in the correct sequences.
- Amino acids are added in to the growing chain of polypeptide by tRNA
- Peptide bonds are formed to link adjacent amino acids
- Elongation of amino acids chain is terminated by stop codons

Note

- **Ribosome** is a small organelle in the cytoplasm, used to synthesize protein
- **A codon** is a sequence of three nucleotide triplet on mRNA
- Each codon encodes specific amino acids

  Example, the codon TAG represents an amino acid called **tryptophan** the codon **GCA** represents an amino acid called **alanine**

  The codon **GCC** represents an amino acid called glycine
- There are 20 naturally existing amino acids but the only possible number of triplet codon is $4^3 = 64$

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- Among the 64 codons, three are stop codons (UAA, UAG and UGA), one codon is initiator / start codon (AUG)
- **Anticodon** is a triplet codon on tRNA which is complementary to a triple codon on mRNA.

Example, if atriplet codon on mRNA is AUC, then the complementary triplet anticodon on tRNA is UAG

The central dogma of life in the molecular genetics is :

$$DNA \xrightarrow{\text{Transcription}} RNA \xrightarrow{\text{translation}} protein$$

Illustrative Examples

23. Which one of the following is initiation amino acid during translation process in ribosome

A. arginine C. cysteine

B. methionine D. serine

Explanation :-

There are a total of 64 codons, among them three are stop codons (UAA, UAG and UGA), and one codon is initiator / start codon (AUG) which represents an amino acid called methionine.

Answer : B

24. DNA triplet codes AAT would code for amino acids carried by tRNA with anticodon of _____ ?

A. AAU B. TTA C. AAT D. UUA

Explanation :-

AAT is codes on DNA ; by transcription process mRNA will have complementary codons UUA ; then codon (UUA) on mRNA and anticodon (AAU) on tRNA are complementary.

Answer : A

25. A protein with 100 amino acids will have a minimum number of _ nucleotide bases representation on mRNA ?

A. 64 B. 33 C. 300 D. 200

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Explanation :-

One amino acid is represented by a triplet codon (3 nucleotide base) on mRNA ; therefore, 100 amino acids will have 100 triplet codons. 100×3 is equal to 300 nucleotide bases.

Answer : C

26. The total number of triple genetic codes used to specify the 20 amino acids, including the initiator and terminator codons are ____ ?

A. 64 B. 61 C. 60 D. 20

Explanation :- naturally there are 20 different amino acids ; if the four nucleotides (A, T, C, and G) are arranged in a triplet, it will form $4^3 =$ 64 triplet codes.

Answer : A

27. The triplet code on tRNA is known as _____ ?

A. codon C. antisense codon

B. anticodon D. sense codon

Explanation :- the triplet nucleotide on mRNA is called **codon** ; while on tRNA is called **anticodon**.

Answer : B

28. Which of the following triplet codons on mRNA is not a terminator (stop) codon ?

A. UAA B. UAG C. AUG D. UGA

Explanation : - among the 64 triplet codes, three code are stop / terminator codon (UAA, UAG, and UGA).

Answer : C

29. Which of the following **is not true** about the process of protein synthesis

A. Synthesis of mRNA from DNA is called transcription

B. The conversion of codons on mRNA into protein is called translation

C. DNA is used as a blue print to synthesize protein by ribosome

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D. One type of tRNA can carries all the 20 different amino acids in to ribosome

Explanation :-

- There are different types of tRNA, which carries unique / specific amino acids
- tRNA that carries methionine will not be useful to carry arginine

Answer : D

30. What does the process of transcription produce ?

A. mRNA B. tRNA C. rRNA D. DNA

Explanation :- Synthesis of mRNA from DNA is called transcription.

Answer : A

31. The immediate result of a mistake in translation would most likely be a

A. Different cell C. different protein

B. different gene D. different sets of allele

Explanation : - If translation process is failed the right type of protein may not be produced.

Answer : C

Match the item to those given in right

32. Contains an anti codon	A. mRNA
33. Carries amino acids to the ribosome	B. t RNA
34. Carries genetic information from DNA to ribosome	C. r RNA
35. A part of ribosome	D. Polypeptide

Answer

32. B 33. B 34. A 35. C

36. Which of the following statements about the genetic code is not true.

A. The genetic code is almost universal

B. The genetic code is double code

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C. Multiple codons may encode the same amino acid

D. Some codons are " stop" or terminator codon

Answer : B

4.5 Mendelian Inheritance

Gregor mendel is famous as the father of genetics. The modern understanding of inheritance began with the work of an Austrian monk named **Gregor Mendel** who studied the inheritance of traits in peas. Mendel established fundamental laws of inheritance that are still used today, which is known as **Mendel's Laws**. These are

The Law of segregation and the law of independent assortment

(i) **The Law of Segregation :** It states that during gamete formation the pair of alleles is separated and become sex cell. Offspring inherit one allele from each parent during fertilization.

Example, if a gene has an allele of "Aa", and then gamete formed after segregation will be ; **'A'** and **'a'**, . During fertilization gametes get together with other gametes to make a new organism.

(ii) **The Law of Independent Assortment :** it states that each pair of alleles sorts itself independently from other pairs of allele. In other words, when a pair of alleles are getting split up and sent to gametes, each pair is doing this process on its own. It is takes place in meiosis cell division at metaphase I.

Note : Mendel established different terminologies used in modern genetics :

1. **Allele** is an alternative version of a gene. For example, pea plants have a gene for flower color (Pp). One allele is for purple flowers, while another is for white flowers.

2. **Genotype and phenotype :**

   The **genotype** is the total of all the genetic make - up of an organism. The genotype is like the total blueprint for the organism. For example, you mention the genetic composition of a trait as 'Bb', 'TT', pure tall, or heterozygous, etc, it is genotypic expression.

3. **Homozygous :** when an organism has two identical alleles for a gene,

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the organism is homozygous for that gene.

4. **Heterozygous :** when the alleles for a gene are different, the organism is heterozygous for that gene.

5. **Pure breeding :** the parental plants that have only one kind of allele for the trait, which is done by self - pollinating the plants for several generations.

6. **Dominant allele :** it is anallele that can be expressed in a heterozygous organism. For example, tallness is a dominant trait that can be represented by alleles of either 'Tt' or 'TT'.

7. **Recessive allele :** is the allele only expressed in a homozygous organism. For example, shortness is a recessive trait expressed by only homozygous allele 'tt'

4. 5. 1 Mendelian Crosses

G. Mendel studied seven /7/ contrasting features in his pea plants. Why Mendel choose garden peas to his study?

Because the plant has : many contrasting traits, usually self - pollinates, and grow easily.

The seven contrasting traits of pea plant are :

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Stem length	Flower position	Flower color	Pod color	Pod shape	Seed color	Seed shape	Contrasting trait
Tall	Axial	purple	Green	inflated	Yellow	Round	Dominant
Short	terminal	white	yellow	constricted	Green	Wrinkled	Recessive
All are tall	All are axial	All are purple	All are green	All are inflated	All are yellow	All are round	F1 phenotypic ratio
3 : 1	3 : 1	3 : 1	3 : 1	3 : 1	3 : 1	3 : 1	F2 phenotypic ratio

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1. The parental generation is identified as P1. P1 plants are always pure **breeding** for the trait that is being studied.
2. The cross between two parental (P1) produces the first filial generation called the **F1 generation.**
3. The cross between two members of the F1 generation produces the second filial generation, which is called **the F2 generation.** The F2 phenotypic ratio is always 3 : 1. In order to determine the mechanism of inheritance of the 7 contrasting traits from parent to offspring, mendel conducted different genetic crosses. These are : **monohybrid and dihybrid crosses.**

   (i) **Monohybrid cross :** it is the cross that tracks a single trait at a time.

Illustrative Examples

1. If you cross **pure round** seed with **pure wrinkled**, what will be the phenotypic and genotypic ratio in the F2 generation ?

   Explanation :- Given : P1 plants are pure round and pure wrinkled **steps :**

   ➥ Parental phenotype : round x wrinkled

   ➥ Parental genotype : RR x rr

   ➥ P1 gamete : **R, R, and r, r**

   ➥ Use punnet sequare to get F1 generation. (a punnett square is used to predict the outcome of the genetic cross easily).

♀ \ ♂	R	R
r	Rr	Rr
r	Rr	Rr

   F1 plants are all phenotypically round

   ➥ F2 plant is the product of self - cross between F1 : = F1 x F1

   F2 = Rr x Rr

   ➥ F2 gametes are R, r and R, r

   ➥ use punnet square to get F2

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♀ \ ♂	R	r
R	RR	Rr
r	Rr	rr

The F2 phenotypic ratio is **3 round : 1 wrinkled**, which means 3 : 1

The F2 genotypic ratio is **RR : 2Rr : rr**, which means 1 : 2 : 1

Mendel's conclusion :

☑ Every trait is controlled by two heritable **factors'** - these are what we now call genes. The heritable factors (genes) may take different forms (alleles).

☑ If the two alleles in an individual are different, one is dominant an another is recessive. Dominant traits mask the appearance of recessive traits.

☑ The only physical link between the generations is the **gametes** or sex cells. These must pass the genes from one generation to the next.

☑ Alleles are separated during the gametes (sex cells) are formed. This is Mendel's ' law of segregation'. He also stated that the gametes (sex cells) fuse randomly at fertilization.

Illustrative Examples

Use the diagram of a punnet square to answer the following question (32 and 33)

♀ \ ♂	?	?
?	Rr	Rr
?	rr	rr

37. What are the genotypes of the parents represented in this cross ?

A. Rr and Rr

B. RR and rr

C. Rr and rr

D. rr and rr

Explanation :- the only parental genotypes that produces the offspring listed in the punnet square are Rr and rr (just refer te above example of monohybrid cross).

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Answer : C

38. Alternative version of a gene is known as _____ ?

A. Sister chromatids C. DNA

B. homologous chromosomes D. Alleles

Explanation :- see the terminologies established by G. Mendel.

Answer : D

39. What genotypic ratio is expected in the offspring in this cross ?

♀ \ ♂	R	r
r	Rr	rr
r	Rr	rr

A. 1 : 1 B. 1 : 2 C. 1 : 3 D. 1 : 4

Explanation :- the genotypic ratio of the offspring is 2Rr : 2rr, which is equals to 1 : 1

Answer : A

40. Assume that you mated two individuals heterozygous for each of two traits and obtained 80 offspring. How many of them would be expected to look like their parents phenotypically ?

A. 25 B. 45 C. 60 D. 80

Explanation :- assume that the two heterozygous parental individuals are Tt and Tt

	T	t
T	TT	Tt
t	Tt	tt

The total number of individuals, TT + 2 Tt + tt = 80 ; among the offspring 3/4 are phenotypically look like their parents. Therefore, 3/4 × 80 = 60.

Note : The phenotype of TT and Tt are similar.

Answer : C

41. Assume that you started a genetic cross between pure tall pea plant and

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short pea plant and end up by producing 100 plants in the F2 generation. What is the chance (% of getting heterozygous tall plants in the F2 generation ?

A. 25 B. 100 C. 50 D. 75

Explanation :- the parental (P1) plant are pure tall (TT) and short (tt) ; the cross between the two P1 plants produces heterozygous (Tt) in F1 ; again the self - cross between F1 individuals (Tt x Tt) produces F_2 plants that are (TT : 2Tt : tt).

There fore, the total number of heterozygous plant is 1/2 (Tt) x 100 = 50.

Answer : C

42. What low states that the inheritance of one trait has no effect on he inheritance of another ?

A. The law generation

B. The law of universal inheritance

C. the law of independent assortment

D. the law of dominant

Explanation :- G. Mendel formulated two laws : the law of independent assortment and the law of segregation. According to the law of independent assortment, each pair of alleles sorts itself independently from other pairs of allele.

Answer : C

4.5.2 Dihybrid Cross

Dihybrid cross is a genetic cross that involves inheritance of two characteristics at the same time. Mendel followed essentially the same procedure as with his monohybrid experiments.

Example

The cross between plants that were homozygous for round, yellow seeds with plants homozygous for wrinkled, green seeds, then what will be the phenotypic ratio of F_2 ?

Explanation : - it is a dihybrid cross, because it involves two traits :

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The seed color and the seed shape.

Steps :

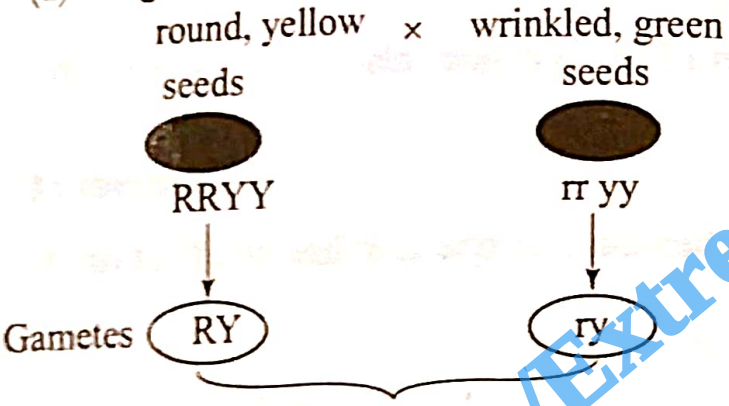
(i) The phenotype of P1 is **round, yellow seeds x wrinkled, green seeds**

(ii) The Genotype is **RRYY x rryy**

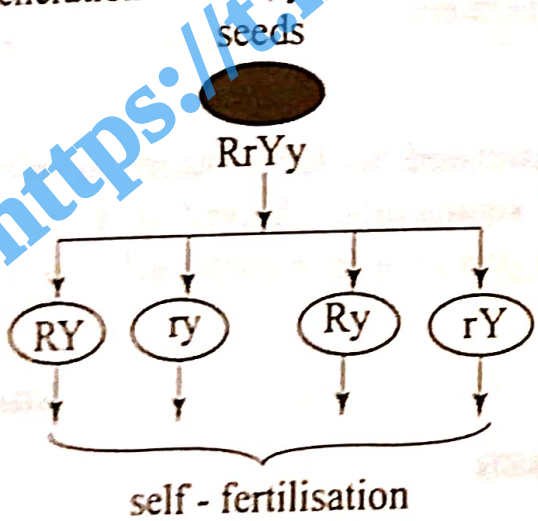
(iii) The Gamete formed are **RY**, and **ry**

(iv) The F1 phenotype are all are **Rr Yy**

(a) P generation



(b) F_1 generation



(c) F_2 generation

	RY	ry	Ry	rY
RY	RRYY	RrYy	RRYy	RrYY
ry	RRYy	rryy	RrYY	rrYy
Ry	RRYy	Rryy	RRyy	RrYy
rY	RrYY	rrYy	RrYy	rrYY

F_2 Phenotype ratio
9 round, yellow: 3 round green
3 wrinkled, yellow: 1 wrinkled, green
9: 3: 3: 1

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(v) The F2 generation is the product of the cross between

$F_1 \times F_1 = F_2$ = **RrYy x Rr Yy**

(vi) Gametes : there are four possible gametes

RY, Ry, rY, ry x RY, Ry, rY, ry

- When these gametes are fertilized, 16 progenies / offspring are formed in the F2 generation. (Look the above Fig.)
- The four phenotypes that appeared in the F2 are : round and yellow, round and green, wrinkled and yellow, are wrinkled and green. The F2 phenotypic ratio is 9 : 3 : 3 : 1

43. From the above problem, the percentage of Rr Yy in the F2 generation is ?

A. 18.75 B. 6.25 C. 25 D. 37.5

Explanation : - there are 16 progenies in the F2 generation ; among them, there are 3 individuals which has the genotype Rr Yy ; which means the percentage is calculated as 3 / 16 x 100 = 18.75.

Answer : A

44. What characteristic is described in the statement "the dog coat is brown" ?

A. Pedigree C. phenotype

B. genotype D. dominance

Explanation :- phenotype is the way the organism appears (its physical) and functions because of its genes. The phenotype is what is built from that genotype. For example, if you call a trait of an organism as 'black', or 'tall' it is phenotypic expression.

Answer : C

4.5.3 Test Cross

A test cross involves crossing an individual of unknown dominant genotype with a homozygous recessive individual. The offspring will exhibit certain phenotypes that will allow you to determine if your unknown is either homozygous dominant or heterozygous.

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For example :

The cross between unknown genotype of purple (P -) pea and white (pp) Pea produces all purple, and then what will be the genotype of the parents?

A. pp B. PP C. Pp D. none

Explanation :- since purple is a dominant trait of pea, the possible genotype is either **PP or Pp ;** where as white flower is pp. let us make a genetic cross and compare the result. The cross between **Pp x pp** & **PP x pp**

PP	p	p
P	Pp	Pp
P	Pp	Pp

Pp	p	p
P	Pp	Pp
p	pp	pp

Therefore, the correct parental genotype is **pp.**

Answer: B

4.6 Sex Determination

In human sex is determined by the **X and Y** chromosomes. Males have one X chromosome and one Y chromosome, where as females have two X chromosomes. Because males have two different sex chromosomes, they are called the **heterogametic sex**, whereas females are the homogametic sex. The ratio of being males or females is always 50 % (1 : 1).

There are several other systems of sex - determination in animals. For example :

- In birds it is females who are the heterogametic sex and males the homogametic sex. They have W and Z sex chromosomes. **Females are ZW and males ZZ.**
- In some reptiles, such as **alligators and crocodile,** sex is determined by the temperature at which the egg is incubated.
- Some snails start out male, and then become female !
- In tropical clown fish, the **dominant individual** in a group becomes female while the other ones are male.
- Some species have no sex - determination system - they are hermaph-

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rodite (have both male and female sex organs). Hermaphrodites include the common earthworm and some species of snails.

- In some insects such as male bee, sex is determined by parthenogeneis (haplodiploidy) ; the male bee is hatched from unfertilized haploid egg, then chromosomes become doubled (diploid).

4.7 Non - Mendelian Inheritance

Sometimes, the inheritance of a trait behaves differently from what you would expect based on Mendel's laws.

Does that mean Mendel got it wrong ? Not really. The laws that Mendel developed based on his observations are still believed to be correct. It's just that Mendel didn't see everything there was to see in his experiments.

Some of the examples are : incomplete dominance, codominance, multiple allelism, Rh factor inheritanace, environmental effect on phenotype and sex - linked inheritance.

I. Incomplete Dominance

The effects of two different alleles combine to create a phenotype that is some where in between the dominant phenotype and the recessive phenotype.

- Alleles are neither dominant nor recessive ; one allele is not "stronger" than the other.
- Phenotypes are blended (mixed) together.

Example : crossing a red flower (RR) with a white flower (WW) creates pink flowers (RW).

II. Co - dominance

The effects of two different alleles both contribute equally to the phenotype.

- Both of the phenotypes are visibly seen
- Phenotypes do not blend, they are both seen.

Example 1 : In a breed of chickens,the allele for black feathers is co - dominant with the allele for white feather. Blank (BB) father and White (WW) mother produced black and white (speckled) baby chicks.

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Example 2 : The ABO human blood type are an excellent example of co - dominance ; in **AB** blood type, allele A and B are equally dominant, express themselves equally.

III. Multiple allelism

Are there always two alleles of a gene ? The short answer is no.

- *Some genes have more than two alleles, and we call this situation* ***multiple allele*** *inheritance.*
- *An example of multiple allele inheritance is the ABO blood groups.*
- *Individuals can have only two alleles for each type of blood ; but at population level the total number of genes are three (A, B and O).*
- *There are three alleles involved in the inheritance of these blood groups (A, B and O)*
- *Allele A and B are dominant, but allele O is recessive*
- *The possible genotype and phenotype of the blood group are :*

Genotype	Blood group
$I^A I^A$, $I^A I^O$	A
$I^B I^B$, $I^B I^O$	B
$I^A I^B$	AB
$I^O I^O$	O

- *There are four major human blood groups :* **A, B, AB, and O.**

What determine the blood type of an individuals ?

- *Each blood group is classified by the presence or absence of a protein marker known as* ***antigen*** *on the RBC.*
- *There are two types of* ***antigen*** *on the surface of red blood cells. These are antigen A and B.*

What are the relationship between antigen and antibodies in the human blood types ?

Antigen is any foreign protein substance that trigger / stimulate an immune response

Antibody is a protein that is produced by white blood cells, and binds on antigen to fight.

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People with blood type AB have antigen A and B

- So that can receive any type of blood
- Will not make antibodies against any red blood cells
- Thus ; people with type AB blood are called **universal recipient.**

People with blood type A have antigen A

- They can receive blood type A or type O blood
- If blood type B, with B antigen is introduced into the body, the immune system of type A people will make antibodies against the B antigen.

People with blood type O, do not have any A and B antigens on their surface

- So they can donate blood to people of any blood type
- Thus, people with blood **type O** are called **universal donors.**

People with blood type B have antigen B

- They can receive blood type B or blood type O.
- If a blood with **A** antigen is introduced into the blood type **B** people, it will make antibodies against the A antigen.

People with blood type O can receive only from blood type O.

- If any blood with B or A antigen is introduced into the body, the immune system of type O people will make antibodies against these antigens

Blood type	antigen	antibody
A	A	Anti - B
B	B	Anti - A
O	Neither	Anti - A and anti - B
AB	Both A and B	Neither

For a blood transfusion to be successful, the donor and recipient blood must be compatible. If a patient receives a wrong type of blood, the patient's antibodies will react and causing them to agglutinate. Agglutinated red blood cells can clog blood vessels, blocking circulation.

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IV. Rh factor

Another factor that affects the compatibility of human blood is the Rhesus factor, a protein marker on RBC.

- People who carry the Rh protein on RBC are called **Rh - positive**, while people without the protein are called **Rh - negative.**
- Rh - person can donate blood to Rh - and Rh+ person, but receive only from Rh-.
- Rh+ person can donate blood to only to Rh+, but receive from Rh - and Rh+.

Note :

- People who are Rh^+ have a genotype of either (Rh^+ Rh^+) or (Rh^+ Rh^-)
- People who are Rh^- have a genotype of Rh^- Rh^-.
- Therefore, a parent who is Rh^+ can pass either **Rh+** or **Rh** - allele to his / her children.
- However, a parent who is Rh^- can pass only Rh^- allele to his / her children
- Rh^+ allele is dominant over Rh^-

The anti - Rh antibody is not always present in the blood. It is manufactured after an exposure to the Rh protein maker.

Therefore, the first transfusion of Rh - positive donor blood into an Rh - negative recipient is usually safe, since the development of anti - Rh antibodies will take time, 2 - 4 months after the transfusion.

For example : If Rh^- mother and Rh^+ father conceive Rh^+ child, the fetus RBC may pass through placenta to the mother's circulatory system. As a result, the mother produces anti - Rh antibodies to fight and destroy the child's RBCs and put all subsequent pregnancies at risk.

- The solution is to give the mother an **anti D vaccine** / injection containing **anti - Rh antibodies** in her first pregnancy or just after 72 hr of giving a birth.
- The anti - Rh antibodies /vaccinedestroys any of the child's red blood cells that have crossed the placenta before the mother's immune system

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begins producing antibodies

Illustrative Example

45. The cross between red snap dragon flower and white snap dragon flower produces pink flower. This types of dominancy is called ___ ?

A. complete

B. incomplete

C. co - dominance

D. over dominance

Explanation :- Incomplete dominance

The effects of two different alleles combine to create a new phenotype that is some where in between the dominant phenotype and the recessive phenotype. The new phenotype here is **pink.**

Answer : B

46. Explain how Rh^- mother carrying an Rh^+ baby could encounter no complications during her first pregnancy, but face blood compatibility problems during a second pregnancy with an Rh+ baby?

Explanation :- if she is in her first pregnancy, the development of anti - Rh antibodies against her child will take - time after the transfusion. But later, the mother will have the anti - Rh antibodies to fight against her child's antigen.

47. Which one of the following is not true about sex determination in animal ?

A. male bird is ZW

B. human female is homogametic (XX)

C. male honey bee is haplodiploidy

D. some snail are hermophrodite

Explanation :- different animals have different sex determination mechanism. For example, male bird ZZ, female ZW ; human male XY, female XX ; male honey bee is haplodiploidy (partinogenesis).

Answer : A

48. Which one of the following is not true about the blood of Rh^- person ?

A. Can donate blood to Rh positive

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B. can receive blood from Rh positive

C. can donate blood to Rh negative

D. can receive blood from Rh negative

Explanation :-

- Rh^- person can donate blood to Rh^- and Rh^+ person, but receive only from Rh^-.
- Rh^+ person can donate blood to only to Rh^+, but receive from Rh^- and Rh^+.

Answer : C

49. The blood type of Rh positive person can have a genotype of

A. $Rh^- Rh^-$

B. $Rh^+ Rh^-$

C. $Rh^+ Rh^+$

D. B and C

Explanation :-

- People who are **Rh^+** have a genotype of either **(Rh^+ Rh^+)** or **(Rh^+ Rh^-)**
- People who are **Rh^-** have a genotype of **$Rh^- Rh^-$.**

Answer: D

50. Explain how A child born from Rh positive mother and father, can be an Rh negative ?

Explanation :-

- Look at the above explanation, people who are **Rh^+** can be a genotype of **Rh^+ Rh^-**
- So that, the child may inherit Rh negative allele from both parents and become $Rh^- Rh^-$

51. Which of the following blood type of a person is a universal donor, to all blood type individuals ?

A. O^- B. AB^+ C. AB^- D. O^+

Explanation :- **O^-** blood type has no **antigen** (A and B) and **Rh factor.** Therefore, **O^-** blood type can be donated to all blood types.

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Answer: A

52. The cross between red bull (RR) and white cow (rr) produces roan calf (Rr). If you cross two roan cattle, what would be the phenotypic ratio ?

A. 1 : 1 B. 1 : 2 : 1 C. 3 : 1 D. 2 : 1

Explanation :- it is an example of co - dominance ; thus the F2 phenotypic ratio will be 1 red : 2 roan : 1 white.

Answer : B

♀ \ ♂	R	r
R	RR	Rr
r	Rr	rr

53. How many genes / alleles for blood type does each person carry ?

A. 1 B. 2 C. 3 D. 4

Explanation :-

Multiple allelism is a condition for some gene to have more than two allele. The best example is **ABO** blood type. At population level there are three genes (A, B and O) for blood type ; but, each person has only two genes.

Answer : B

54. If a man has type AB blood and he has a child with a women who has type O blood, what are all the possible blood genotypes of the child ?

A. AO B. BO C. AA D. A and B

Explanation :-

The blood type of a father is AB ; and the mother is OO.

	O	O
A	AO	AO
B	BO	BO

The possible blood genotype of the child in the punnet square is either AO or BO.

Answer : D

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55. There are 3 genes for blood type : A, B and O. Which gene (s) are dominant and mask the recessive gene ?

A. B B. A C. O D. B and A

Explanation :- the dominant genotype of the blood type of humans are A and B ; while the O genotype is recessive

Answer: D

56. The possible genotype of blood type B is ?

A. OB B. AB C. BO D. OO

Explanation :- the possible blood type of B is either BB or BO.

Answer : C

57. What is the genotype of a person with blood type A, who has a father with blood type O ?

A. AA B. AO C. BO D. OO

Explanation :- Blood type A means (AO or AA) genotype. The only chance this person can have is AO blood type ; because half of the allele is inherited from the father (OO).

Answer : B

58. The possible blood type of a child born from a woman of blood type O and father of blood type AB ?

A. OO B. AO C. BO D. B and C

Explanation :- The cross between OO blood type and AB will be either AO or BO.

Answer : D

	O	O
A	AO	AO
B	BO	BO

The possible blood genotype of the child is AO or BO

59. A woman of blood type B and husband of blood type O have children with blood type O. The most probable genotype of the woman is ?

A. BB B. OO C. BO D. AB

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Explanation :- the husband blood type is OO and the woman blood type is B, which is either (BB or BO). However they have a child with blood type OO. The child inherited half of the blood gene from mother and half from father. Therefore, the only genotype of the mother that gives the child's OO blood type is BO.

Answer : C

	O	O
B	BO	BO
O	OO	OO

V. Sex - linked inheritance in humans

The inheritance of a traits / genes that are linked with X and Y chromosomes in human are called **sex linked** inheritance. These traits are inherited differently in males and females. The main characteristics of sex linked traits are :

The disease can skip generations.

In other words, parents who don't have the disease can produce children who do.

☑ The disease is much more common in males than in females.

- Because males have one X and Y, whereas females have two X chromosomes

☑ For a boy to be born with sex - linked trait, only his mom needs to give him the defective allele.

☑ For a girl to be born with a sex - linked trait, both parents need to give a defective allele

☑ The disease is never passed from father to son.

- Sons get their Y chromosomes from their fathers, so the disease isn't carried on the Y chromosome.

☑ Examples of sex linked inheritance in human are : **hemophilia, and red - green color blindness**

☑ hemophilia, and red - green color blindness are inherited due to a recessive alleles.

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The possible genotypes and phenotypes are :

- X^BY - normal male
- X^bY - affected male
- X^BX^B - normal female
- X^BX^b - carrier female (phenotypically normal)
- X^bX^b - affected female

Illustrative Example

60. What is the expected genotypic ratio of hemophiliac children born from normal (carrier) mother and normal father ?

A. 25 % B. 50 % C. 100 % D. 75 %

Explanation :- hemophilia is X - linked trait, thus the genotype of the mother is X^BX^b and the father is X^BY

	X^B	Y
X^B	X^BX^B	X^BY
X^b	X^BX^b	X^bY

From the punnet square the probability of getting hemophiliac children is 25 % .

Answer : A

61. From the above question, the chance of getting carrier daughter is

A. 25 % B. 50 % C. 100 % D. 75 %

Explanation :- 50% of the daughter is normal and the other 50 % is carrier.

Answer : B

4.8 Environmental Effects on Phenotype

Although the **genetic makeup / genotype** of an organism plays a major role in determining the phenotype, there are also several **environmental factors,** such as temperature, oxygen availability, and presence of mutagens, which can alter an organism's phenotype.

- **A phenocopy** is an environmentally induced phenotype of an individual

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For example, the effect of temperature variation observed in Himalayan rabbits ; in moderate temperature, the phenotype of rabbit coat is white where as at cold temperature, the coat color is black.

What is Pedigree

A pedigree is a chart that shows the genetic relationships between individuals in a family. The pedigree chart is constructed using a standard procedure with specific symbols and definition.

What are the symbols used when drawing human pedigrees ?

- Males are shown as squares, females as circles . ⬭ , ▭
- A mating is shown by a line drawn between a female and a male .
- The children of a mating are drawn below the parents in birth order
- The symbol for a person who has the trait that is being studied is shaded. ⬭ ▭

The row shows the generation

- The oldest generation is on the top row, while the youngest raw is the bottom

Analyzing pedigree can reveal the following :

- Allele ; dominant and recessive
- Type of chromosome : autosomal or sex chromosome
- Genotype of the family
- The probability of phenotype of future generation

Note : Using a pedigree chart and Mendelian genetics, scientits can determine whether the allele responsible for a given condition is **dominant, recessive, autosomal, or sex - linked.**

List some examples of monogenetic/ single disorders in human ?

Monogenetic disorders are disorders caused by the defect on one particular gene / single gene. These are :

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1. Autosomal Recessive Inheritance

Autosomal recessive disorders are expressed in **homozygous** condition. It is not specific to the sex of the person. One example of such a disorder is **albinism disease.**

- Albinism is a genetic condition in which the eyes, skin, and hair have no pigment
- Individual who is an albino, lack one of the enzymes required to produce **melanin.**
- Other example is cystic fibrosis

2. Autosomal Dominant Inheritance

Autosomal dominant disorders are expressed in heterozygous condition. It can happen in both sexes, male and female equally.

- One example is Huntington disease ; it is a lethal disorder in which the brain progressively deteriorates over a period of time. Early symptoms include irritability and mild memory loss, followed by involuntary arm and leg movements.

3. X - Linked Recessive Inheritance

X - Linked inheritance is the genetic conditions and disorder that can be carried on the sex chromosomes, especially the **X chromosome.**

- For example, **hemophiliais** an X - linked inheritance pattern that is characterized by the blood's inability to clot.
- Red - green color blindness, inability to differenciate red and green color.

Illustrative Examples

62. Albinism is an autosomal recessive condition. Suppose that George with albinism marries Mary who is not affected. They have a daughter, Frances, who has albinism. Frances marries Jack, whose mother and sisters both have albinism. Frances and Jack have a daughter, Anne, who is not affected.

a. Construct a pedigree of the family

b. Identify the genotype of each person in the family

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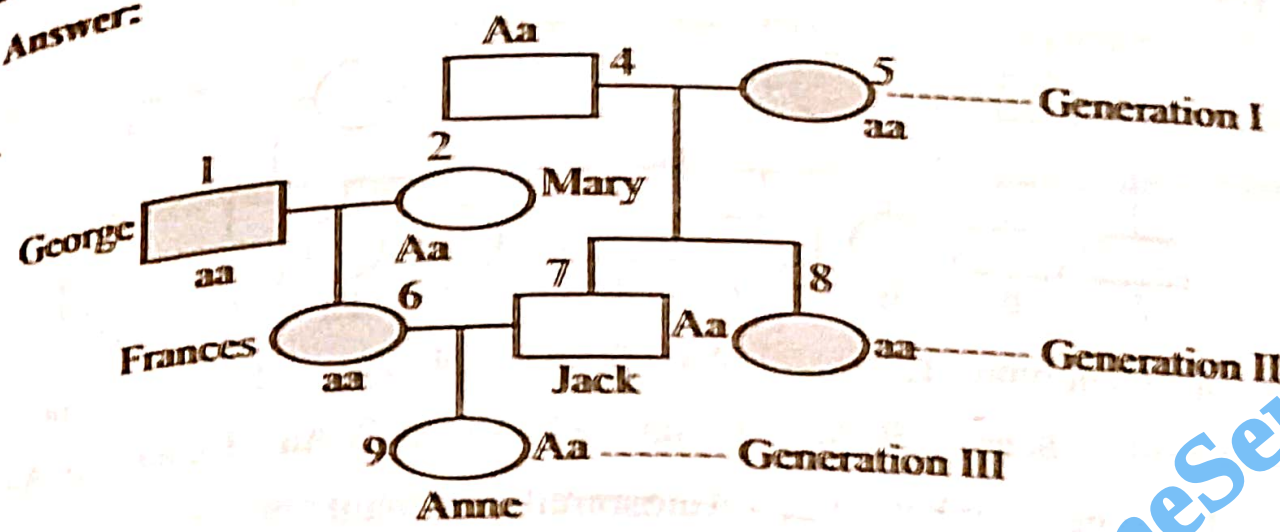
c. How many generations are seen on the pedigree

Answer:

a.



b. 1 = aa, 2 = Aa, 4 = Aa, 5 = aa
6 = aa, 7 = Aa, 8 = aa, 9 = Aa

c. 3 generation

63. In the pedigree (I - III) below, determine whether the inheritance is sex linked dominant, **sex linked recessive, autosomal dominant or autosomal recessive ?**

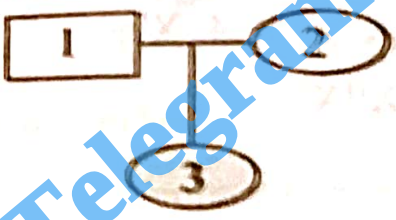
a. genotype ?

b. Type of inheritance ?

c. example of disease ?

Answer:

(i)



a. Genotype, 1 = Aa, 2 = Aa, 3 = aa

b. Type inheritance = Autosomal recessive

c. Example

➥ Albinism, cystic fibriosis

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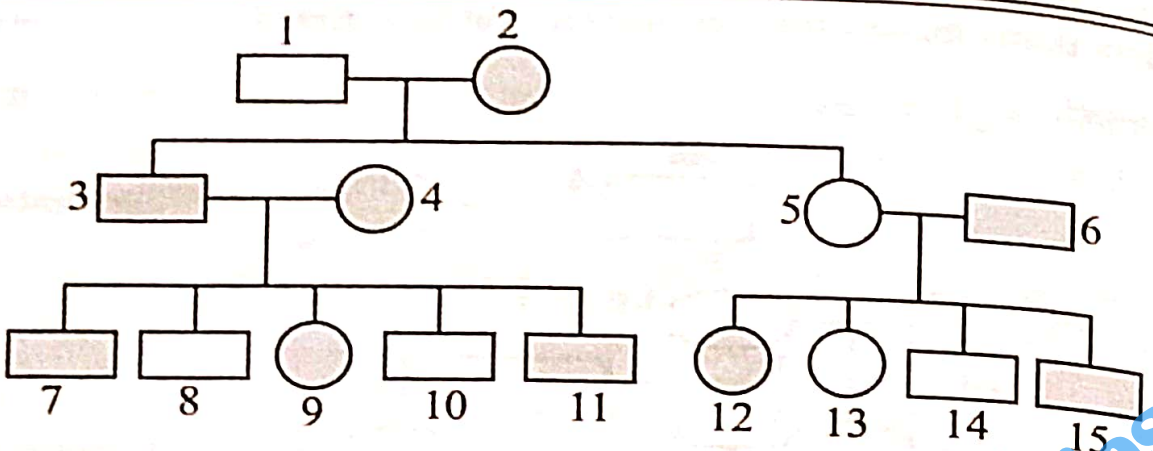
(ii)



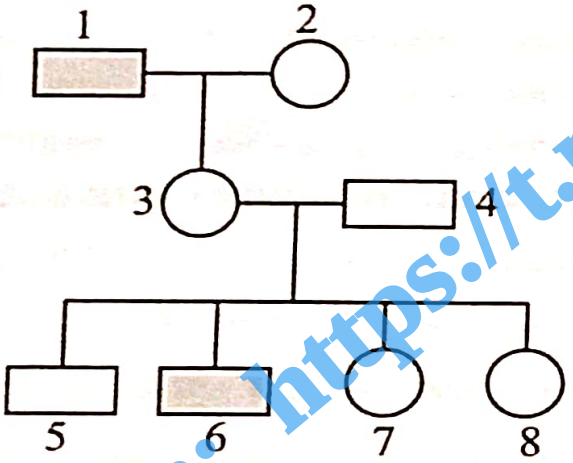
a. Genotype 1, aa 2, Aa 3, Aa 4, Aa 5, aa 6, Aa 7, Aa 8, aa 9, Aa 10, aa 11, Aa 12, Aa 13, aa 15, Aa

b. Types of inheritance - Autosomal dominant

c. Example

➥ Huntington disease

(iii)



a. Genotype, 1, X^HY 2, X^HX^h 3, X^HY 4, X^HY 5, X^HY 6, X^hY 7, X^HX^H 8, X^HX^h

b. Type of inheritance

Sex - linked (x - linked)

c. Example

➥ Hemophilia

➥ red - green color

64. The phenotype of an individual is always determined by its own genotype ?

A. True B. False C. Both D. none

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Explanation : - Although the **genetic makeup/ genotype** of an organism plays a major role in determining the phenotype, there are also several **environmental factors**, such as temperature, oxygen availability, and presence of mutagens, which can alter an organism's phenotype.

Answer : B

65. Which of the following genetic disorder is due to autosomal dominant inheritance ?

A. Albinism
B. Huntington disease
C. hemophilia
D. color blindness

Explanation : - Examples of autosomal dominant inheritance include **Huntington disease** ; it is a lethal disorder in which the brain progressively deteriorates over a period of time.

66. Indicate how a female could inherit complete red - green color blindness

Explanation : - Color blindness is an X - linked recessive inheritance. A female to be color blind, she must have two recessive alleles on her X chromosomes (X^hX^h). One allele is inherited from mother and the other is from the father.

Chromosomal Disorder

Chromosomal disorder is a disorder due to significant changes in the physical structure of chromosomes ; it can occur either **spontaneously** or when cells are **irradiated** or exposed to certain **chemicals** and **error during cell division**. Essentially, there are four means by which these changes can happen : deletion, duplication, inversion, and translocation

1. Duplication

Duplication occurs when a gene sequence is repeated one or more times within one or several chromosomes.

2. Inversion

When an inversion occurs, a certain gene segment becomes free from its chromosome momentarily before being reinserted in the reverse order. This completely changes the position and order of the chromosome's genes, and can alter gene activity.

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3. Translocation

In translocation, part of one chromosome changes places with another part of the same chromosome or with part of another, non - homologous chromosome.

4. Deletion

In a deletion, a portion of the chromosome is lost. There are several factors that can trigger a deletion. Viruses, radiation, and chemicals can cause pieces of a chromosome to be broken off.

5. **Aneuploidy :** occurrence of one or more extra or missing chromosomes in a cell or organism (2n- 1, 2n+1, 2n+2). For example, Down syndrome is a genetic disorder occur due to trisomy of chromosome number 21.

4.9 Genetic Counseling and Testing

Genetic counseling is the process of checking a family with regard to the medical history and medical records, ordering genetic tests and recording, help parents understand and reach decisions about what to do next.

- For example, a couple who have a family history of a genetic disorder may want to consult a genetic counselor before the conception of their next child.
- A genetic counselor is a medical professional who gathers detailed information through interviews, blood tests, and discussions with geneticists.
- The counselor then constructs family pedigrees, and then predicts the probability of the next child having a genetic disorder.
- And communicates the level of risk to the parents so they can make their own decision about where to conceive another child.
- **Genetic testing** is an important process used to identify/examine the type of disorder from individuals /couples.

To determine whether the developing fetus is affected or not,

- Pregnant woman may choose to have a test known as an **amniocentesis** (sample is taken from amniotic fluid) or

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- A woman interested in obtaining results sooner may opt for a procedure called **chorionic villi sampling** (sample is taken from chorion/ placenta)

Examples of genetic disorder are : Down syndrome, cancer, diabetes, turner syndrome, hemophilia, etc.

Illustrative Example

67. List some of the major causes of structural changes in chromosomes ?

Explanation :- occur when cells are irradiated or exposed to certain chemicals and error during cell division.

68. A kind of chromosomal disorder that involves an exchange of chromosomal segment in between non homologous chromosomes is ___ ?

A. deletion
B. translocation
C. inversion
D. aneuploidy

Explanation : - In translocation, part of one chromosome changes places with another part of the same chromosome or with part of another, non - homologous chromosome.

Answer: B

69. The process of checking a family with regard to the medical history on genetic disorder ; ordering genetic tests and recording to help parents understand and reach decisions about what to do next is known as __ ?

Explanation : - Genetic counseling

70. A kind of diagnosis / test used to determine whether the developing fetus is affected by genetic disorder or not in pregnant woman is __ ?

Explanation : - the type of test is known as **amniocentesis or chronic villi sampling.**

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4.10 Gene Theraphy

Gene therapy is a medical procedure in which normal or modified genes are transferred into the defective cells of an individual. In theory, the transferred genes will allow the recipient's cells to begin functioning normally by giving them the instructions for producing the missing molecules, thus reversing the symptoms of the genetic disorder.

There are two types of gene therapy, these are : somatic gene therapy and germ line gene therapy.

1. **Somatic gene therapy :** refers to transferring a section of DNA to somatic / body cells ; the effect will not be passed on to the parents children
2. **Germ line gene therapy :** refers to transferring a section of DNA to sex cells ; the effect will be passed on to the parents' children and subsequent generation.

What are the different vectors used to carry curative genes into human cell ?

- **Plasmid DNA**
- **Viral vector**
- **Bacterial vector**
- **Human gene editing technology** is used to disrupt or repair mutated / harmful genes
- **Patient - derived cellular gene therapy products use** cells removed from the patient, genetically modified then returned them to the patient.

What are the risk factors / challenges related to the gene therapy ?

- Avoiding the immune system ; when new gene or antigen is introduced in to the body.
- Delivering the new gene into the right place and making it functional (making gene on).
- Making sure the new gene doesn't interfere the function of other normal genes.

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4.11 Bioinformatics Introduction

Bioinformatics is a science that blends computing, biology, and information technology to organize and analyze the large amounts of information that are being generated by biologist all around the world. Bioinformatics requires good communication skills and experience with computers, software applications, and the internet.

The field of bioinformatics incorporates three main areas : **genomics, proteomic,** and **system biology**

- **Genomics** is the study of entire genome of organisms. By studying all of the DNA is regulated in cells.
- **Proteomics** is the study of the entire body of proteins in a cell and how they interact with each other. The types of proteins found in different cells are compared in order to look for patterns common to certain cell types
- **System biology** is the study of interactions and behavior of the components of biological entities, including molecules, cells, organs and organisms. **For example,** illustration of brain functioning by using computer images, path way of enzymes and its metabolites.

4.12 Breeding

Breeding is an application of genetic principles, in animal husbandry, agriculture, and horticulture to improve desirable qualities.

This can be done through **selective breeding** or **natural selection.**

Selective breeding is a kind of breeding where organisms or plants are chosen to breed with a specific outcome in mind.

In animal it can be of two types : **cross - breeding** and **inbreeding**

1. **Cross - breeding** is breeding between unrelated breed of a species. This breeding method is used to increase overall productivity. Commercial cattle farmers may cross - breed animals for two reasons :
   - To take advantage of **hybrid vigour** (bred that have extra qualities).
   - To take advantage to improve market suitability

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Example, the breeding between cattle of Boren and Holstein Friesian to get hybrid vigour

2. **Inbreeding** is breeding between closely related / similar breed of a species.
   - ➥ Aimed to produce **pure bred or inbred,** those have similar appearance, behavior, etc.
   - ➥ Sometimes, it has negative consequences, undesired recessive trait in the family is more likely

Plant Breeding

Plant breeding is the science of change the trait of plants in order to produce desire characteristics. Its aim is to produce crop variety with different qualities, such as :

- ➥ Superior quality, to improve quality nutrition for human and animal
- ➥ The production plants that are stress tolerance, such as drought, temperature, salinity, pest, and flooding resistant.
- ➥ Improve crop yields, taste, biological molecules (protein, sugar, lipid, vitamin, fibers).
- ➥ Improve the ease of processing (harvesting, milling, baking, malting, blending, etc)

Plant breeding could be as simple as propagation of selected desired qualities, and as complex as using molecular techniques. Some of the examples of selections are : mass selection, pure line selection, and bulk selection.

1. Mass selection methods :

It is the simplest form of plant breeding, practiced by early humans and inexperience farmers.

A large number of phenotypically similar plants with superior qualities are selected, cross pollinated and harvested in bulk, and the new varieties of the seeds are used to reproduce for the next generation.

2. Pure line selection :

Selection of the best trait among the pure bred / true breeding (self - pollinated individuals after a long time)

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3. Bulk method :

In bulk method, F2 and subsequent generation are harvested in bulk to rise the next generation until the genotype attains homozygousity. Natural selection increases the frequency of superior genotype.

Illustrative Examples

71. The main difference between inbreeding and cross breeding is that, cross breeding is __?

A. Produce pure bred

B. Produce hybrid vigour

C. is breeding between close relatives

D. Produce inbred

Explanation :- the comparison between cross breeding and inbreeding

cross breeding	Inbreeding
Breeding between distantly related organisms	Breeding between close relatives organisms
Produce hybrid vigour	produce pure bred / inbred

Answer: B

72. List the three types of plant breeding selection for propagation ?

Explanation :- The three types of plant breeding selections are : mass selection, pure line selection, and bulk selection.

73. Which of the following is not true about the gene therapy ?

A. Somatic gene therapy can't pass to the offspring

B. germ line gene therapy can pass to the offspring

C. it is used to transfer curative gene to the patient

D. curative genes are transferred by needle.

Explanation :- Curative genes are carried thorough plasmid DNA, viral vector, bacterial vector etc.

Answer : D

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74. The field of bioinformatics, which studies about sequencing of DNA is

A. Systemic biology

B. genomics

C. proteomic

D. none

Explanation :- Genomics is a field of bioinformatics, which studies about the entire genomes of organisms. Such as DNA sequencing discovering new protein

75. Bioinformatics require a skill of all of the following, except ?

A. Computer / information technology

B. good communication

C. software application

D. all

Explanation :- just look at the definition of bioinformatics on page 155

Answer: D

76. What is the comparison between germ line gene therapy and somatic gene therapy ?

Explanation :- there are two types of gene therapy. These are : germ line gene therapy and somatic gene therapy.

1. Somatic gene therapy : refers to transferring a section of DNA to somatic / body cells ; the effect will not be passed on to the parents children
2. Germ line gene therapy : refers to transferring a section of DNA to sex cells ; the effect will be passed on to the parents' children and subsequent generation.

Review Exercises

I. Multiple choices

1. During protein synthesis, tRNA

A. Produce a new RNA

B. Carries an amino acids into the site of protein synthesis

C. Produce protein sub unit by translating the codon on mRNA

D. Deliver instruction for protein synthesis to ribosome.

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2. Each alternative version of a gene is called a (n)
A. trait B. allele C. genotype D. phenotype

3. In Mendel's first experiment with pea plants, the phenotypic ratio of contrasting traits in the F2 generation was
A. 1 : 0 B. 1 : 1 C. 2 : 1 D. 3 : 1

4. The phenotype of an organism
A. Can't be seen
B. is the physical result of its gene
C. exactly matches its genotype
D. occurs only in pure trait

5. Mendel obtained his P_1 generation by forcing pea plant to
A. segregate C. cross - pollinate
B. self - pollinate D. assort independently

6. The passing of traits from parent to offspring is called
A. Heredity C. assortment
B. probability D. reproduction

7. What is the expected phenotypic ratio resulting from a cross of a homozygous dominant parents with a heterozygous parents ? Asume Mendelian inheritance
A. 1 : 3 : 1 B. 1 : 2 : 1 C. 2 : 1 D. 1 : 0

8. The number of gametes formed from genotype of Aa Bb Cc
A. 3 B. 4 C. 8 D. 6

9. In the cross between AA BB X aa bb, the chance of producing Aa Bb individual in the F2 is ___ ?
A. 1 / 16 B. 3 / 16 C. 4 / 16 D. 2 / 16

10. From question 9, if 1000 plants are produced in F2, then the chance of getting aa BB is ___ ?
A. 63 B. 125 C. 188 D. 25

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11. Separation of homologous chromosomes and move to opposite pole during ____ phase of meiosis

A. Metaphase I
B. anaphase I
C. anaphase II
D. metaphase II

12. Which one of the following is true about sex determination in animals?

A. Human male is homogametic (XX)
B. human female is hetrogametic (XY)
C. Male honey bee is developed from fertilized egg
D. some snails are hermaphrodite

13. If two heterozygous individuals of two traits are crossed and produces 200 offspring by Mendel. How many of them would be phenotypically look like their parents ?

A. 100 B. 200 C. 150 D. 50 A

14. If a child of blood O is born from a mother of blood B, which one of the following men whose blood types are given, would you exclude not as a possible father of the child ?

A. A blood type
B. B blood type
C. AB blood type
D. O blood type

15. In a cross between round yellow (dominant) and wrinkled green (recessive) seed plants. In F1 all are round seed but their color is some yellow and some are green. The genotype of the dominant parent was

A. RR YY
B. Rr Yy
C. Rr YY
D. RR Yy

16. Which one of the following is not true about individual of blood of Rh positive ?

A. Can donate blood to Rh positive
B. can donate blood to Rh negative
C. can receive blood from Rh positive
D. can receive blood from Rh negative

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17. Which one of the following types of dominancy is not studied by G. Mendel ?
A. complete
B. incomplete
C. codominance
D. B and C

18. Trisomy at chromosome number 21, due to chromosomal disorder (aneuploidy) causes ___ ?
A. Down syndrom
B. turner syndrom
C. hemophilia
D. klinefiltersyndrom

19. A chromosomal disorder occurs due to rearrangement of chromosomal segment from one location to another ?
A. Inversion
B. Aneuploidy
C. Translocation
D. deletion

20. Which one of the following genetic disorders is sex linked ?
A. hemophilia, color blindness
B. Huntington disease, Albinism
C. albinism, cystic fibrosis
D. Hemophilia, albinism

21. Pure tall pea is crossed with pure short, gives F1 all tall pea 'the type of dominance shown is ____
A. Codominance
B. Complete dominance
C. Over dominance
D. incomplete dominance

22. Which one of the following is not genotypic expression ?
A. BB
B. pure black
C. Heterozygous black
D. black

23. What is the best example of trait in human, which is inherited due to multiple allelism ?
A. Blood type
B. hemophilia
C. skin color
D. Huntington disease

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24. Which one of the following includes the others ?

A. Genes
B. Alleles
C. Chromosomes
D. DNA

25. What do we call the type of dominance in which a single allele of the dominant trait produces the same effect as the two dominant alleles ?

A. Co dominant
B. Incomplete dominance
C. Intermediate dominance
D. Complete dominance

26. Suppose a pure monohybrid cross produces 40 F2 plants. What is the number of plants that are expected to have the dominant and recessive in the F2 phenotypes respectively ?

A. 30 ; 10
B. B.20 ; 20
C. 25 ; 15
D. 35 ; 5

27. Which one is mendelian test cross

A. tall x short
B. Tt Rr x Ttrr
C. Aa x DD
D. purple axial x purple axial

28. Which one is not true about crossing over ?

A. is a source of variation
B. Occurs in mitosis
C. Occurs in between bivalent
D. Occurs at Prophase I

Short answer

29. Knowledge of the mechanisms of inheritance is the result of __ work on pea plants ?

30. The __ is used to determine the outcome of various genetic crosses ?

31. The __ refers to the genetic make - up of an organism ?

32. A chromosome is made up of smaller units called ___ ?

33. The transfer of curative genes into the patient to treat genetic disorder is called __ ?

34. Each human somatic cell contains __ number of chromosomes ?

35. The ABO blood group in humans is an example of ___ inheritance ?

36. The triplet code on mRNA is called ____, while in tRNA it is called _

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37. Color blindness in human is an example of ___ inheritance ?

38. A cross that involves two traits and their associated alleles is called a ___ cross ?

39. In RNA, adenine always pairs with ___ ?

40. A computer based science that uses soft ware to analyze and interpret a large biological data is ___ ?

41. Write the two process of protein synthesis ?

42. Write the three stop codons found on mRNA ?

43. The total number of genetic code that are needed to specify the 20 amino acids are ___

44. The initiation codon on mRNA that used to represent an amino acid methionine is called ___ ?

45. A chart that shows the genetic history of a family in different generation is ___

46. Which stage of meiosis cell division is similar to mitosis ?

47. Write the 3 main stages in interphase ?

48. A condition in which both alleles are expressed equally rather than a dominant allele taking a complete control over a recessive allele ?

49. The classification and blood transfusion of human blood is based on ___ and ___ ?

50. Wrong type of blood transfusion between individuals in human causes ___ ?

51. List the different /means of introducing curative genes in to the patient

52. Classify the following genetic disorder as : X - linked, autosomal recesive, autosomal dominant

a. hemophilia,
b. huntigton disease
c. color blindness
d. albinism
e. cystic fibriois

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Esay type

53. Write the paragraph that summarizes the relationship between the chromosome and gene ?

54. In tomatoes, red fruit (R) is dominant over yellow fruit (r). If a heterozygous red fruit is crossed with a yellow fruit,
    a. What is the appearance of the F1 generation ?
    b. What are the genotypes of the F2 generation if two plants from the F1 generation are crossed ?
    c. What are the phenotypes of the F2 generation ?

55. What is the expected genotypic ratio of children born from normal mother whose father was hemophiliac and her husband is normal ?

56. If a normal sighted female whose father was color blind married a color blind man, what is the probability that the couple will have colored blind son ? And color blind daughter ?

57. Discuss the main reason why a live birth is possible from Rh negative mother and Rh positive father if it is the first pregnancy ?

58. The comparison between somatic and germ line gene therapy ?

59. Explain the conditions how Rh sensitization occur in pregnant woman and the name of vaccine taken to solve the problem ?

Answer to Review Exercise

I. Multiple choice

1. B	6. A	11. B	16. B	21. B	26. A
2. B	7. D	12. D	17. D	22. D	27. A
3. D	8. C	13. C	18. A	23. A	28. B
4. B	9. B	14. C	19. A	24. C	
5. B	10. A	15. D	20. A	25. D	

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II. Short answer

29. G. Mendel
30. Punnett square
31. genotype
32. gene
33. gene therapy
34. dipliod
35. multiple allele
36. codon, anticodon
37. X - linked
38. dihybrid
39. Uracil
40. Bioinformatics
41. transcription, translation
42. UAA, UGA, UAG
43. 64
44. AUG
45. pedigree
46. meiosis II
47. G1, S, and G2
48. co - dominance
49. antigen, antibody
50. agglutination/ blood clump
51. plasmid DNA, bacterial, virus vector, human gene editing technology, patient derived cellular gene therapy product use
52. X - linked (hemophilia, color blindness) Autosomal recessive (albinism, cystic fibriosis), autosomal dominanat (huntington disease)

III. Essay type

53. Chromosomes are thread like structure found in the nucleus, which is used as a storage of genetic information of an organism. Gene is a unit of chromosome, determine specific protein.

54.
   a. all are red
   b. F2 genotypic ratio is 1 : 1
   c. F2 phenotypic ratio is 1 : 1

55. 50 % son is X^HY
    50 % son is X^hY
    50 % daughter is X^HX^H
    50 % daughter is X^HX^h

56. 50 % hemophiliac son
    50 % hemophiliac daughter

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57. The first transfusion of Rh - positive donor blood into an Rh - negative recipient is usually safe, since the development of anti - Rh antibodies will take months after the transfusion.

58.

1. **Somatic gene therapy :** refers to transferring a section of DNA to somatic / body cells ; the effect will not be passed on to the parents children.
2. **Germ line gene therapy :** refers to transferring a section of DNA to sex cells ; the effect will be passed on to the parents' children and subsequent generation.

59. Rh sensitization occur in pregnant woman occur if she is Rh negative and her husband is Rh positive. The name of vaccine / drug is anti D **vaccine** / injection containing anti Rh antibodies in her first pregnancy or just after 72 h of giving a birth.

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UNIT 5

5. The Human Body System (I)

The human body is a biological machine made of body system that work together to sustain life.

It is composed of **eleven body systems,** Thoese are integumentary system, skeletal system, muscular system, lymphatic system, respiratory system, digestive system, nervous system, endocrine system, cardiovascular system, urinary system and reproductive system.

5.1 Human Musculo Skeletal System

The human musculoskeletal system is a complex network of muscle, bones, joint and cartilage, tendon, connective tissues, ligaments. It is also known as a human **locomotor system** that gives human the ability to move.

- It is made of **muscular and skeletal system :**
- It provides form, support, stability, and movement to the body, protect internal organs, produce blood cells, store minerals and fat

Types of Muscles in Human

Muscles are one of the basic soft tissues in the body ; and have a variety of functions; it is located in different part of the body.

There are three main types of muscle. These are **skeletal, cardiac** and **smooth** muscles.

I. Skeletal muscles :

Skeletal muscles are involved in the movements of the bone of the skeleton and in athletic activities such as in arm, legs, chest, shoulder, and abdomen.

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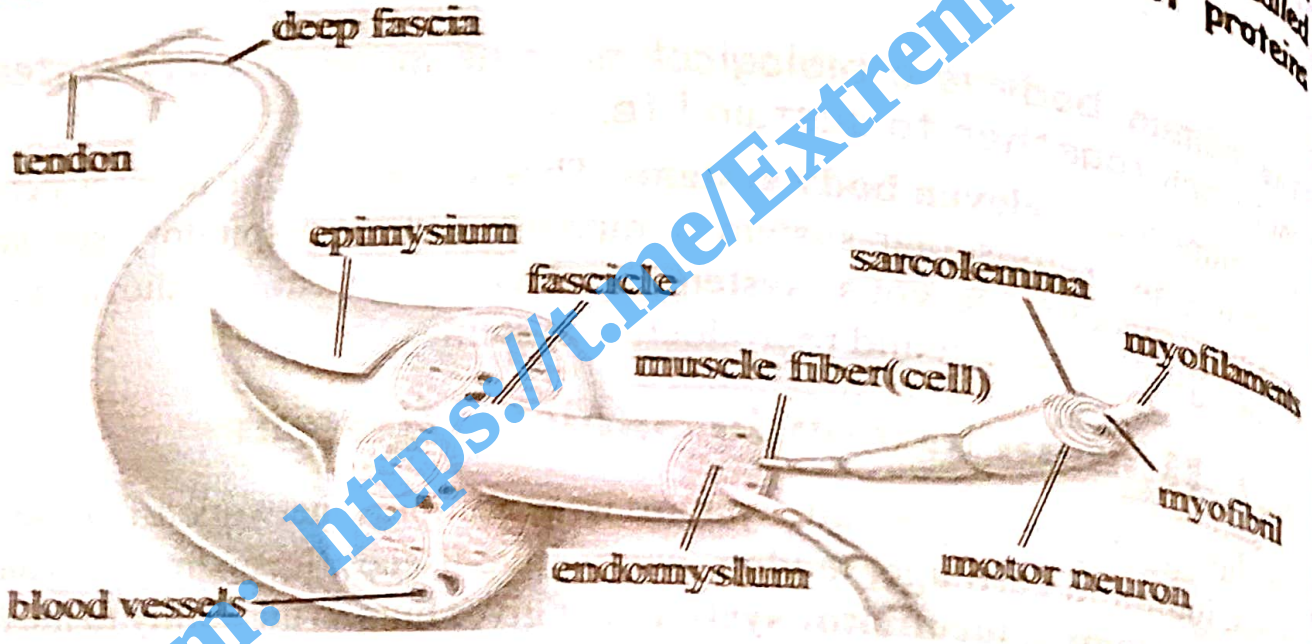
Some of the characteristics of skeletal muscles include :

- It makes up the bulk of your body (30 – 40 %)
- It is voluntary muscles, controlled by the conscious mind and innervated by a single motor axon.
- Muscle cells are striated and multinucleated

Note : Skeletal muscles are made of bundles of muscle fibers (muscle cells), which are composed of many threads like myofibrils. Myofibril contains two types of filaments : myosin (the thick filament protein) and actin (thin filaments protein). Myosin and actin filaments are organized in to repeating units called sarcomeres. The thin filament of actin is made of protein known as Troponin and tropomysin.



II. Cardiac muscle

Cardiac muscle is found exclusively in the heart, pumps blood from heart throughout the body. The main characteristic of cardiac muscle includes:

- They are not connected to any bones
- It is involuntary, can't be controlled by he conscious mind
- The cells are uninucleated, straited and branched

III. Smooth muscle

Smooth muscle is found in the hollow wall of internal organs such as intestine, arteries, esophagus, respiratory tract, stomach, bladder, uterus, reproductive system, lymph vessel, eyes, and erectile muscle of hair in the skin. Some of the characteristics of smooth muscles include :

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- It is involuntary, can't be controlled by the conscious mind
- The cells are non - striated, uninucleated and spindle shaped.

Illustrative Examples

1. What is the human musculoskeletal system ?

A. is made of muscular and skeletal system

B. it provides form, support, and movement to the body

C. network of muscle, bones, joint and other tissues

D. all

Explanation :- Musculoskeletal system :

- It is a human locomotor system that gives human the ability to move
- made of muscular and skeletal system (muscle, bone, joint, cartilage tendon and ligaments)
- provides form, support, stability, protect internal organs, produce blood cells, store minerals and at.

Answer : D

2. A type of muscle that helps the movement of the body by pulling on bones so that we walk, dance, and chew to ate ?

A. Skeletal

B. Cardiac

C. Smooth

D. All

Explanation :- Skeletal muscles : are muscle on the bone of skeleton : involved in athletic activities such as in arm, legs, stomach to move, dance and chew food

Answer : A

3. Which one of the following is **not** the characteristic of cardiac muscle ?

A. involuntary

B. striated and branched

C. uninucleated

D. located in the wall internal part of the body

Explanation :- Cardiac muscles are exclusively found in the heart, it is involuntary muscle that can't be controlled by conscious mind. The cardiac muscle cells are striated and branched and have one nucleus (uninucleated).

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4. What type of muscle in our body is used to move food along the gut and push blood along blood vessels ?

A. Skeletal
B. Smooth
C. Cardiac
D. All

Answer : D

Explanation :- Smooth muscles are found in the wall of hollow internal bod parts : (respiratory tract, blood vessel, reproductive system, etc). It helps to push food, blood, egg, etc along the gut / vessel/ tube.

5. Compare and contrast the three types of muscles ?

Answer : B

Explanation :

Types of muscle	Unique feature / descriptions
Skeletal muscles	➥ Located on leg, arm, shoulder bones, stomach, etc ➥ Voluntary muscle ➥ Cell are striated and multinucleated
Cardiac muscle	➥ Located on wall of heart ➥ Involuntary muscle ➥ Cells are uninucleated, striated and branched
Smooth muscle	➥ Located n the hollow wall of internal body parts ➥ Involuntary muscle ➥ Cells are non - striated, uninucleated and spindleshaped

5.1.1 Mechanism of action of Skeletal Muscle

The mechanism of actions of skeletal muscle is all about activation of muscle cells to contract (shorten) to move our body parts. According to the **sliding filament theory**, the mechanism of skeletal muscle contraction is explained in the following steps :

I. Muscle activation

➥ A **motor neuron** carries impulse (signal) from CNS to muscle fibre to generate an **action potential.**

➥ The action potential cause the release of **calcium ion** from **sarcoplasmic**

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reticulum.

II. **Muscle contraction**

- Calcium ion binds to **troponin** on actin, thus exposing the binding site for myosin.
- The myosin head binds to actin, forming a **cross bridge,**
- Then the **sliding of myosin over actin** filaments
- This causes **power stroke** to pull the actin filaments
- The energy is continuously supplied to myosin from hydrolysis of **ATP**
- **Sarcomere** is shortened, thus muscle is contracted

There are **four types of muscle contractions**. These are :

1. **Isomeric :** a muscular contraction in which the length of the muscle does not change
2. **Isotonic :** a muscular contraction in which the length of the muscle changes **(concentric or eccentric).**
3. **Concentric :** a muscular contraction in which the length of the muscle shortened
4. **Eccentric :** a muscular contraction in which the length of the muscle lengthened

III. **Recharging**

Further supply of **ATP** to myosin, allows a strong state of forming a cross bridge and causes a power stroke.

How does **muscle relaxation** is achieved ?

- The motor neuron stops sending an impulse (signal) to muscle fibre
- Calcium ion is returned to sarcoplasmic reticulum
- Inactivation of the actin. Thus tropomysin blocks the attachment site for myosin.
- The cross bride between myosin and actin is broken down.
- ATP binds to myosin head and detaches it from actin
- Sarcomere lengthened, thus muscle is relaxed.

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5.1.2 The Human Axial and Appendicular Skeletons

Humans are born with over 300 bones ; but remain 206 bones in adult skeleton. This is because many bones are fused during maturity. The bones that make up the skeletal system are divided in two types : axial and appendicular skeleton.

1. Axial skeleton

Axial skeleton is the part of the skeleton along the axis of the body. It contains 80 bones out of a total of 206 bones. It is composed of skull, sternum, ribs and vertebral column.

- **Skull** is the bone of the head that is composed of cranial and facial bones. Cranium is the head bones that fit together by joints (suture) and protects the brain. The facial bones provides structure for mouth cavity (jaw bone) cheeks, nose, the ear and the eye.
- **Sternum** - is the bone found on the middle of the rib cage.
- **Rib cage** (thoracic cage) is flat bones that are connected to the sternum at the middle of the chest. It consists of 12 pairs of ribs with costal cartilage and the sternum. It provides protection for the delicate organs of the body such as heart and lung.
- **Vertebral column** - It is also known as back bone, which extends from the back of skull to pelvis (hip) ; it provides support for the body and protects the spinal cord.

Parts of the axial skeleton

Parts of the axial skeleton	Number	Functions
Skull (Cranium 8 + Facial 14)	22	It holds and protects the brain
Ossicles (both ear, 3 + 3)	6	Auditory
Hyoid (neck)	1	Holds the head
Thoracic (rib cage)	24	It protects the heart and lungs
Thorax (sternum)	1	Bind and handle the abdomen
Vertebral column	26	Holds up human body
Total bone	80	

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2. Appendicular Skeleton

Appendicular skeleton is the appendage of the skeletal system, which consists of all the bone (appendages) that are connected to the axial skeleton. It contains 126 bones out of a total of 206 bones in the body. It consists of the bones of **shoulder, pelvis** (hip) and **limbs** (fore ad hind limb).

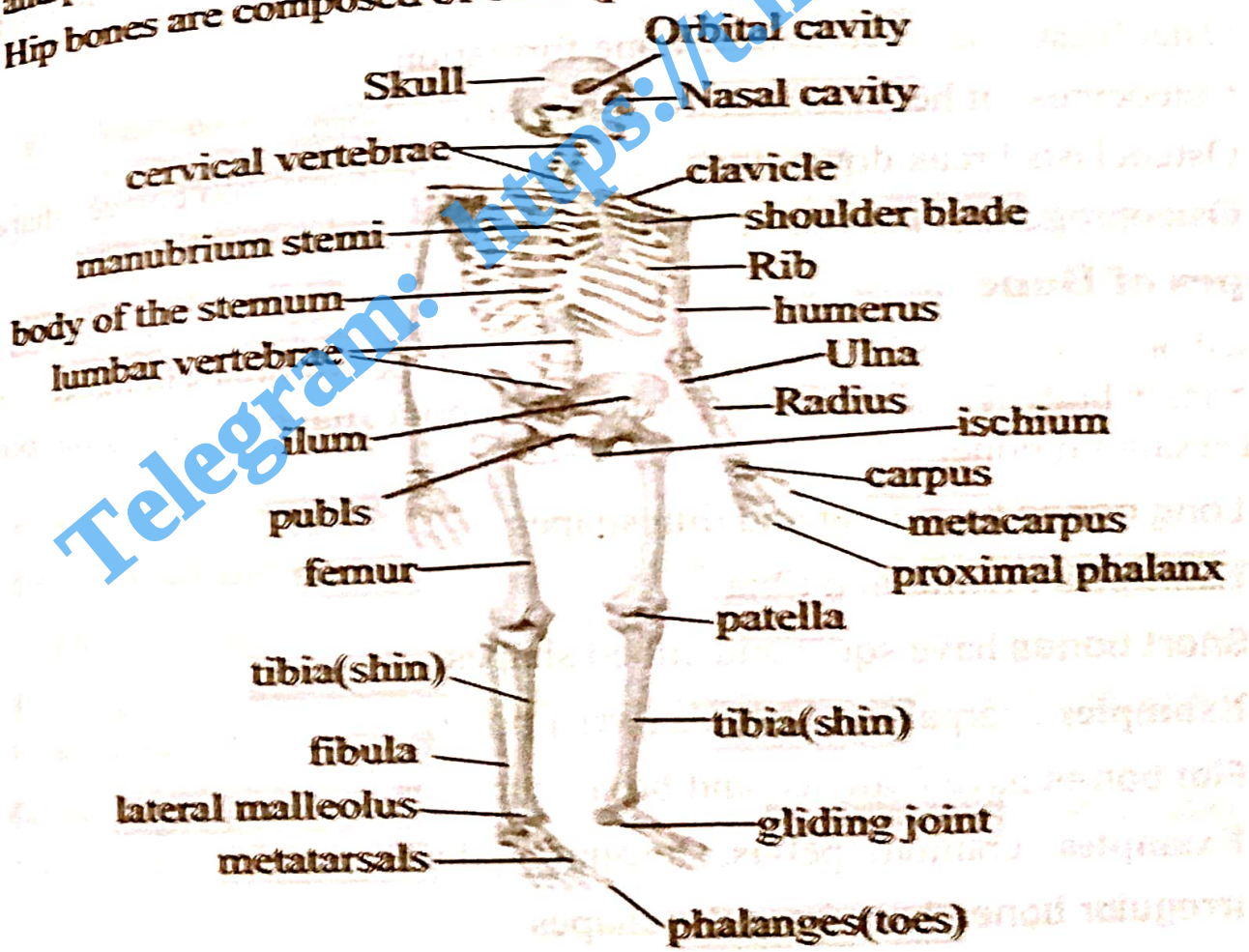
a) **Shoulder bone is composed of clavicle and scapula**

- **Clavicle** - is connected to the sternum on the front
- **Scapula** - is a bone at the back of the shoulder

b) **Limbs : it consists of fore and hind limbs**

- The fore limbs consist of Humerus (upper arms), radus (inner lower arm), ulna (out outer lower arm), carpals (wrists), metacarpals (hands) and phalanges (fingers). They are used for manipulation and grasping.
- The hind limbs consist of femur (upper leg or thigh), tibia and fibula (lower leg or shin), patella (knee cap), tarsal (ankles), metatarsal (feet), and phalanges (toes). They are used for locomotion.

c) Hip bones are composed of coxa (pelvic girdle) and sacrum.



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Parts of the appendicular skeleton

appendicular skeleton parts	Number of bones
Shoulde girdle	4
Pelvis	2
Arms and forearms	6
Thighs and legs	8
hands	54
Feet and ankle	52
Total bone	126

5.1.3. The General Characteristics Bone

Bones have three different tissue layers made from different kinds of cells. Bone is a combination of a protein called **collagen** and a molecule called **calcium** and **phosphate,** which makes it strong and light weight structure. Functions of bones are movement, support, protection, produces blood cells, storing and releasing mineral and fat. Bone is composed of four types of cells. These are :

- Osteoblasts : involve in new bone formation
- Osteocytes : it helps to mature bones of newborns
- Osteoclasts break down bones and help them form into correct shapes
- Osteoprogenitor are important in repair of fracture

Types of Bone

Based on their shapes, there are five different types of bones in the skeleton of human body. These are : long bone, short bone, flat bone, irregular bone, and sesamoid bone.

1. **Long bones** have long and thin shapes

   Examples : femur and tibia

2. **Short bones** have squat and cubed shapes

   Examples : carpals and tarsals (wrist bone)

3. **Flat bones** have flattened and broad surfaces

   Examples : cranium, pelvis, sternum, and ribs.

4. **Irregular bones** have irregular shapes

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Examples : vertebrae, sacrum, coccyx, mandible, ilium, pubis

5. **Sesamoid bones** are small, flat bones and are shaped similarly to a sesame seed, embedded in muscle or tendon

Examples : patella (knee cap)

5.1.4 Joints

Joint are the parts of the body where two or more bones meet with binding tissues ; they are useful for organisms in various ways :

- To keep bones far apart to prevent bones rubbing one another
- To hold the bones in place

Types of joints

Joint can be classified as : immovable joints and movable joints

Immovable joints (fixed joints) : are bones that are fixed in place and they do not make any movement at all. Examples are bones of sternum, pelvis, etc.

Movable joints are joints that allow twisting or bending of the body at different angle. Movable / synovial/ joints can be classified in to different types :

1. **The ball and socket joint :**
   - Makes movement in all directions
   - They are formed when the ball shaped end of one bone fits in to a cup shaped section of other bone.

   Examples : shoulder and hip joints.
2. **Hinge joints** : a joint that opens and closes in one direction along plane only, like a door.

   Examples : elbow and knee joints.
3. **Condyloid joints :** joints that do not rotate but allow movements.

   Example : finger and jaw.
4. **Pivot joints** are rolary joints in which one bone swivels in a ring.
   Examples : ulna, radius and neck.
5. **Gliding joints :** allows adjacent bones to slide over one another ; it allows only limited movement.

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Examples : wrist joint.

6. Saddle joint: enables movement back and forth, side to side, and cannot rotate. Example : thumb base.

Illustrative Examples

6. Which one of the following is not true about muscle contraction ?
   A. Myoin head and actin bind to form a cross bridge
   B. Calcium is released from sarcoplasmic reticulum and binds on tropomysine
   C. Sarcomere is shortened
   D. Energy is supplied to myosin from ATP

Explanation : - Steps of muscle contraction :

- Calcium ion binds to **troponin** on actin
- The myosin head binds to actin, forming a **cross bridge**
- **Power stroke** to pull the actin filaments
- Energy is supplied to myosin from **ATP**
- **Slidin of myosin over actin** filaments
- **Sarcomere** is shortened, thus **muscle is contracted**

Answer : B

7. Which one of the following is not a part of an axial skeleton ?
   A. Femur and tibia
   B. Vertebrae
   C. Cranium
   D. Rib - cage

Explanation :-

- Parts of axial skeleton are : skull, sternum, ribs and vertebral column. Femur and tibia are the part hind limb / leg, which is a part of appendicular skeleton.

Answer : A

8. The smallest and largest number of bones from appendicular skeleton is located in ___ and ___, respectively ?
   A. hands / pelvis
   B. legs / hands
   C. shoulder / legs
   D. pelvis / hands

Explanation :- the number of bones of bones in hands is 54, legs (thigh and legs) are 8, shoulder girdle is 4, and pelvis is 2.

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Answer : D

9. A component of the appendicular skeleton is the
   A. Skull
   B. Femur
   C. rib cage
   D. vertebral column

Answer : B

10. The **incorrect** combination of bone shape and its example is _____ ?
   A. Flat bone - pelvis
   B. irregular bone - vertebrae
   C. long bone -femur
   D. short bone - ribs

Explanation :- based on the shape, bones are classified as :

- **Long bones :** Examples : femur and tibia
- **Short bones :** Examples : carpus (wrist bone)
- **Flat bones :** Examples : cranium, pelvis, sternum, and ribs
- **Irregular bones :** Examples vertebrae, sacrum, coccyx, mandible, palatine, nasal, temporal
- **Sesamoid bones :** bones imbedded in the muscle. Example : patella

Answer: D

11. Which of the following joint bone is fixed / immovable ?
   A. the knee
   B. the elbow
   C. the skull
   D. the neck

Explanation :- Immovable joints (fixed joints) : are bones that are fixed in place and they do not make any movement at all. Examples are bones of sternum, pelvis, etc.

- The skull (cranium) has bones which are fused in to suture, which is immovable.

Answer : C

12. Saddle joints are located in _____ parts of the bone ?
   A. Thumb
   B. knee
   C. hip
   D. shoulder

Explanation :- there are 6 types of movable joints in our body. These are :

- **The ball and socket joint :** Examples : shoulder and hip joints
- **Hinge joints :** Examples : elbow and knee joints

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- **Condyloid joints :** Example : finger and jaw
- **Pivot joint Examples :** ulna, radius an neck
- **Gliding joints :** Examples : wrist joint
- **Saddle joint :** Example : thumb base

13. A freely movable, fluid - filled joints is a (n) ____ joint

Answer: A

A. Otolithic
B. hydrostatic
C. Synovia
D. Golgi

Explanation :- In Synovial joints, ligaments bind the two bones togther, and forming a capsule containing lubricating Synovial fluid

14. The thick filaments in a sarcomere are composed mainly of

Answer : C

A. Calcium
B. Myosin
C. actin
D. rhodopsin

Explanation :- Each Sarcomere contains two sorts of protein filament .There are :

- Thin filament of **action**
- Thick filament of **Myosin**

15. The contractile portion of a skeletal muscle fibril is the ____

Answer : B

A. myofibril
B. thick filament
C. thin filament
D. sarcomere

Answer: A

5 . 2 The Reproductive System

The human reproductive system includes. The male reproductive system which functions to produce and deposit sperm . The female reproductive system which function to produce egg cells, and to protect and nourish the fetus until birth.

5 . 2 . 1 The Male Reproductive System

The male reproductive system consists of the external genital organs (penis, testes, and the scrotum), and the internal parts (prostate gland,

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Cowper's gland, seminal vesicle, vas deferens, and urethra). It is important to produce sperm, sex hormone, and other component of semen.

The male external genital organs include :

1. The penis :

It is the male copulatory organ used deposit sperm into the vagina ; contains soft, spongy tissue as well as muscles, fibrous tissue, veins, arteries, and the urethra. These allow the penis to perform **urinary and sexual function**

- The sexual function of penis has two stages **erection and ejaculation.** An erection is the stiffening of the penis caused by sexual arousal and / or physical stimulation. An erection occurs when there is an increased flow of blood through arteries into the corpus cavernosa and **corpus cavernosa** and **corpus spongiosum,** causing the penis to enlarge.
- The engorgement compresses the veins through which blood usually exits the penis. This "traps" the blood and helps sustain the erection.

2. Urethra

The urethra is tube that connects the urinary bladder to remove urine from the body ; in males, the urethra allows the passage of urine and sperm / semen to outside during pee and ejaculation, respectively.

3. The scrotum

The scrotum is a sack of thick skin that protects the testes and controls the temperature of the testes for suitable sperm creation.

- It contains seminal vesicles, vas deferens, testicles (testes), and prostate gland that constitute all the remaining components of the male reproductive system.

4. Testes

Testes are oval pair bodies found in the scrotum ; responsible for the production of testosterone (male sex hormones) and sperm cells.

- It contain **seminiferous** tubule, cells to produce immature sperm cells

The male internal genital organs include :

1. **Epididymis :** matures and stores sperm cells in coiled tubules
2. **Vasdiference :** carries sperm from the epididymis to the urethra
3. **Glands / vescle :**

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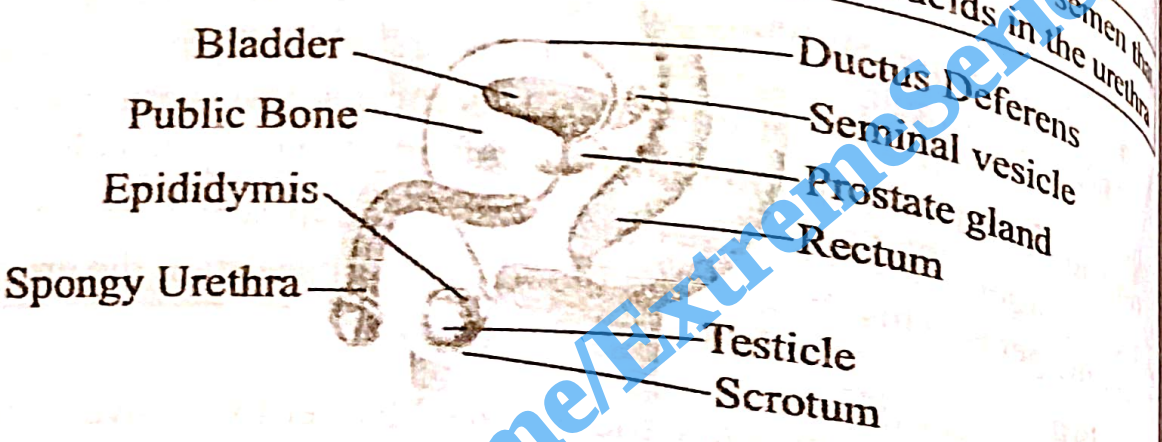
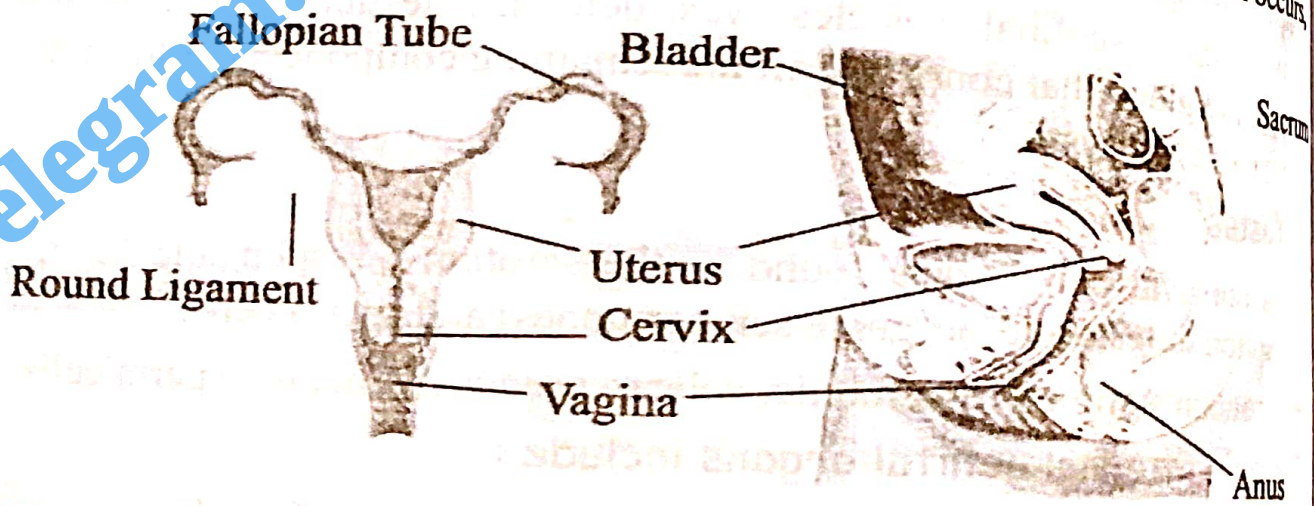
Glands / vesicle	Functions
seminal vesicle	secretes fructose into the semen, which provides energy for the sperm
prostate gland	secretes an alkaline buffer into the semen to protect the sperm from the acidic environment of the vagina
Cowper's gland	secretes mucus - rich fluids into the semen that may protect the sperm from acids in the urethra



5 . 2 . 2 The Female Reproductive Systems

The female reproductive system includes the fallopian tube, ovaries, uterus, vagina, accessory glands, and external genital organs. Its functions include producing gametes called eggs, secreting sex hormones (such as estrogen), and providing a site for fertilization, gestating a fetus if fertilization occurs, giving birth to a baby.



The female external reproductive organs include :

- **Labia majora :** The labia majora : ("large lips") enclose and protect the other external reproductive organs. During puberty, hair growth occurs on the kin of the labia majora, which also contain sweat and

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oil - secreting glands.

- **Labia minora :** The labia minora ("small lips") can have a variety of sizes and shapes. They lie just inside the labia majora, and surround the openings to the vagina (the canal and urethra). This skin is very delicate and can become easily irritated and swollen.
- **Bartholin's glands :** It is located next to the vaginal opening on each side and produces a fluid (mucus) secretion.
- **Clitoris :** is a small, sensitive protrusion that is comparable to the penis in males. Located in where the two labia minora are meet ; it is covered by a fold of skin called the **prepuce**, which is similar to the foreskin at the end of the penis. Like the penis, the clitoris very sensitive to stimulation and become erect.

The female internal reproductive organs include :

- **Vagina :** The vagina is a canal that joins the cervix (the lower part of the uterus) to the outside of the body ; it provides a passage way for sperm and menstrual flow. It is also known as the birth canal.
- **Uterus (womb) :** The uterus is a hollow, pear - shaped organ that is the home to a developing fetus. Is divided into two parts : the cervix and the corpus.
- **The cervix:** which is the lower part that opens into the vagina ; separate the vagina from the uterus, hold the fetus in place during pregnancy and dilate during birth.
- **Ovaries :** The ovaries are small, oval - shaped glands that are located on either side of the uterus. The ovaries produce eggs and hormones called estrogen and progesterone.
- **Fallopian tubes / oviduct :** This are narrow tubes that are attached to the upper part of the uterus. It serves as pathways for the ova (egg cells) to travel from the ovaries to the uterus, it also a site of fertilization of an egg by sperm. The fertilized egg then moves to the uterus, where it implants into the uterine lining.
- **Fimbria :** Sweep the ovum into the oviduct following ovulation

Illustrative Examples

16. A vesicle or gland that produces fructose to supply energy to the sperm cells ?

A. Prostate gland B. Bartholin gland

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C. Cowper's gland D. Seminal vesicle

Explanation :-

Glands / vesicle	Functions
Seminal vesicle	Secretes fructose into the semen, which provides energy for the sperm
Prostate gland	Secretes an alkaline buffer into the semen to protect the sperm from the acidic environment of the vagina
Cowper's gland	Secretes mucus - rich fluids into the semen that may protect the sperm from acids in the uethra
Bartholin gland	Produce a fluid (mucus) secretion in the vagina

Answer : D

17. A male reproductive part through which Semen and urine pass to outside is _____ ?

A. Vas difference C. Testes

B. Urethra D. Seminiferous tubule

Explanation:- Urethra is a male reproductive tube, ejaculation of sperm/ semen and urine is removed to the outside.

Answer : B

18. A part of a female reproductive part, where fertilization is takes place ?

A. Ovary C. Uterus

B. Oviduct D. Cervix

Explanation : Fallopian tube / oviduct is a narrow tube, attached to the upper part of the uterus. A pathway of ove (egg cells) from ovaries to uterus, and a site of fertilization of an egg by sperm.

Answer : B

19. Clitoris of the female is _____ ?

A. Sensitive protrusion C. Covered by prepuce

B. Comparable to penis D. All

Explanation :- Clitoris is a small sensitive protrusion that is comparable to the penis in males and become erect ; located at the two labia minora meet ; it is covered by a fold of skin, **prepuce** similar to the foreskin at

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